Measuring the inboard side scrape-off layer of DIII-D plasmas using Swing-Probes

by

Cedric Tsui

A thesis submitted in conformity with the requirements
for the degree of Doctor of Philosophy

Graduate Department of Aerospace Engineering
University of Toronto

© Copyright by Cedric K Tsui 2014

Measuring the inboard side scrape-off layer of DIII-D plasmas using Swing-Probes

Cedric Tsui

Doctor of Philosophy

Graduate Department of Aerospace Engineering
University of Toronto

2014

Abstract

The scrape-off layer (SOL) plasma of a tokamak often has a complicated spatial dependence. The temperatures, densities and flow speeds can vary significantly on the same magnetic flux tube at different poloidal locations. To fully understand the plasma variation, we must make active measurements along the full length of the flux tubes by expanding our diagnostic capability to include the critically under-diagnosed inboard side. To accomplish this, a new pair of in-situ reciprocating Mach probes called Swing-Probes have been developed and deployed on the DIII-D centerpost. This design is unique in that the probe swings vertically through the SOL plasma, taking measurements along a 180° arc with a 20 cm radius. Two electrodes maintain a Mach-pair orientation throughout the swing and provide measurements of saturation current, electron temperature, and parallel flow speeds. The probes can handle very high heat fluxes and have taken measurements up to the Last Closed Flux Surface (LCFS) in high-powered H-Mode at 10 MW.

The Swing-Probe temperature and density measurements have been verified against the floor Langmuir probes, the core Thomson scattering and the divertor Thomson scattering systems in DIII-D for conditions where poloidal variation are expected to be small. Measurements have been taken across a wide range of plasma conditions and

provide informative relationships between the plasma parameters at the entrance of the inner divertor and the crown of the plasma.

The low-turbulence plasmas on the inboard scrape-off layer make it possible to clearly quantify the sheath-expansion around Langmuir probes. An I-V fitter has been developed which can account for sheath-expansion in a theoretically consistent way, improving the reliability of Langmuir probe data analysis.

In an inner-wall limited experiment in DIII-D requested by ITER, the Swing-Probes made the first Langmuir probe measurements of an enhanced heat flux feature just outside the LCFS. These measurements match those made using infrared camera images of the centerpost in DIII-D and in other tokamaks and confirm that this feature should be accounted for in the designs of plasma facing surfaces in ITER.

Acknowledgments

This doctoral project would have been impossible without a tremendous amount of support. I would like to thank everyone…

…for their work on the design or construction of the Swing-Probes:

Doug Taussig, Matt Watkins, David Ayala, Jim Kulchar, Armando Chavez, Matt Hansink, Chris Murphy, Rick Lee, Steve Celle, Doug Nelson, Chris Chrobak, Kurt Holtrop, and the rest of the DIII-D Vacuum Committee, Jim Leuer, Karl Lapinska, David Piglowski, Chris Stacy, Matt Madruga, and the GA Machinists.

…for their part in design reviews:

Jose Boedo, Jon Watkins, Dmitri Rudakov, Brian LaBombard, Rick Moyer, Michael Schaffer, Neil Brooks, Steve Lisgo, Al Hyatt, Arnold Kellman, Pete Taylor, Gary Jackson, Nick Eidietis, Adam McLean, Eddy Gonzales, and Howard Grunloh.

…for their help, guidance, and mentorship:

Steve Allen, Max Fenstermacher, David Hill, Rajesh Maingi, David Eldon, Charles Lasnier, Clement Wong, Mike Makowski, Tom Osborne, Rich Groebner, Dan Thomas, Mathias Groth, Gary Porter, and Jeannie Buchert.

…to the entire DIII-D team,

… for financial support:

The Natural Sciences and Engineering Research Council of Canada, the Ontario Graduate Scholarship, and General Atomics (San Diego).

… to my advisors:

Rejean Boivin, Tony Leonard, Clinton Groth, David Elder, and Jim Davis.

…and to my supervisor Peter Stangeby.

Table of Contents

Acknowledgments iv

Table of Contents v

List of Tables ix

List of Abbreviations x

List of Figures xi

Chapter 1 Introduction 1

1.1 Magnetic Confinement Fusion 1

1.2 Motivations for Study 1

1.3 Thesis Objectives 2

1.4 DIII-D 3

1.5 In-Vessel Probes 5

Chapter 2 Swing-Probe Design Requirements 6

2.1 Design Process 6

2.2 The DIII-D Vessel, Centerpost, and Vacuum Environment 7

2.3 Risks to DIII-D 9

2.3.1 Jamming in the path of the plasma 10

2.3.2 Thermal Quench in a Worst Case Disruption 11

2.3.3 Halo Currents, and Short-Circuits to Plasma Currents 12

2.3.4 Runaway Electrons 13

Chapter 3 Swing-Probe Physical Design 15

3.1 Overall Design 15

3.2 Probe Head and Electrodes 18

3.3 Probe Arm 24

3.4 Signal Cables 27

3.5 Drive Train 27
3.5.1 Conservative Designs 27
3.5.2 Drive Train Performance Issues in the 2012 Campaign 29
3.5.3 Horizontal Drive Shaft 31
3.5.4 Bushings and Bearings 32
3.5.5 Gears 34
3.6 Spring Assembly 35
3.7 Actuators 37
3.7.1 Winding 38
Chapter 4 Swing-Probe Control and Data Acquisition 40
4.1 Coordinates 40
4.2 Coil Back-EMF 42
4.2.1 Using the Back-EMF to Determine the Probe's Motion 42
4.3 Physical Model 44
4.3.1 Eddy Current Drag 44
4.3.2 Assembling the Physical Model 45
4.4 Electronics and Data Acquisition 50
4.5 Control systems and Interlock Integration with DIII-D 54
Chapter 5 Langmuir Probe Data Analysis 56
5.1 Reliable Langmuir Probe Fitter 56
5.2 Sheath Expansion 57
5.3 Child-Langmuir sheath expansion function 60
5.3.1 Evaluating c 61
5.3.2 Comparing I-V Fitting Models 64
5.3.3 Motivating the ¾ power scaling 70

5.3.4 Conclusion 71
5.4 Alternate Sheath expansion Models 72
5.4.1 I-V Fitter #2 uses separate ion and electron collection areas 72
5.4.2 I-V Fitter #3 includes sheath contraction for $V > V_f$ 73
5.4.3 Comparing the three sheath expansion fitters 74
5.4.4 Conclusion 78
5.5 Swing-Probe Specific Data Analysis 78
Chapter 6 Validation of Swing-Probe Measurements 80
6.1 Validation Against the Divertor Target Langmuir Probes 80
6.2 Validation Against the Divertor Thomson Scattering 83
Chapter 7 2013 Campaign Experiments 85
Chapter 8 L-mode Divertor Detachment Experiment 88
8.1 The Simplest Shot 89
8.2 EFIT Uncertainties 91
8.3 Trends Across the Density Scan 93
Chapter 9 Inner Wall Limiter Heat Load Experiment 96
9.1 Background 96
9.2 The DIII-D IWL Experiment 99
9.3 Swing-Probe Results 105
9.4 IR Camera Results 114
9.5 Results from Other Diagnostics 116
9.6 Core Thomson Scattering Results 121
9.7 Conclusion for ITER 125
9.8 Discussion and Hypotheses 126
9.9 Further Work 128

Chapter 10 Conclusions and Contributions 130

10.1 Conclusions 130

10.2 Original Contributions 132

References 134

List of Tables

Table 5-1 The magnitude of the sheath expansion effect tabulated for the condition where $V-V_f = -100$ V. Sheath expansion is affected most strongly by the density. 64

Table 7-1: A list of Swing-Probe participation in the 2013 campaign. Useful data was collected in the tasks highlighted in green. 86

Table 8-1: Shifts to the location of the LCFS for shot 152835 and the resulting T_e and n_e values for the LCFS. 92

Table 8-2: The power values used to calculate input power to the SOL, the shifts to the location of the LCFS for the shots in the density scan calculated using power balance and the resulting T_e and n_e values for the LCFS. 93

Table 9-1: The shots from the 2013 DIII-D IWL experiment. Shots and times where the Swing-Probe operated are shown in bold. 100

Table 9-2: Parameters for best double-exponential fits to the Swing-Probe $q_{||}$ profiles calculated using individual values of J_{sat} and values from the fit to T_e. The uncertainty values in the averages are two standard deviations from the column. 112

List of Abbreviations

B_t	Toroidal Magnetic Field
EFIT	The magnetics Equilibrium FITting of the plasma flux surfaces, i.e. the poloidal expression of the shape of the plasma and its individual flux surfaces.
ELM	Edge Localized Mode
eV	Electron Volt
HFS	High Field Side (aka inboard side)
I_{sat}	Ion saturation current
J_{sat}	Ion saturation current density
LCFS	Last Closed Flux Surface aka the 'Separatrix'
LFS	Low Field Side (aka outboard side)
$M_{\|\|}$	Mach Number
n_e	Electron Density
OEDGE	Osm Eirene Divimp edGE. A code used to simulate and investigate the physics in scrape off layer plasmas.
PFZ	Private Flux Zone
$q_{\|\|}$	Parallel Heat Flux
RCP	Reciprocating Probe
sep	Separatrix or Last Closed Flux Surface. This abbreviation is often used as a subscript to variables defined at the separatrix, i.e. R_{sep}.
SOL	Scrape Off Layer
T_e	Electron Temperature
TS	Thomson Scattering
VDE	Vertical Displacement Event, a type of disruption where the plasma moves upwards or downwards in an uncontrolled manner.
V_f	Floating Potential

List of Figures

Fig 1-1 A view from inside the DIII-D tokamak. 4

Fig 1-2 A poloidal cross-section of the DIII-D vessel overlaid with the EFIT of a lower single null plasma. 5

Fig 2-1 A horizontal cross-section of the DIII-D centerpost with attached armour tiles. The tiles are approximately 13 cm x 13 cm x 5.5 cm. Taken from [4]. 8

Fig 2-2 A poloidal cross-section of the DIII-D centerpost with attached armour tiles. 8

Fig 2-3 A VDE and accompanying halo current in Alcator C-Mod. 13

Fig 3-1 The locations of the main active Scrape Off Layer diagnostics in DIII-D mapped over a poloidal projection of a typical plasma shape. The swing probes measure along the red and blue arcs. 15

Fig 3-2 The upper Swing-Probe with the custom centerpost tiles installed over top. The drive coils are hidden behind the seam left of the slot. 16

Fig 3-3 (right) A schematic of the Swing-Probe components. The blue oval represents the slot in the tiles visible in figure Fig 3-2. (left) A photograph of a Swing-Probe on the DIII-D centerpost with the graphite tiles and the actuator cover removed. A temporary stainless steel shield covers the probe head and electrodes to protect them during installation. 17

Fig 3-4 2012 model of the probe head, which is ¾'' in diameter at the widest part of the interface shield (transparent). 20

Fig 3-5 The electrodes crumbled during the 2012 campaign, electrically shorting to each other and to the probe head. 21

Fig 3-6 An exploded view of 2013 design for the probe head 22

Fig 3-7 2013 model of the probe head including the carbon interface shield (not shown in Fig 3-6) 22

Fig 3-8 A photo of the 2013 probe head as installed in the completed assembly 22

Fig 3-9: The spring-catch. Six layers of Inconel foil are interleaved such that they partially overlap in the center. The impact energy is dissipated by friction between the sheets as the slide past each other, which reduces how high the probe bounces upon hitting the spring catch. The rainbow coloured deposits are (roughly) within line of sight of the plasma. 23

Fig 3-10 (left) A photo of the probe arm as it is installed in DIII-D with the tiles removed. 25

Fig 3-11 (right) A schematic of the probe arm. 25

Fig 3-12 A photo of the alumina sandwich blocks connecting the probe arm to the main driveshaft. Notice the Inconel spring holding the pyrolytic graphite washers in place. 26

Fig 3-13 A schematic of the horizontal drive shaft with part of the mated probe arm. The bearing and their mounts are not shown, nor is the spring assembly crank wheel that mounts on the end of the shaft on the right. For scale, the hub of the miter gear is 1.0'' is diameter, and the pyrolytic graphite washers are 0.25'' tall. 32

Fig 3-14 The bushings used in the 2012 campaign coated in Dicronite (left) and the custom ball bearings that replaced them (right) 33

Fig 3-15 A photo of the zirconia ball bearings and their mounting hardware. The axial tolerances were adjusted using washer shims set between the bearing and the collar (bottom right). Actuator coils are shown with and without the bearing installed. Note that the bushing mount has been slid back to reveal the bearing, and does not normally come in contact with the coil bracket. 34

Fig 3-16 The spring assembly and the motion of the crank wheel. 36

Fig 3-17 A schematic of the actuator coil design. 37

Fig 3-18 One of the actuator coils mounted in the Swing-Probe frame. 39

Fig 4-1: The toroidal coordinates. 41

Fig 4-2: The coil angle ϕ_{coil} 41

Fig 4-3: The probe angle θ_{probe} 41

Fig 4-4 The eddy current paths in the actuator coil frames. κ_{loop} represents the current path in the top image and κ_{face} represents the one in the bottom image. 45

Fig 4-5 A typical current control waveform for the fireing sequence of the lower Swing-Probe. The coils hold the probe against the armed limit switch until 100ms, drive the probe forward from 100ms to 260ms, changing polarity as necessary. After 260ms the forward torque is decreased to allow the probe to decelerate. 48

Fig 4-6 The simulated and measured back EMF from a typical swing compared to simulated values from a well-tuned physical model. Differences are due to the backlash in the gears, e.g. the upper actuator transitions from accelerating to decelerating at 250ms, and the shift within the gears causes a jump in the back-EMF. After the probe lands at 300ms it bounces several times causing the back-EMF to flip polarity rapidly. 49

Fig 4-7 The probe angle simulated in the model and calculated from the back EMF..... 49

Fig 4-8 The probe's angular velocity simulated in the model and calculated from the back EMF. 50

Fig 4-9 The electronics setup for the DIII-D Swing-Probes..... 52

Fig 4-10 A photo of the machine pit rack's communication panel. The yellow boxes at the top are the fibre optic links that transmit data to the Annex from the data processing and control circuit board in the center of the photo. At the bottom, the first of five Kepco power supplies is visible. 54

Fig 5-1 (top) a schematic of the Swing-Probe head and electrodes. (bottom) The collection surface area of the Swing-Probe electrode. The black area represents the projected geometrical surface area A_0 while the yellow area represents the additional collection area due to the sheath. 58

Fig 5-2 An example of an I-V characteristic where the ion saturation branch does not actually saturate due to sheath expansion. The data are digitized at 500 kHz. It takes < 500 μs to collect the data shown here. 59

Fig 5-3 Sheath expansion example from typical ohmic shot 152838 sweep#90. 62

Fig 5-4 Sheath expansion example from typical ohmic shot 152838 sweep #98. 62

Fig 5-5 Sheath expansion example from low I_p plasma shot 155650 sweep#87. 62

Fig 5-6 Sheath expansion example from low I_p plasma shot 155650 sweep#91. 62

Fig 5-7 Sheath expansion example from high power low B_t shot 153836 sweep#360. 63

Fig 5-8 Sheath expansion example from high power low B_t shot 153836 sweep#85. 63

Fig 5-9 An I-V characteristic example with a significantly non-saturating ion branch. .. 66

Fig 5-10 A moderately noisy I-V characteristic example. 67

Fig 5-11 An I-V characteristic example with significant plasma fluctuations. 68

Fig 5-12 A very low-noise I-V characteristic example. 69

Fig 5-13 An example of an independent power-fit to the ion branch from -100V to -250V. For this example,the fit (blue line) has the values I_{sat}^{+} = 0.025 A, a = 1.4e-4, and b = 0.87, see eq. 5-9..... 71

Fig 5-14 An exaggerated schematic showing the ion collection area increasing with the $3/4^{th}$ power of $(V - V_f)$. Simple Sheath Expansion Fitter (solid black), I-V Fitter #2 (dashed red), I-V Fitter #3 (dashed blue). 74

Fig 5-15 An I-V characteristic example with a significant sheath expansion effect. 75

Fig 5-16 A high temperature noisy I-V characteristic example. 76

Fig 5-17 A noisy I-V characteristic example. 77

Fig 6-1: The path of the lower Swing-Probe (red) and the locations of the fixed Langmuir probes in the lower divertor (black). Note that the Swing-Probe crosses the poloidal location of the highest fixed probe..... 81

Fig 6-2: Agreement is shown between the Swing-Probe data (orange) and the fixed Langmuir probes in the target (black). The rightmost fixed probe datapoint at $R-R_{sep}$ = 4.2 cm shares the same R,Z coordinates as the Swing-Probe when it passes. The error-bars for the fixed-probe data represent one standard deviation. 82

Fig 6-3: The plasma EFIT overlaid with the lower Swing-Probe path (green and orange) and the divertor Thomson measurement locations (blue). Note that the Swing-Probe and the TS share measurements on many flux surfaces both inside and outside the LCFS. 83

Fig 6-4: The averaged values for the Divertor TS (one value per channel) and for the Swing-Probe (1 mm bin averaged). We see good agreement between the electron temperature and the density measurements made by the two diagnostics. 84

Fig 8-1: The EFIT for the divertor detachment experiments where the lower Swing-Probe reached the LCFS. The locations of some of the spatially localized diagnostics used in this experiment are shown. 89

Fig 8-2: A comparison between the core Thomson scattering (1 average value per channel) and the lower Swing-Probe (1mm bin averages) for the lowest density shot of the L-mode detachment study. The uncertainty-bars show the scatter in the data. The Thomson T_e data shows significant scatter between 1 and 3 cm due to fluctuations in the plasma. 90

Fig 8-3: 1mm bin averages of the parallel flow velocity (left) and the floating potential (right) measured by the swing probe in the lowest density shot of the L-mode detachment study. Positive flow is in the $+\hat{\phi}$ direction and towards the inner target..... 91

Fig 8-4: Comparison of T_e and n_e between the core Thomson scattering (black) and the lower Swing-probe (red) close to the LCFS. The error bars represent the scatter in the data. Averages were taken over the range 0.15 cm ≤ R-Rsep ≤ 0.4 cm. TS data have been shifted using power balance . Each datapoint represents one shot. 94

Fig 8-5: Comparison of T_e and n_e between the core Thomson scattering (black) and the lower Swing-probe (red) in the far SOL. The error bars represent the scatter in the data. Averages were taken over the range 3.0 cm ≤ R-Rsep ≤ 3.5 cm. TS data have been shifted using power balance. 94

Fig 8-6: Comparison of p_e between the core Thomson scattering (black) and the lower Swing-probe (red) near the LCFS (left) and in the far SOL (right). TS data have been

shifted using power balance. 95

Fig 9-1: Measurements of the parallel heat flux of (a) an inner wall limited and (b) an outer wall limited plasma in JET mapped to the outer-midplane, inferred from infrared thermography Taken from [54]. 97

Fig 9-2: The elongated (κ ≅ 1.6) plasma shape. 99

Fig 9-3: The round (κ ≅ 1.25) plasma shape (vertical location varied from shot to shot). The diagnostics used in this experiment are shown. 99

Fig 9-4 Data from the lower-half arc of 155646 101

Fig 9-5: The J_{sat} (left) and T_e (right) data for the lower-half swing of 155646. The trend line is from an exponential best fit T_e = 30.785 EXP[-25.14(R-Rsep)] 102

Fig 9-6: $q_{||}$ calculated from the smooth T_e values and averaged into 1mm bins are shown in red. For comparison, the unbinned data are shown in grey. 103

Fig 9-7: Example Least Squares Fit 104

Fig 9-8: Example Far-SOL Fit 104

Fig 9-9: Swing-probe $q_{||}$ fit from shot 155645 @ 2000ms lower swing 106

Fig 9-10: Swing-probe $q_{||}$ fit from shot 155645 @ 2000ms upper swing 106

Fig 9-11: Swing-probe $q_{||}$ fit from shot 155646 @ 2000ms lower swing 107

Fig 9-12: Swing-probe $q_{||}$ fit from shot 155646 @ 2000ms upper swing 107

Fig 9-13: Swing-probe $q_{||}$ fit from shot 155648 @ 3500ms lower swing 108

Fig 9-14: Swing-probe $q_{||}$ fit from shot 155648 @ 3500ms upper swing 108

Fig 9-15: Swing-probe $q_{||}$ fit from shot 155649 @ 3500ms lower swing 109

Fig 9-16: Swing-probe $q_{||}$ fit from shot 155649 @ 3500ms upper swing 109

Fig 9-17: Swing-probe $q_{||}$ fit from shot 155650 @ 2000ms lower swing 110

Fig 9-18: Swing-probe $q_{||}$ fit from shot 155650 @ 2000ms upper swing 110

Fig 9-19: Swing-probe $q_{||}$ fit from shot 155651 @ 2000ms lower swing 111

Fig 9-20: Swing-probe $q_{||}$ fit from shot 155651 @ 2000ms upper swing 111

Fig 9-21: $q_{||}$ profiles inferred from IR camera data from the DIII-D centerpost from the toroidal angles with the clearest signals (close to the limiter apexes) as a function of R-R_{sep}. 114

Fig 9-22: A comparison between the $q_{||}$ profiles for 155648 normalized to 1 at the LCFS. The normalized IR camera data are shown in black. The normalized Swing-Probe data are shown in green for the lower swing and blue for the upper swing, which is closer to the target. The two diagnostics show similar short features. 115

Fig 9-23: The electron temperature and density data from the Core Thomson, Divertor Thomson, and Swing-Probe for shot 155645 at ~2000 ms (elongated shape). We note the excellent agreement in T_e and that the density has good agreement in magnitude but not necessarily in the profile shape. 116

Fig 9-24: The electron temperature and density data from the Core Thomson, Divertor Thomson, Midplane Reciprocating Probe, and Swing-Probe for shot 155646 at ~2000 ms (elongated shape). We see good agreement among all of the diagnostics and do see signs of a short feature in the MP-RCP and perhaps in the LFS TS. However, without the SP data, the data from the other diagnostics would have too much uncertainty to confirm the existence of the short exponential component. 117

Fig 9-25: The electron temperature and density data from the Core Thomson, Divertor Thomson, Midplane Reciprocating Probe, and Swing-Probe for shot 155648 at ~3500 ms (round shape). The TS data is sparse for this shot. The MP-RCP data has an inflection point, but it is well inside the LCFS 118

Fig 9-26: The electron temperature and density data from the Core Thomson, Divertor Thomson, Midplane Reciprocating Probe, and Swing-Probe for shot 155649 at ~3500 ms (round shape). The data from the TS and MP-RCP are sparse but in excellent agreement with the SP data. 118

Fig 9-27: The electron temperature and density data from the Core Thomson, Divertor Thomson, Midplane Reciprocating Probe, and Swing-Probe for shot 155650 at ~2000 ms (elongated shape). Only the Swing-Probe shows a short feature. 119

Fig 9-28: The electron temperature and density data from the Core Thomson, Divertor Thomson, Midplane Reciprocating Probe, and Swing-Probe for shot 155651 at ~2000 ms (elongated shape). There is insufficient data in the far-SOL to infer the shape of the parallel heat flux profile for this shot. 119

Fig 9-29: Core Thomson Scattering T_e and n_e data 1 mm bin averages for 155645 from 2800-4100 ms 121

Fig 9-30: The 1 mm binned $q_{||}$ values for 155643 from 2800-4100 ms. 122

Fig 9-31: T_e and n_e bin averages for 155646 from 2800-4100 ms. There is a hot channel at R-Rmaj = 0.75 cm which we ignore for the $q_{||}$ calculation. 123

Fig 9-32: The 1mm binned $q_{||}$ values for 155643 from 2800-4100ms. 123

Fig 9-33: The Core Thomson scattering $q_{||}$ data combined over multiple shots (4 shots for the 1.2 MA data, 3 shots for the 0.6 MA data) and bin averaged. For κ ~ 1.6 (left) the dashed lines are the best-fit single exponential to the data from -2 cm to 2 cm. For κ ~1.25 (right) the dashed lines are the best-fit single exponential to the data from 0 mm to 6 cm. 124

Fig 9-34: A comparison of the λ_{short} values from Swing-Probe $q_{||}$ profiles and the $r_{D^+}^{pol}$ values at the OMP calculated assuming Ti=3Te. 127

Chapter 1 Introduction

1.1 Magnetic Confinement Fusion

Nuclear fusion promises to be an abundant, clean, and safe, energy source for the future. Fusion energy is effectively limitless since the fuels such as hydrogen and helium are the most abundant elements in the universe. Unlike most green power generation, fusion energy will be capable of providing reliable 'baseload' electricity. Fusion reactors will not produce CO_2 or any chemical exhaust and there is no long-lived radioactive waste burden left behind by a fusion reactor. Fusion reactors will be inherently safe since the reactions cannot take place when a fusion reactor loses power, so there is no possibility for an uncontrolled chain reaction. Fusion energy will have all of the necessary characteristics to replace fossil fuel energy generation. When nuclear fusion becomes commercially available, it will change the world.

1.2 Motivations for Study

A tokamak confines a hydrogen plasma using magnetic fields in the shape of a torus. The magnetic field lines that close on themselves without being interrupted by the vessel walls form nested closed flux surfaces that define the shape of the confined plasma. Outside of this region we find the scrape-off layer (SOL) where the flux surfaces intersect with the walls and the charged particles flow rapidly towards the walls where they recombine into neutrals. The boundary between these the confined plasma and the SOL is called the last closed flux surface (LCFS) or the separatrix. In Fig 1-2, the LCFS is the flux surface that forms the 'X' or x-point. While the confined plasma can largely be described using 1D equations, the plasma in the scrape-off layer is a more complicated region where plasma, neutral gasses, and solid surfaces all interact. Geometrically, the SOL must be treated with 2D or 3D equations. The complexity of the

SOL combined with our incomplete physics understanding means that the only tractable computational models are partly empirical, and lack predictive power. We continue to develop our computational models by validating them against experimental results [1].

Our lack of physics understanding of the SOL is one of the largest roadblocks impeding the design of power plant scale fusion reactors. A more thorough understanding of SOL physics would improve our ability to control the SOL plasma, providing a more hospitable environment for the plasma facing surfaces. An even larger obstacle is that we lack the ability to accurately predict what the SOL conditions will be when designing future devices. This large unknown is very challenging for engineering teams to contend with so it is very important that we continue to develop the physics basis for the SOL.

The toroidal geometry of a tokamak presents a fundamental challenge to studying the SOL. While the outboard side of a tokamak has an accessible vessel wall that is filled with diagnostics ports, the inboard side is difficult to access. (See Fig 1-1) There is no space behind the central column (aka the centerpost) for access ports, and even spectroscopic measurements are more challenging here since the inboard SOL is typically viewed by looking through the rest of the plasma. As a result the outboard SOL has been studied with far more detail than the inboard SOL.

Without constraining measurements in the inboard SOL, there is nothing to validate SOL codes against for nearly half of the simulation space. To understand the SOL plasma as a whole, we must study the inboard SOL, and to study the inboard SOL, we must measure it.

1.3 Thesis Objectives

Researchers at Alcator C-Mod addressed the need for active characterization of the inboard SOL by developing reciprocating Langmuir probes that are installed on the centerpost [2,3]. The location of these probes gives them direct access to the inboard

SOL. This thesis focuses on the development and deployment of a set of swinging probes installed on the DIII-D centerpost that are broadly based on those used in Alcator C-Mod. The research objectives are:

- To establish the design constraints and requirements for the Swing-Probes presented by the DIII-D environment as well as those necessary to minimize the risks presented to the tokamak and to ensure the smooth operation of its research program.
- To conceive, design, construct and install a pair of swinging probes on the DIII-D centerpost capable of measuring electron temperature T_e, density n_e, floating potential V_f, and parallel flow Mach number $M_{||}$.
- To develop the electronics, control systems, and physical control model needed to operate the Swing-Probes.
- To develop a novel method for rapid reliable analysis of Langmuir probe data that accounts for sheath expansion.
- To validate the new Swing-Probe data against existing diagnostics to prove their reliability.
- To take measurements over a wide range of conditions and compare them against DIII-D's existing diagnostics.
- To make preliminary investigations made of the physics of the inner SOL in DIII-D
- To use the Swing-Probes to investigate the existence of a narrow heat flux feature near the LCFS of inner wall limited plasmas hypothesized to be observable in the inboard SOL.

1.4 DIII-D

DIII-D is the largest tokamak in the US Department of Energy fusion program [4]. It is operated by General Atomics in San Diego California. DIII-D has a very versatile experimental capability combined with a comprehensive set of diagnostics. While it is a

medium size tokamak, (R = 1.695 m, a = 0.681 m, a total height from floor to ceiling of 2.72 m, centerpost radius = 1.01 m, outermost wall radius = 2.38 m) it is one of the world's leading fusion research facilities. It has a maximum toroidal field strength of $B_t = 2.1$ T and a maximum plasma current $I_p = 3$ MA. DIII-D's extensive suite of diagnostics makes it the ideal machine to deploy this type of inboard side probe in order to obtain a comprehensive spatial mapping of as many parameters as possible across the whole SOL.

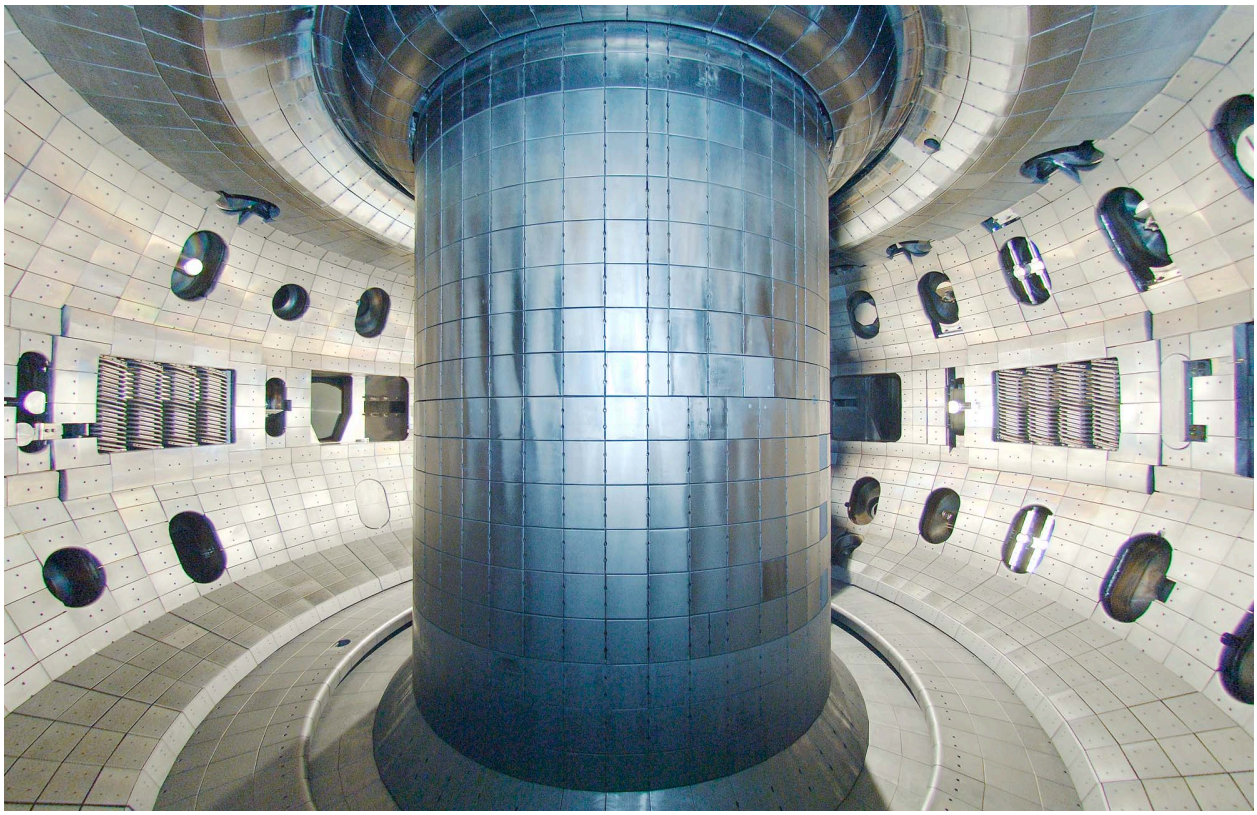


Fig 1-1 A view from inside the DIII-D tokamak.

Since a tokamak is toroidally symmetric, they are typically depicted using their poloidal cross-sections. An example of such a cross-section is shown in Fig 1-2. The inner wall (centerpost) is on the left side of the cross-section, and this will be true for all poloidal cross-sections in this thesis. The magnetics Equilibrium FITting (EFIT) of the plasma flux surfaces is overlaid to show the shape of the plasma for a typical lower single null plasma. The LCFS is the flux surface forming the 'X' aka the x-point. The closed flux surfaces are contained within.

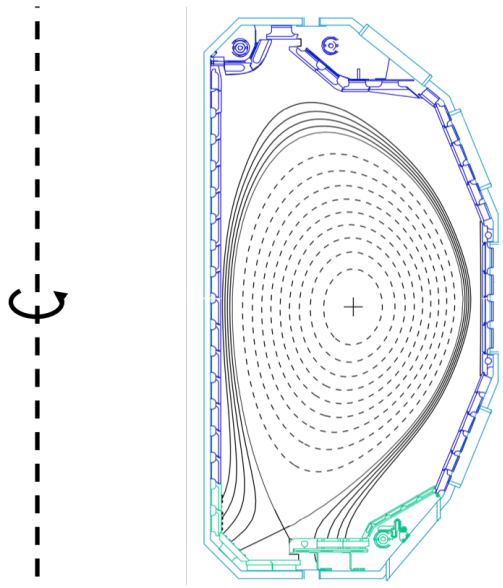


Fig 1-2 A poloidal cross-section of the DIII-D vessel overlaid with the EFIT of a lower single null plasma.

1.5 In-Vessel Probes

The DIII-D Swing-Probes are broadly based on in-vessel reciprocating probes that have been deployed in ASDEX Upgrade [5,6], Alcator C-Mod [2,3], RFX [7], and Tore Supra [8]. The motion of these in-vessel probes is powered using magnetic coils that provide torque in the presence of the toroidal magnetic field. The coils also provide the means to determine the position of the probes by integrating the back-EMF that is induced when the coils rotate in the field.

The C-Mod In Vessel Probe (IVP) and Wall Actuated Scanning Probe (WASP) designs were particularly relevant as they too are installed on the centerpost to measure the inboard SOL. Their drive mechanisms are designed to fit in the very limited space within the centerpost armour tiles and have plunges of ~12 and 17 mm respectively. DIII-D however is a much larger tokamak and can have a much larger gap between the centerpost and the LCFS of 100-300 mm. Therefore a unique design was needed for the DIII-D Swing-Probes.

Chapter 2 Swing-Probe Design Requirements

2.1 Design Process

The Swing-Probe design turned out to be far more complex than anticipated. A number of engineers, designers, and physicists were involved over different aspects of the design process spanning two years, six design reviews, and one failed deployment attempt. Diagnostics engineer Doug Taussig in particular was heavily involved in the design process. He was responsible for all of the nuts-and-bolts details of the Swing-Probe assemblies, and the 3D CAD models shown in this thesis are predominantly his work. Diagnostics engineer Matt Watkins was responsible for the majority of the electronic control systems. Many other people contributed to the design, testing, and/or installation. Every component of the Swing-Probes was designed with the input of many collaborators. See the Acknowledgements section for a full list. The main contributors will be listed for each system in Chapters 2 through 4.

The Swing-Probes were first used in the 2012 campaign, but several components failed as the electronics control systems were being developed. While the probes were operated in vacuum successfully, no plasma data were collected. After the 2012 campaign, the probes were removed and inspected. Some parts were redesigned and repaired and the Swing-Probes were reinstalled for the 2013 campaign following an upgrade design review.

In hindsight and after speaking with some DIII-D engineers, I believe it was overly optimistic to have expected the Swing-Probes to work correctly during their first campaign. This is in no small part due to the numerous technologies that were pioneered. The main contributors are listed in square brackets.

- In-vessel moving parts - With the exception of a pneumatic shutter, the Swing-Probes are the only moving parts inside DIII-D that are not actuated through a vacuum feedthrough. [D Taussig, C Tsui]
- Zirconia bushings and ball bearings – These are custom designs that have never been used anywhere before. These are the first ball bearings ever used inside the DIII-D vessel, though some metal bearings are used in the gate valves just outside the vessel. Future use is already planned for DIII-D's hot-tile station. [C Tsui]
- Nextel insulation over copper wire with a nickel flashing. – custom design, never used anywhere before [D Taussig, C Tsui]
- Grade A boron nitride and grade ZSBN boron nitride – approved specifically for this application and never used in DIII-D before [C Tsui]
- Magnetic coil actuators [D Taussig, M Hansink (engineer)], and their use as encoders [C Tsui] – never used in DIII-D before
- Gall resistant gears [C Tsui]– never been used inside DIII-D before
- Vacuum lubricants, MoS_2, WS_2, Dicronite –never been used inside the DIII-D main vacuum vessel before [C Tsui]

This list does not include several novel technologies that were rejected during design reviews or by the Vacuum Committee.

2.2 The DIII-D Vessel, Centerpost, and Vacuum Environment

From the center and moving outwards, the DIII-D centerpost consists of the toroidal field coils, the poloidal field shaping coils, an actively cooled vacuum vessel wall, and finally, graphite tiles which face the plasma. The graphite tiles extend approximately 5.5 cm out from the steel wall and are held in place using large nut-bars that clamps the tile feet securely onto a layer a grafoil, a soft graphite 'gasket' which is used to ensure good thermal and electrical contact as it conforms to any unevenness on the vessel wall. See Fig 2-1 and Fig 2-2 for the layout of the tiles. The entire probe assembly must fit

within the graphite tiles without compromising their thermal load handling capability. This means that no changes can be made to the vessel wall, and the tiles need to remain at least 2.5 cm thick, leaving a gap ~3 cm deep to work with between the tile nutbars.

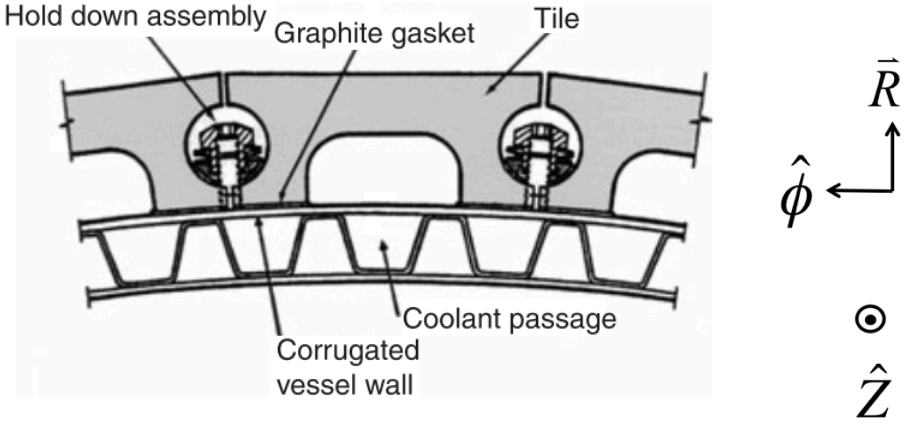


Fig 2-1 A horizontal cross-section of the DIII-D centerpost with attached armour tiles. The tiles are approximately 13 cm x 13 cm x 5.5 cm. Taken from [4]**.**

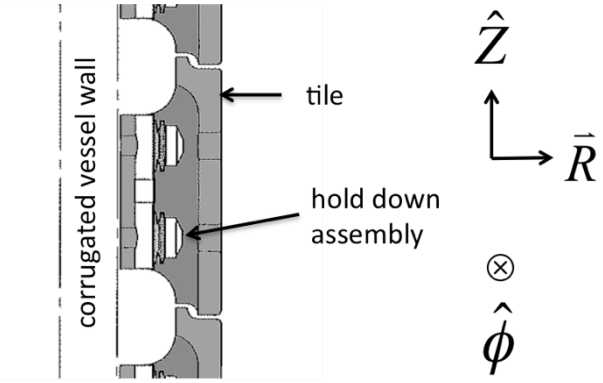


Fig 2-2 A poloidal cross-section of the DIII-D centerpost with attached armour tiles.

The typical distance between the centerpost and the LCFS is between 10-30 cm. A probe using a linear 'pop up' motion could at best reach 5 cm towards the plasma, while a horizontal rotating probe would require a long horizontal gap in the graphite tiles. A horizontal gap is not a viable option since it would allow plasma to penetrate deeply into the gap, exposing the metal structures and the probe mechanisms to plasma contact. The remaining option then was to design a probe that rotates (or swings) in the

vertical plane so that the toroidal width of the gap is kept small, minimizing plasma penetration. However, in-vessel coils that torque against the toroidal magnetic field can only generate torque about the $\hat{Z}$ axis. A drive train with a gearbox is needed to transmit the coil torque about the $\hat{Z}$ axis to the rotational axis of the probe, which is roughly the $\hat{\phi}$ axis.

Further design restrictions arise from the fact that a clean vacuum environment and a low impurity plasma must be maintained. The vessel is regularly baked and the centerpost reaches a temperature of 400°C. Useable materials are limited to a short list of ceramics and metals which are non-magnetic, have low vapour pressures and maintain their strength at high temperature. This excludes the use of aluminum, kapton, polyamide, Vespel, lead oxide based electrical insulation and a range of vacuum compatible circuits. Any plasma facing components can only be made out of carbon or boron nitride. This list of materials and their approved uses [9] is maintained by the DIII-D Vacuum Committee. New materials were tested and approved specifically for these probes.

2.3 Risks to DIII-D

The first priority taken into account when designing the Swing-Probes was risk management for the DIII-D machine. The reliability of the Swing-Probes was given second priority, while versatility came third. Compromises in favour of the first two priorities are evident throughout the design.

Unfortunately, reciprocating probes can and do fail catastrophically. If a probe explodes, depending on what it was made out of, the lingering impurities can drastically degrade plasma performance until the tile surfaces are grit blasted.

Since they are inaccessible without a manned entry to the DIII-D vessel, the Swing-Probes are inaccessible for the entire operational campaign, so it was important to make them as reliable as possible.

The Swing-Probes were therefore designed considering the following potential modes of failure.

2.3.1 Jamming in the path of the plasma

Having a Swing-Probe jam in the path of the plasma was a particularly large concern due to the experimental nature of their actuator system and the fact that the moving parts are under vacuum. Even if the probe took no damage when it jammed, DIII-D could be forced to cease operations for several months if a manned entry was necessary to move the probe. If the jam goes undetected, depending on the location of the probe with respect to the plasma, the probe could be destroyed, spreading impurities throughout the vessel.

The probes swing a full 180° through the plasma instead of swinging in, reversing, and swinging out. The fear was that if the probe stops in the plasma, it is more likely to stay stopped in the plasma. Most reciprocating probes are able to control how deeply they enter the plasma, this ability was sacrificed for the Swing-Probes to decrease this risk. A redundant spring system pulls the probe in the same direction as the actuator coils through the plasma. More details are in section 3.6.

The drive train, gears, bushings and ball bearings were tested in vacuum to the point of failure to ensure that they would perform well under load and to evaluate the risks from jamming, and galling. See section 3.5

Interlocks[1] were necessary to detect if the probe was out of place before the start of each shot. See section 4.5. Finally, in the event that the probe did become stuck, the actuator coils were over-engineered to be able to force the jammed system to move.

[1] Simple electronic circuits that are used to ensure safe operating conditions exist and prevent operation in the case when those conditions aren't met, e.g. a switch to ensure a door or panel is closed.

2.3.2 Thermal Quench in a Worst Case Disruption

If plasma confinement is suddenly lost, then a 'disruption' occurs and all of the energy stored in the plasma is quickly dumped into the walls or any probes that happen to be in the way. While the probe need not remain in operating condition, it needs to be able to survive the heat flux from a worst-case disruption.

The probe arm's metal support structure is armoured with 4mm thick pyrolytic graphite washers that are highly resilient to thermal shock. See section 3.3. The carbon will ablate rapidly, cooling the plasma and keeping the heat flux low. We will show that the total amount of carbon lost will not expose bare metal to the plasma. We follow [10] closely here.

Assume a 2 MJ plasma disrupts and quenches completely into the lower divertor[2]. The energy divides evenly over the entire plasma wetted areas of the inner and outer strikepoints[3]. During a disruption, the energy widths grow to $w_{sol} \approx 6cm$. The total wetted area is therefore:

$$A_{wet} = \pi D \cdot w_{sol} \cdot (2 \text{ strikepoints}) = \pi (2 \cdot 1.5m) \cdot .06\ m \cdot 2$$
$$A_{wet} = 1.1\ m^2 \qquad \textbf{eq. 2-1}$$

The Swing-Probe intercepts some of the flux surfaces. The field is almost parallel to the floor while the probe faces the plasma head-on. The wetted area projected along the field line can be as much as 30x smaller than the physical wetted area. The energy deposited per unit projected surface area is

---

[2] The SOL is often pinched off and diverted into a part of the vessel designed to accommodate the heat and particle flux. The region where the SOL is directed is called the 'divertor.' In the DIII-D vessel walls, there are two divertors. One makes up the 'floor' and one makes up the 'ceiling.'

[3] A diverted plasma produces two independent bands of heat and particle flux in the divertor where the LCFS intercepts the wall. These are called the 'strikepoints'

$$\frac{2\ \text{MJ}}{\left(1.1\ m^2 / 30\right)} = 53\ MJ / m^2 \qquad \textbf{eq. 2-2}$$

If we assume that 100% of the energy deposited goes into the vaporization of graphite, then we can determine the maximum amount of material lost in a disruption.

The heat of vaporization of graphite is 75 kJ/g and the density of graphite is approximately 2 g/cm^2, so

$$\left(\frac{5.3\ \text{kJ/cm}^2}{75\ \text{kJ/g}}\right)\frac{1}{\left(\sim 2\ g / cm^3\right)} = 0.35\ mm / disruption \qquad \textbf{eq. 2-3}$$

The probe arm is armoured with 4mm of graphite, and should be able to survive one or more disruptions without exposing metal to the plasma.

2.3.3 Halo Currents, and Short-Circuits to Plasma Currents

Another type of disruption is a Vertical Displacement Event (VDE), which occurs when loss in vertical position control causes the plasma to move upward or downward before any appreciable cooling of the plasma has occurred [11]. As the plasma continues to move towards the limiter, it decreases in volume. The plasma current previously flowing through closed flux surfaces must now close through the vessel walls. See Fig 2-3. These 'halo currents' can place enormous IxB forces on limiter surfaces, especially those that protrude in towards the plasma such as the limiter circled in Fig 2-3. If the probe was inserted into the SOL during a VDE, it could become the path of least resistance.

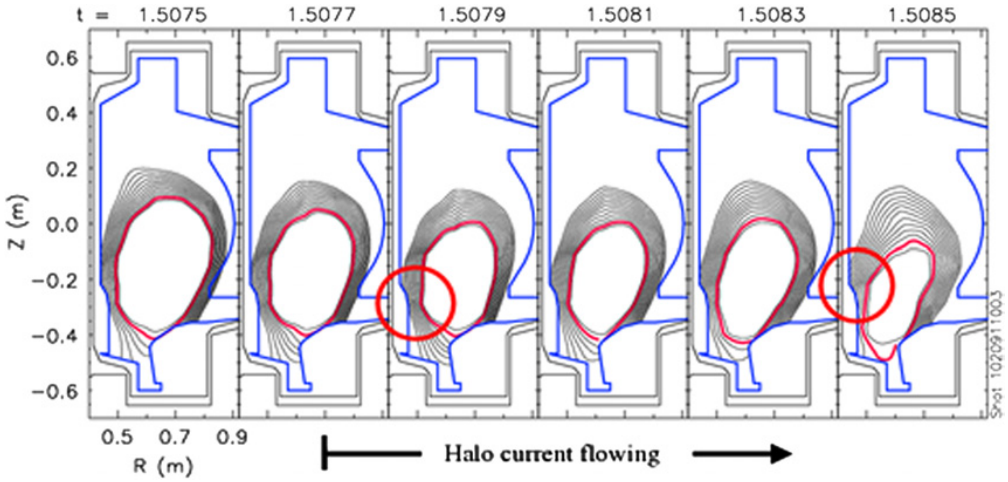


Fig 2-3 A VDE and accompanying halo current in Alcator C-Mod. Taken from [11]**.**

The probe arm therefore needs to be isolated from vessel ground by two separate layers of insulation each with a minimum breakdown voltage of 6 kV. Additionally, to prevent currents from shorting across two separate flux surfaces along the probe arm, the pyrolytic graphite washers are isolated from each other and from the probe arm to a minimum breakdown voltage of 1 kV.

2.3.4 Runaway Electrons

During a disruption, it is possible for the plasma current to drop rapidly, causing a large increase in the toroidal loop voltage. This in turn can create beams of runaway electrons. As the electrons accelerate, their collision cross section decreases allowing them to continue to accelerate unchecked until they reach relativistic velocities. Once a population of runaway electrons forms, Coulomb avalanche multiplication rapidly increases the population in tightly focused filaments that can contain as much as 70% of the original plasma current [11].

Generally speaking, the filaments will eventually run into the wall or some other solid surface. Runaway electron beams can deposit a tremendous amount of energy at a single location. There are no design changes that can protect a probe if it is hit by runaway electrons. The Midplane Probe was struck by runaways during the 2012

campaign, destroying all of the electrodes, insulating housing and copper contacts. Surprisingly, the beam appears to have passed around or through the graphite shield with almost no effect. We need to be careful to avoid experiments where disruptions and runaway electron production are likely.

Chapter 3 Swing-Probe Physical Design

3.1 Overall Design

Two identical Swing-Probes were installed at the entrances of the upper and lower divertors. See Fig 3-1. The probes swing vertically taking measurements along poloidal arcs. The addition of the Swing-Probes to DIII-D's extensive suite of diagnostics now allows us to actively characterize the entire SOL, albeit with some large gaps.

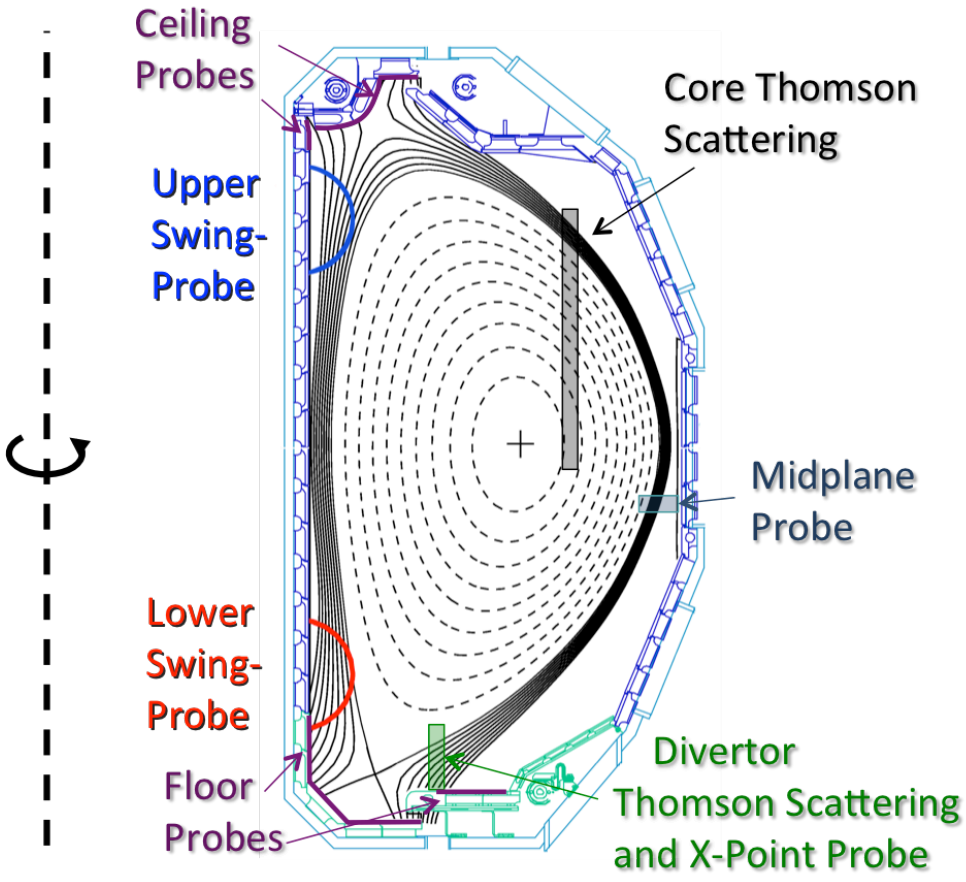


Fig 3-1 The locations of the main active Scrape Off Layer diagnostics in DIII-D mapped over a poloidal projection of a typical plasma shape. The swing probes measure along the red and blue arcs.

An axle behind the centerpost tiles rotates, swinging a probe arm 180° through a slot in the centerpost tiles. See Fig 3-2. Electrodes at the tip of the probe arm make Mach-probe measurements of the plasma. Engineer Chris Murphy designed the centerpost tiles covering the Swing-Probes.

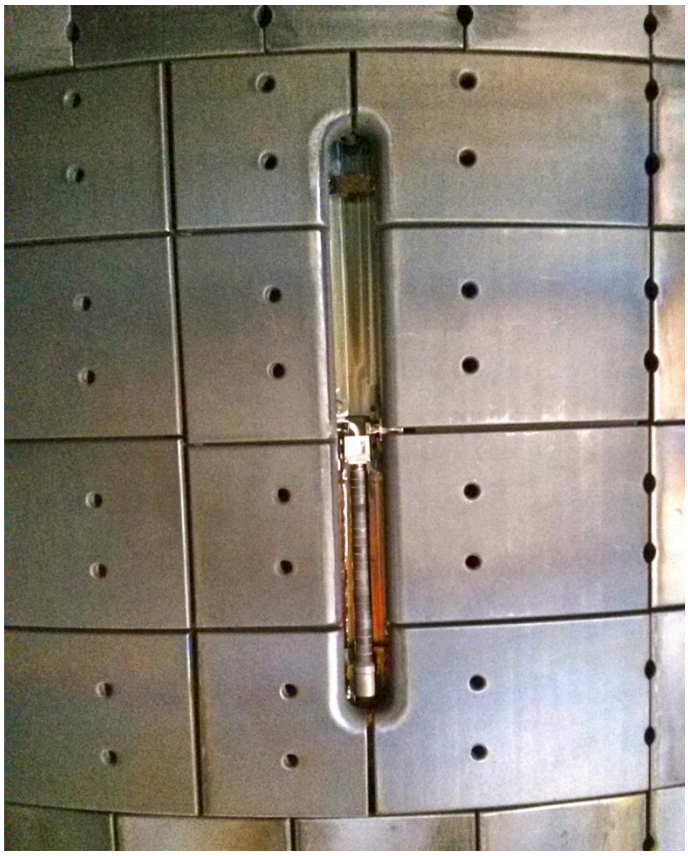


Fig 3-2 The upper Swing-Probe with the custom centerpost tiles installed over top. The drive coils are hidden behind the seam left of the slot.

The Swing-Probes consist of two actuator coils that torque against the toroidal magnetic field. The coils rotate in opposite directions such that one is perpendicular to the magnetic field when the other is parallel to it. They transmit their torque through a set of anti-galling miter gears to the horizontal axle which rotates the probe arm into the plasma. A spring assembly helps to pull the probe arm through the plasma.

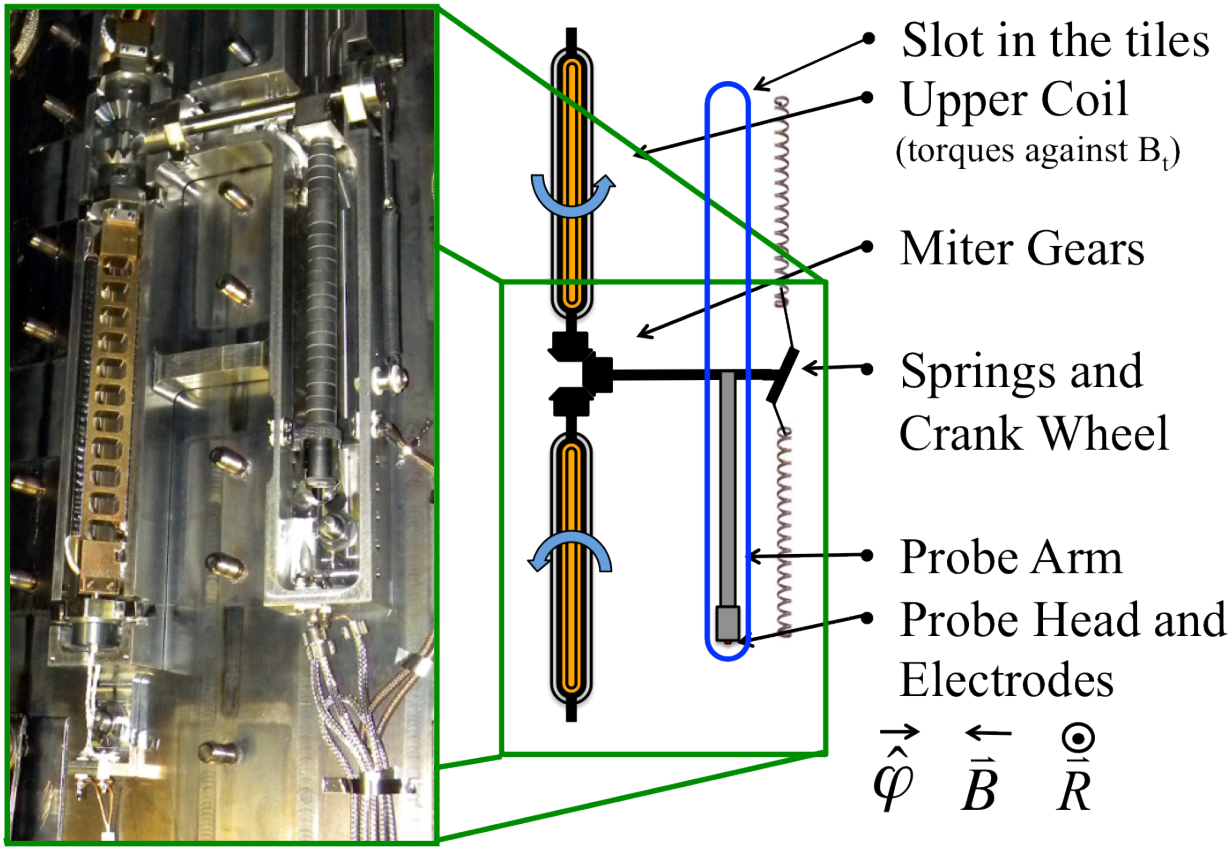


Fig 3-3 (right) A schematic of the Swing-Probe components. The blue oval represents the slot in the tiles visible in figure Fig 3-2. (left) A photograph of a Swing-Probe on the DIII-D centerpost with the graphite tiles and the actuator cover removed. A temporary stainless steel shield covers the probe head and electrodes to protect them during installation.

During the B-field ramp up, when there is no plasma in the vessel, the coils rotate the probe, stretching the springs until the probe reaches its 'armed' limit. Here, a catch made of interlaced spring sheets softens the probe arm's landing, and acts as a limit switch indicating the probe has successfully reached the armed position. When the

probe is within 25° of the armed position, the spring assembly pulls the probe towards the armed limit. The coils too are programmed to hold the probe at the armed position, though the springs have sufficient torque to keep the probe safely armed. The interlock system checks that the probe is armed 100 ms before the start of the shot. Should the check fail, DIII-D's power-breakers are tripped under load, preventing plasma breakdown from occurring. When the desired plasma conditions are established, the probe is 'fired.' The actuator coils pull the probe 180° through the plasma to the 'stowed' position where another catch and limit switch exist to then check that the probe arrived safely. Should the probe fail to arrive at the stowed limit switch on time, the interlock system will open the power-breakers, this time causing a controlled disruption. This control system and interlock design limits the probes to one swing per shot. Such limitations were considered to be necessary given the experimental nature of the drive system.

The overall design is predominately the work of C Tsui. The assembly and fasteners are predominately the work of D Taussig.

3.2 Probe Head and Electrodes

Plasma measurements are made with a pair of electrodes at the tip of the probe. They face opposite directions along the field forming a Mach pair capable of measuring the parallel flow Mach number $M_{\parallel}$, the ion saturation current density j_{sat}^{+}, the floating potential V_f, the electron temperature T_e, and the electron density n_e.

The probe head design was based on those used in reciprocating probes at DIII-D [12,13], Alcator C-Mod [3], and NSTX [14], with input from the probe experts J.A. Boedo, J.G. Watkins, D.L. Rudakov, and B LaBombard. With so many probe experts, the probe head underwent a particularly large number of design iterations. The final two designs – used in the 2012 campaign and 2013 campaign respectively – are discussed here.

The electrodes have a split design similar to the C-Mod probes, which is space efficient. This also allows the electrodes to stand 'proud,' protruding from the tip of the probe so that they can sample plasma that has not been disturbed by the presence of the probe shaft. This is not normally possible for Mach-probe electrodes, since there is normally an insulator dividing the two electrodes that extends further than the electrodes. These electrodes were machined by Graphite Machining Inc. out of pyrolytic graphite. The carbon crystal sheets in pyrolytic graphite are aligned within 1°, which makes the material highly non-isotropic. Along the graphite planes, the thermal conductivity is extremely high; 400-1700 $W/m\cdot K$ depending on the grade [15]. For comparison, pure copper has a thermal conductivity around 385 $W/m\cdot K$ [16]. This allows the electrodes to quickly pull heat away from the surface, and keeping the surface temperature low decreases secondary electron emission and prevents arcing. Below the surface, the body of the electrodes are made as large as possible to provide a larger thermal mass. The electrodes have a combined projected surface area of 4.4 mm^2 and the pair form a 45° cone, with a rooftop style top at 6°.

For the 2012 design, (see Fig 3-4) the electrodes were plasma sprayed with 0.005 inches of alumina to electrically isolate them from each other and from the carbon head. Concerns were raised that carbon deposits could cause a short circuit between the electrodes and the carbon head, so a boron nitride ring with a hidden notch or labyrinth was added. This does not prevent carbon deposits from shorting the electrodes to each other, but as they are kept at the same potential, this will only affect the measurements if the resistance between them drops below 1Ω. The electrical terminals for the electrodes were brazed into a hole at the bottom of the electrodes. Unfortunately, the quality of the plasma sprayed alumina was very poor. The coating did not adhere strongly to the graphite, and the application process caused damage to the electrodes. The construction process required 3 contractors and 14 weeks, so there wasn't time before the 2012 campaign to rebuild them. The damaged electrodes were used as is, and half of them crumbled partway through the campaign rendering one of the two probes unusable. Fig 3-5 shows the crumbled electrodes after they were removed.

The carbon parts in the probe head (other than the electrodes) are made from ATJ carbon for its toughness and high tensile strength. Pyrolytic graphite is strong along the carbon planes but has very low tensile strength across the planes, so it is too fragile for these intricate parts that are not held under compression. 3D carbon-reinforced-carbon was also tested, and while it is stronger than ATJ carbon, the porous carbon that binds the fibres together was found to be too reactive and too easily eroded. Test pieces crumbled when baked in air at 900°C.

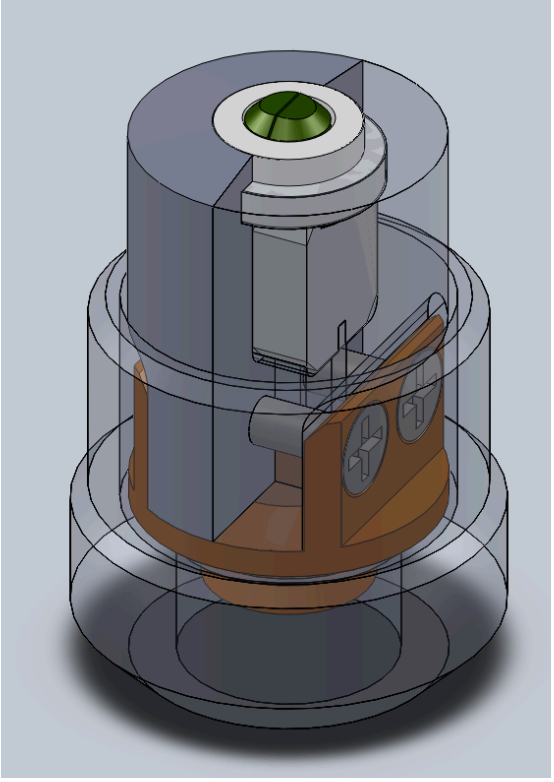


Fig 3-4 2012 model of the probe head, which is ¾″ in diameter at the widest part of the interface shield (transparent).

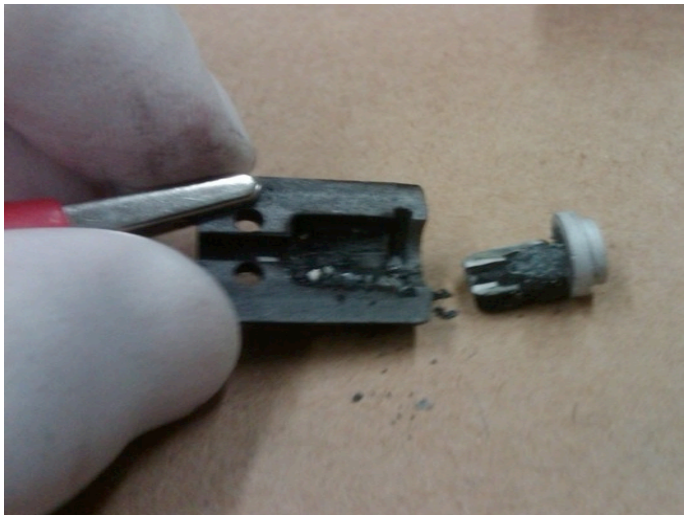


Fig 3-5 The electrodes crumbled during the 2012 campaign, electrically shorting to each other and to the probe head.

For the 2013 campaign, the design for the probe head was greatly simplified. See Fig 3-6 through to Fig 3-8. The Vacuum Committee approved a new grade of boron nitride for plasma facing surfaces. Grade A boron nitride uses a boric oxide binding agent and is 3-5 times stronger and 5 times harder than the high purity grade AX05 boron nitride [17]. This was strong enough to be used in place of carbon in the sandwich blocks. A piece of Inconel[4] shim was brazed to the electrodes to act as strong electrical lead and to add mechanical strength across the pyrolytic graphite planes. A stainless steel tube crimped around the contact wire could then be spot-welded to the tab at the bottom of the Inconel shim, which resulted in a much less fragile connection. The brazing was done in house using a vacuum furnace at 1000°C and Cusil-ABA braze.

[4] Inconel is a family of austenitic nickel-chromium based 'super-alloys.' It was used here because the Inconel-Graphite brazes available will stay bonded up to very high temperatures of ~800 °C, and the Inconel will not become brittle after repeated thermal cycling.

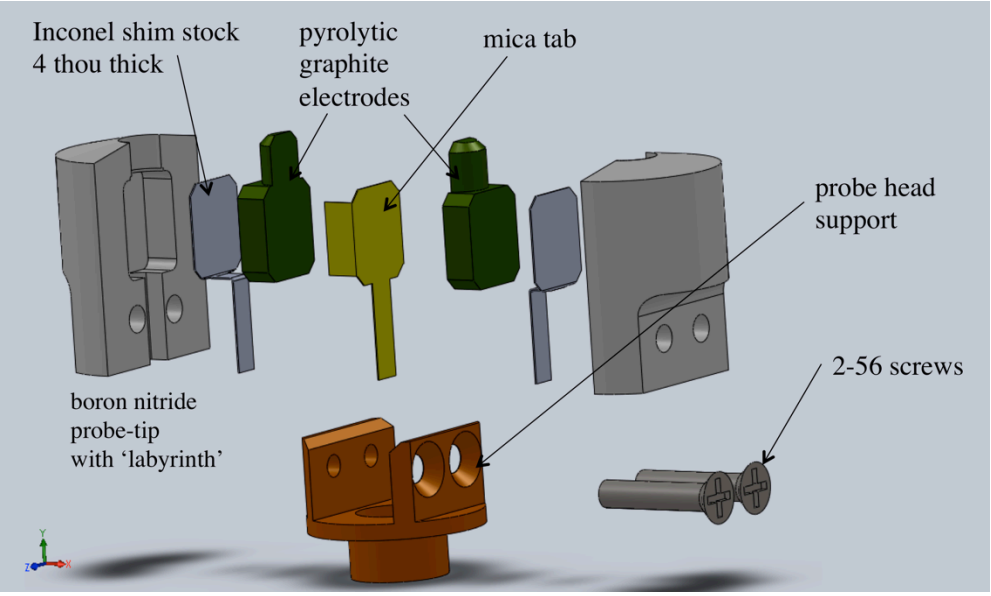


Fig 3-6 An exploded view of 2013 design for the probe head

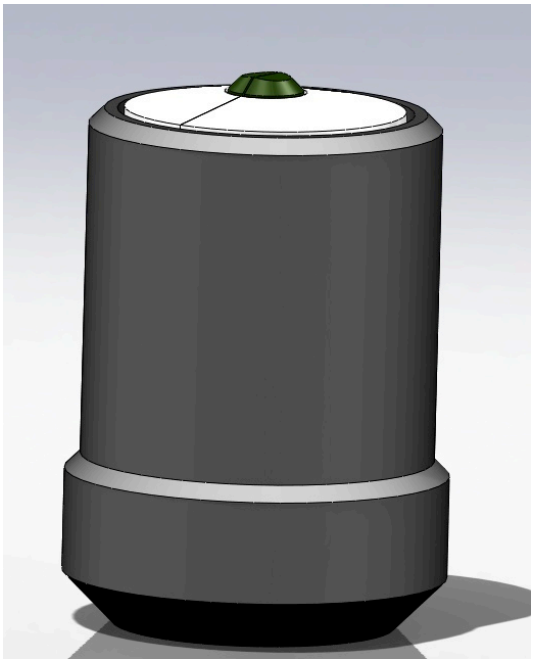


Fig 3-7 2013 model of the probe head including the carbon interface shield (not shown in Fig 3-6)



Fig 3-8 A photo of the 2013 probe head as installed in the completed assembly

The carbon parts of the probe head are electrically connected to the Inconel tube which in turn is connected to a dedicated signal cable. The flange on the bottom of the probe head's interface shield comes in contact with the spring catches. The spring catches are isolated from vessel ground using ceramic hat washers, and are connected to signal cables. Electrical continuity between the probe head and the spring catches form the actual limit-switches. When the probe is in motion, the probe-head is disconnected from the limit-switch circuitry, and can be used as a large V_f sensor or as the large return electrode for an asymmetric double-probe. The spring catches themselves are made of sheets of hardened spring Inconel, which does not anneal (losing its spring temper) at the 400 °C baking temperature. See Fig 3-9.

The probe head and electrode design is predominately the work of C Tsui.

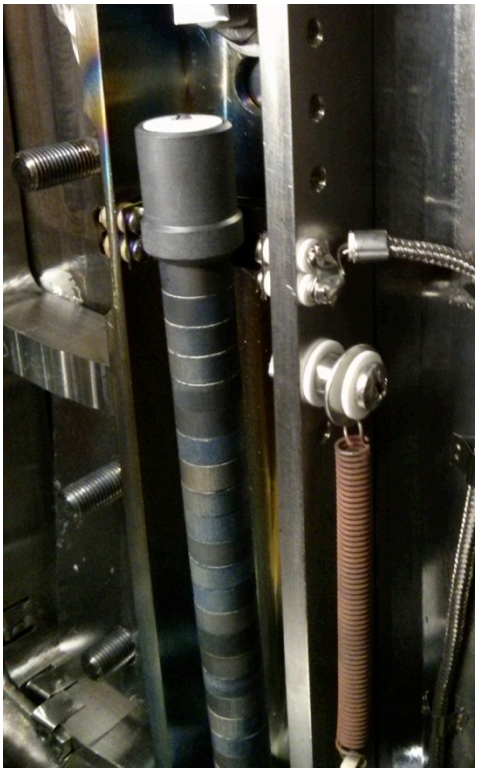


Fig 3-9: The spring-catch. Six layers of Inconel foil are interleaved such that they partially overlap in the center. The impact energy is dissipated by friction between the sheets as the slide past each other, which reduces how high the probe bounces upon hitting the spring catch. The rainbow coloured deposits are (roughly) within line of sight of the plasma.

3.3 Probe Arm

Where possible, design features from the X-point probe shaft [13] were closely replicated for the Swing-Probes since they share some of the same requirements. The arms of both probes can only be accessed during a manned vessel entry so they are both built to be robust. Furthermore, both probes are vulnerable to VDEs and so were required to withstand more extreme conditions than the midplane probe.

Electrical current flowing down the length of the probe arm causes an $\vec{I} \times \vec{B}$ force which acts predominantly in the direction that the probe arm is free to move. Without built in protection, currents can cause large uncontrolled acceleration in the probe arm. In a worst case disruption, the head of the probe may be several kilovolts from machine ground, so two redundant levels of insulation rated to 6 kV were required between the head of the probe and the vessel. Furthermore, insulation along the length of the probe arm was necessary to prevent bridging currents between different flux surfaces of the plasma as well as to stop currents from arcing to the centerpost tiles. See Fig 3-10 and Fig 3-11.



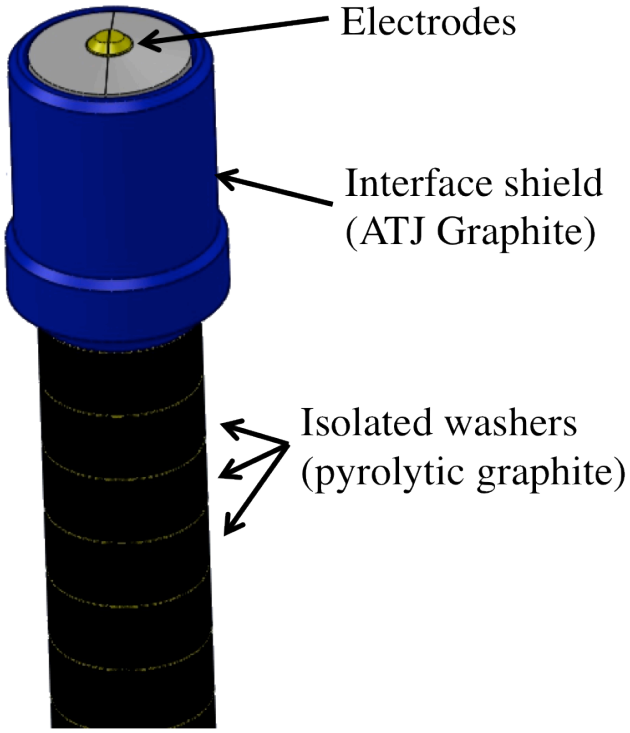


Fig 3-10 (left) A photo of the probe arm as it is installed in DIII-D with the tiles removed.

Fig 3-11 (right) A schematic of the probe arm.

The probe shaft is constructed on top of a 3/8'' outer-diameter Inconel tube weldament that is plasma sprayed alumina. The probe head attaches to the end of the weldament using two 2-56 screws. The probe arm is armoured using pyrolytic graphite washers ¼'' long with an outer diameter of 0.55'' that are insulated on the tops and bottoms with plasma sprayed alumina. The alumina on the Inconel tube and on the washers is ≥ 0.005'' thick. Assuming a minimum dielectric strength of 229 kV/inch for sprayed alumina [18], each graphite washer is electrically isolated from every other one to a breakdown voltage ≥ 2 kV. Carbon deposits will bridge these insulators, but will sublimate if large currents are driven across them. The carbon crystal sheets of the pyrolytic graphite are stacked along the axis of the probe arm so that complete sheets encircle the Inconel tube. This provides excellent thermal shock resistance, toughness and the relatively high cross-plane electrical conductivity also helps to limit currents from flowing along the length of the probe shaft.

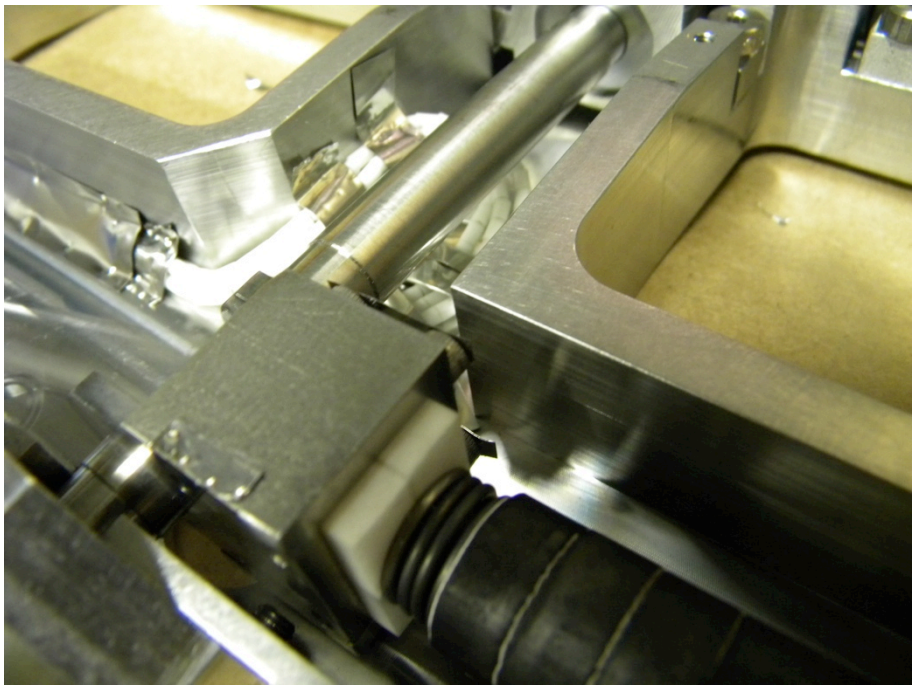


Fig 3-12 A photo of the alumina sandwich blocks connecting the probe arm to the main driveshaft. Notice the Inconel spring holding the pyrolytic graphite washers in place.

The graphite washers are held in place with an Inconel spring that butts up against an alumina sandwich block. See Fig 3-12. This sandwich block mates to a notch in the probe arm and the horizontal driveshaft and provides the first layer of >6 kV electrical isolation. The minimum thickness is 0.03''. Assuming a minimum dielectric strength of 340 V/thou for solid alumina [18] gives us a breakdown voltage ≥10kV. Mechanical tolerances for this part are very small. The gaps need to be small enough so that there is no play in any of the mating components, but need to be large enough so that thermal expansion of the metal parts does not damage the alumina when the vessel is baked and the Swing-Probes reach ~380°C.

The driveshaft is also isolated from vessel ground providing the second layer of > 6kV electrical isolation.

The probe arm was predominately designed by D Taussig, not including the isolated washer design which was copied from the X-point probe.

3.4 Signal Cables

The two electrodes and the probe head each have a signal cable running down the center of the probe arm, through the isolation block and the drive shaft, and then along the housing frame towards a terminal block. These cables must flex 180° at high speed without breaking or shorting. They are made from hand-spun stranded molybdenum wire and are insulated using 2 mm diameter alumina beads. These wires were chosen for their excellent flexibility and for their resistance to hydrogen embrittlement, work hardening, or hardening due to thermal cycling. Since the probe head's large plasma exposed surface area could transmit large currents down this wire should the cable become grounded before reaching the fuses in the electronics rack, an internal fuse was installed at the terminal block. Normally, a vacuum resistor consisting of an alumina rod with a deposited carbon 'coil' is used for this purpose, since the V_f measurements of the probe head are generally insensitive to inline resistance. These resistors failed the approval process for the main vessel as the manufacturer would not disclose the composition of the braze between the wire and the alumina rod. The fuse consists of a short length of 0.25 mm diameter molybdenum wire stretched between two terminal posts. A copper fuse with a wire of the same diameter is rated to fail under 6A. Under normal operating conditions, this cable carries 50mA DC in limit-switch mode, and capacitive currents only when measuring the plasma floating potential.

3.5 Drive Train

3.5.1 Conservative Designs

The design of the drive train was arrived at after careful consideration over many iterations. The use of moving parts and wearing surfaces in vacuum is risky. The thin oxide layer that normally covers metal surfaces wears away, and in the absence of air, is not replenished. The exposed metal surfaces are more susceptible to galling and cold welding [19].

In the 1990s, a complicated drive train was used in TEXTOR to move the ALT-II limiter [20]. The system was extensively tested in a vacuum chamber during the design process [21], but even so, the completed system seized in the vessel [22] and never did reach its design potential [23].

Recognizing that even carefully engineered and tested moving parts in tokamaks had failed, we converged towards an extremely conservative design for the Swing-Probe drive train. The miter gears used are the largest that will fit behind the centerpost tiles without compromising their heat flux handling capabilities. The large size of the gears decreases the contact pressure, but increases the backlash (the amount of slop between the teeth). The axis of rotation for the probe arm is misaligned at 3° in lieu of adding a universal joint since they faired poorly in TEXTOR [22]. This must be taken into account when calculating the locations and the projected surface areas of the electrodes. The shafts, collars and gears are made out of Nitronic-60, a gall-resistant austenitic stainless steel. For the 2012 campaign, the axles and gears slid against bushings made of transform-toughened zirconia. Utilizing three gears instead of two further decreases the contact pressure and balances out the reaction forces. A bench top vacuum vessel was used to run a series of wear tests on the gearbox design. In the most aggressive test, the gearbox was subjected to 3 years simulated motion at 1.2 N-m (two times higher than the peak torque applied for a probe swing) with and without a dry-film lubricant. In both cases, no increase in static friction could be measured using a torque meter with a minimum detectable level of 0.05 N-m. Surface abrasion was visible using a microscope on the gear hubs and at the contact points between the shafts and the bushings, but no discernable difference in roughness could be felt when a fingernail was dragged across the surface. No change was observed on the surfaces of the gear teeth.

To further decrease the risk of seizing, the bushings, gears and collars were coated with Dicronite, which is the commercial name for a tungsten disulphide vacuum lubricant coating applied using a proprietary method that results in a uniform coating even over irregular surfaces. The coating is free from impurities and has excellent wear resistance. This lubricant had never been used within the main DIII-D vacuum vessel before, and

needed to be approved by the DIII-D Vacuum Committee. Research shows that tungsten and molybdenum disulphides have very low vapour pressures and are stable to 930°C in vacuum [24,25]. Vacuum Committee tests confirmed that Dicronite and molybdenum disulphide vapours remained undetectable by quadrapole mass spectrometers when baked to 400°C, so the use of Dicronite was approved for this application.

The complete swing probe assemblies had a static friction = 0.1 N-m, and a dynamic friction of 0.06 N-m was measured in the first toroidal field motion tests which were performed while the DIII-D vessel was still vented with air. These values are higher than those measured in the bench top tests because they did not include the additional weight from the actuator coils. These levels of friction were thought to be reasonable. Unlike graphite, Tungsten disulphide has a lower coefficient of friction in vacuum than it does in air [19], so no problems associated with friction were anticipated for the 2012 campaign. That campaign was in fact plagued with continuing friction problems for reasons that are still not fully understood. Custom ball bearings were developed to address this problem for the 2013 campaign, and the friction levels remained low and constant for that campaign.

3.5.2 Drive Train Performance Issues in the 2012 Campaign

After the initial motion tests in 2012, which were performed using computer control systems in the machine pit, no tests could be run until the electronics control systems were completed several months later. This was partly because the simpler computer control system would not survive the radiation levels if left in the pit, and partly due to the interlock requirements for Swing-Probe used during the campaign. Once the electronics systems had been completed including most notably, the machine pit to annex communication links, we were ready to fully commission the Swing-Probe system. By this time, an electrical short had developed in the upper probe's actuator, which prevented its use during this campaign.

The first time the lower probe was operated we noticed that the dynamic friction in the drive train had increased by a factor of 4 to 0.25 N-m compared to the swings that were performed when DIII-D was still vented with air. This level of friction was large enough to prevent the probe from arming successfully using the original control waveforms. Once we accounted for this difference in the open-loop programming, the lower probe was able to arm and fire as designed, and the dynamic friction decreased gradually over two weeks (one swing per day) to 0.16 N-m. Following a boronization and bake however, the dynamic friction in the system increased sharply to 0.70 N-m. At this level, the spring assembly cannot pull the probe out of the path of the plasma should the actuators fail, and the probes were shelved until the end of the campaign. When plasma operations were completed, several beam-into-gas shots were performed which required the toroidal field but no plasma, and therefore, no risk was presented to DIII-D should the probe get stuck. The lower probe was exercised 4 times with a dynamic friction of ~0.25 N-m.

When DIII-D was vented, the probes were inspected immediately before they were disturbed. The static friction was again 0.10 N-m. The dynamic friction was not measured, since we did not want to energize the toroidal field coils. There were no signs of misalignment or any obvious problems. The same was true when the probes were disassembled. There were no signs of wear on any of the sliding parts, and no indication that anything had gone wrong. Thin carbon depositions were observed on the bushings that are not concealed by the actuator cover, but these were not substantial enough to cause any kind of problem.

The cause of the high and inconsistent friction has not been ascertained. This problem could not be caused by locking parts or by debris falling into the gears since the friction level is roughly constant throughout any individual swing. This also rules out a major misalignment as the cause. The drop in friction when the vessel was vented is consistent with vacuum galling, but the lack of any abrasion to the sliding surfaces makes this an unlikely cause. Misalignment caused by thermal expansion during the bake is unlikely to be the cause since the level of friction appears to be quite constant throughout any

given swing, and only changes gradually over the course of many swings, whereas one would expect a misaligned part to have an effect that would change more dramatically. It is plausible that the boron somehow changes the sliding properties of tungsten disulphide, but we do not have a theory that explains why the friction levels would drop back to normal with the introduction of air. The role that boronization played in the friction problem was not systematically studied because of the risk to DIII-D.

All friction related work was done by C Tsui.

3.5.3 Horizontal Drive Shaft

The horizontal drive shaft mates with the probe arm and disassembles into two parts for installation. The shafts are turned out of Nitronic-60 while the mating parts are milled from 316 stainless steel. Each half was welded together before the milling was completed as welding causes the joints to bend. As a final step, the drive shaft was assembled with an aluminum counterweight in the place where the probe arm is normally installed. The sliding surfaces of the shaft were then turned in a lathe to ensure that they were co-axial and concentric. See Fig 3-13.

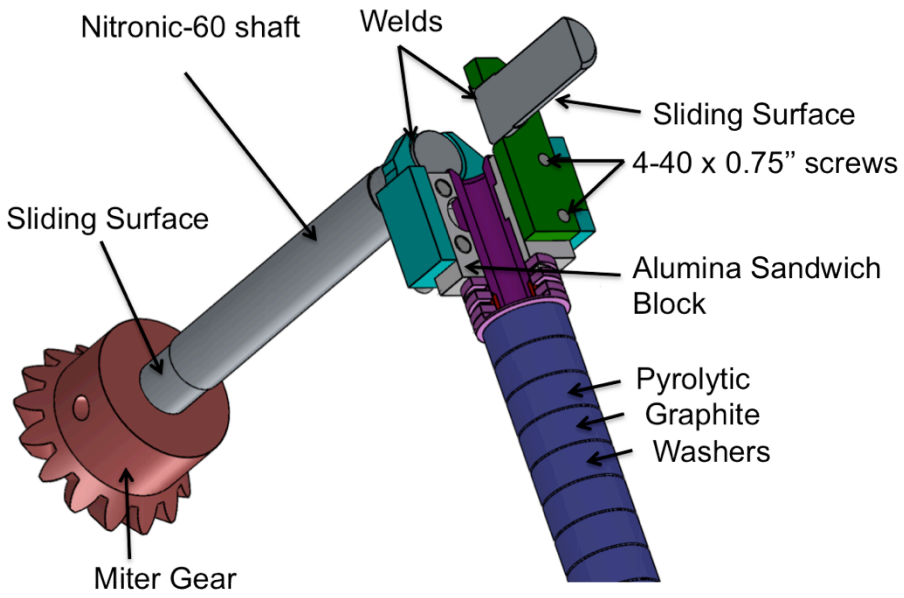


Fig 3-13 A schematic of the horizontal drive shaft with part of the mated probe arm. The bearing and their mounts are not shown, nor is the spring assembly crank wheel that mounts on the end of the shaft on the right. For scale, the hub of the miter gear is 1.0'' is diameter, and the pyrolytic graphite washers are 0.25'' tall.

The horizontal drive shaft was predominately designed by D Taussig and designer S Celle.

3.5.4 Bushings and Bearings

In the 2012 campaign, the drive shafts rotated on transform toughened zirconia bushings that slid up against the rear hubs of the miter gears. To decrease the chances of galling, the bushings were coated in Dicronite and the edges of the holes were chamfered. The bushings have an inner diameter of 3/8'', an outer diameter of 1'' and are 3/8'' thick.

Following the friction problems faced in the 2012 campaign, a set of ball bearings were developed for the Swing-Probes. It was challenging to find a company capable of manufacturing bearings to these specifications, since bearings are generally mass-produced overseas. "All ceramic" bearings are readily available but typically include retainers or shields made of unacceptable materials (Teflon or magnetic metals) which

are necessary to keep the balls from falling out of the bearings. Ortech Ceramics was capable of providing all zirconia ball bearings that use a full complement of balls with no retainers or covers. See Fig 3-14.

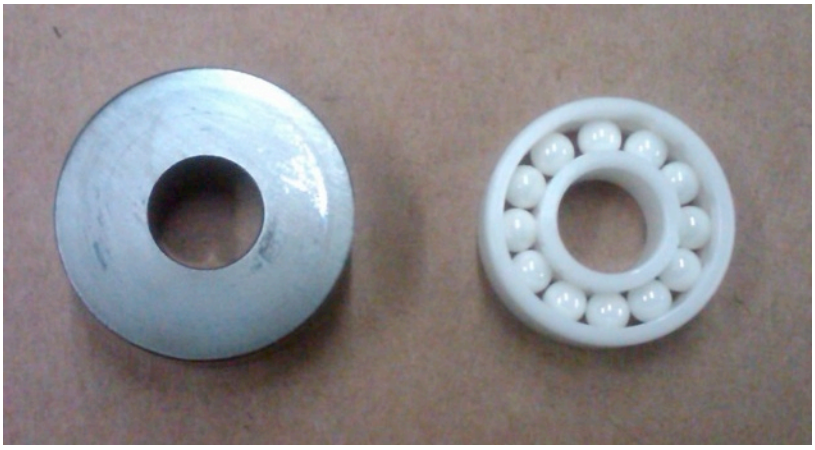


Fig 3-14 The bushings used in the 2012 campaign coated in Dicronite (left) and the custom ball bearings that replaced them (right)

Instead, the races have especially deep grooves that hold the balls safely in place. The deep groves also allows the bearing to be loaded axially, not just radially, meaning they could be used in conjunction with the miter gears, which are typically used with bushings for this reason. While tolerances are tighter in these bearings, tests showed that they are more forgiving to misalignments of the shaft than the bushings. The bearings have an inner diameter of 3/8'', an outer diameter of 7/8'' and are 9/32'' thick. The mounting hardware is shown in Fig 3-15

The bearings were installed without the use of any lubricant for fear that the Dicronite was somehow interacting with boron. Using these bearings decreased the friction to below detectable levels in the dynamic motion tests conducted during the 2013 campaign.

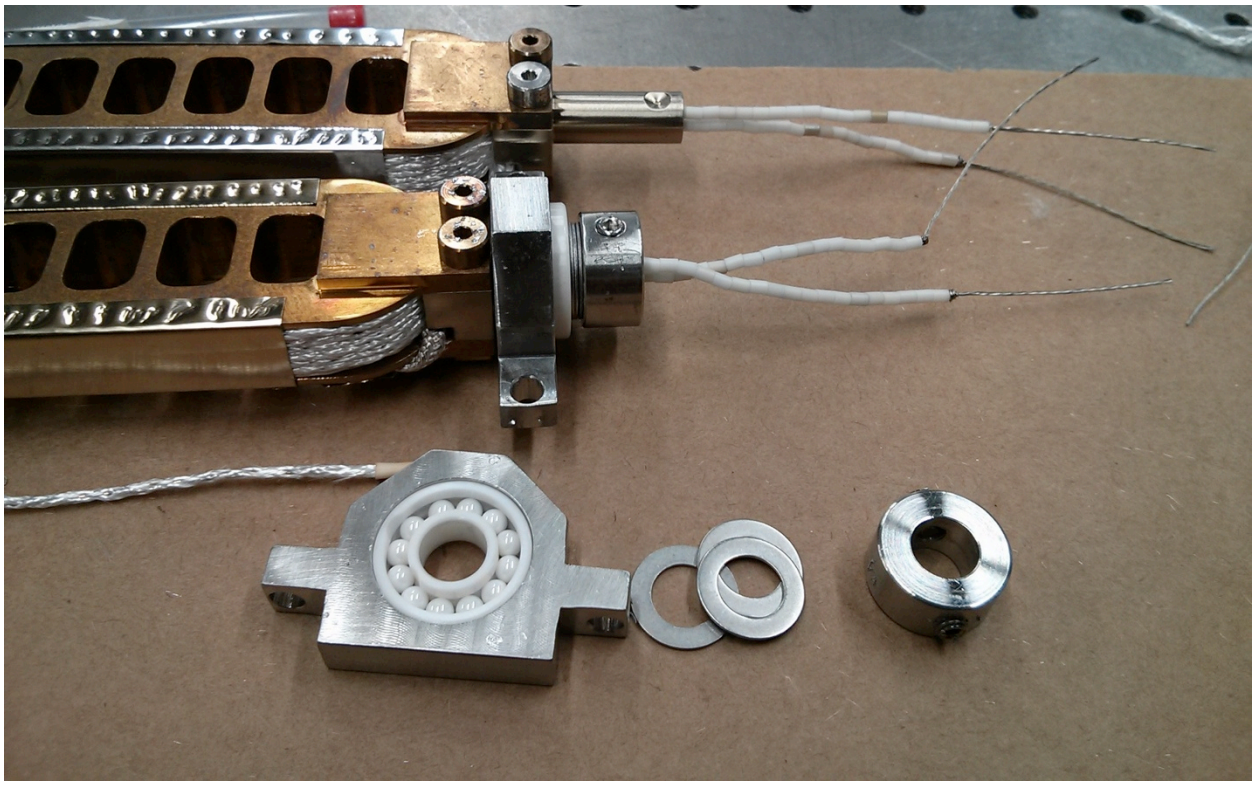


Fig 3-15 A photo of the zirconia ball bearings and their mounting hardware. The axial tolerances were adjusted using washer shims set between the bearing and the collar (bottom right). Actuator coils are shown with and without the bearing installed. Note that the bushing mount has been slid back to reveal the bearing, and does not normally come in contact with the coil bracket.

The designs for the sliding and rolling interfaces are predominately the work of C Tsui. The bearings were designed by Ortech Ceramics.

3.5.5 Gears

The miter gears were custom cut out of Nitronic-60 with 15 teeth and a pitch diameter of 1.25'' and an outer diameter of 1.37''. The large size of the gears maximizes the contact surface area on the teeth, but also increases the backlash or the slop between the gears. The backlash is ≤ 0.23° which corresponds to an uncertainty in the electrode position of 0.8 mm, which is smaller than the width of the electrodes.

The main technique used to avoid galling in metal on metal wearing surfaces is to make them out of dissimilar metals. In this case however, only nonmagnetic type 300 stainless

steels are approved for use in the vessel, and these are not dissimilar enough from Nitronic-60 to be useful. Bronze could have been used, but none have been approved for use in DIII-D. Coating one of the gears in silver by physical vapour deposition would have eliminated the risk of galling, but this cannot generally be done because of the complex geometry of the gear teeth. Electroplating silver was not an option since impurities are trapped in the silver-metal interface, causing the surface to blister when heated in vacuum [26]. Sputtered silver can be used, but not on the irregular surfaces of a gear.

However, there were no visible signs of wear on the surfaces of the gear teeth following any of the in-vacuum stress tests whether the gears were lubricated or bare. The low contact pressure appears to be sufficient to prevent vacuum galling. We therefore assume that Nitronic-60 gears coating in Dicronite pose little risk of galling in DIII-D for this application.

C Tsui and D Taussig designed the miter gears. Friction and wear related work was done by C Tsui.

3.6 Spring Assembly

The spring assembly holds the probe arm firmly against either the armed or stowed limit switches, ensuring good electrical contact. It also helps to pull the probe through the plasma, and is able to pull the probe out of the path of the plasma independently should the coils fail. To accomplish all these tasks, a crank wheel is used with two Inconel springs. The crank wheel is rotated such that it pulls the probe towards the armed limit switch when the probe is within 25° of the armed position. For the rest of the probe angles, the spring assembly pulls the probe towards the stowed position. The crank wheel is directly connected to the horizontal drive shaft and must therefore be electrically isolated from vessel ground. There wasn't enough space around the crank wheel to add electrical insulators, so the anchor points of the springs have been isolated

using ceramic washers. The spring assembly is nearly symmetric, so the forces on the axle cancel out providing a true torque[5]. See photos of the assembly in Fig 3-16.

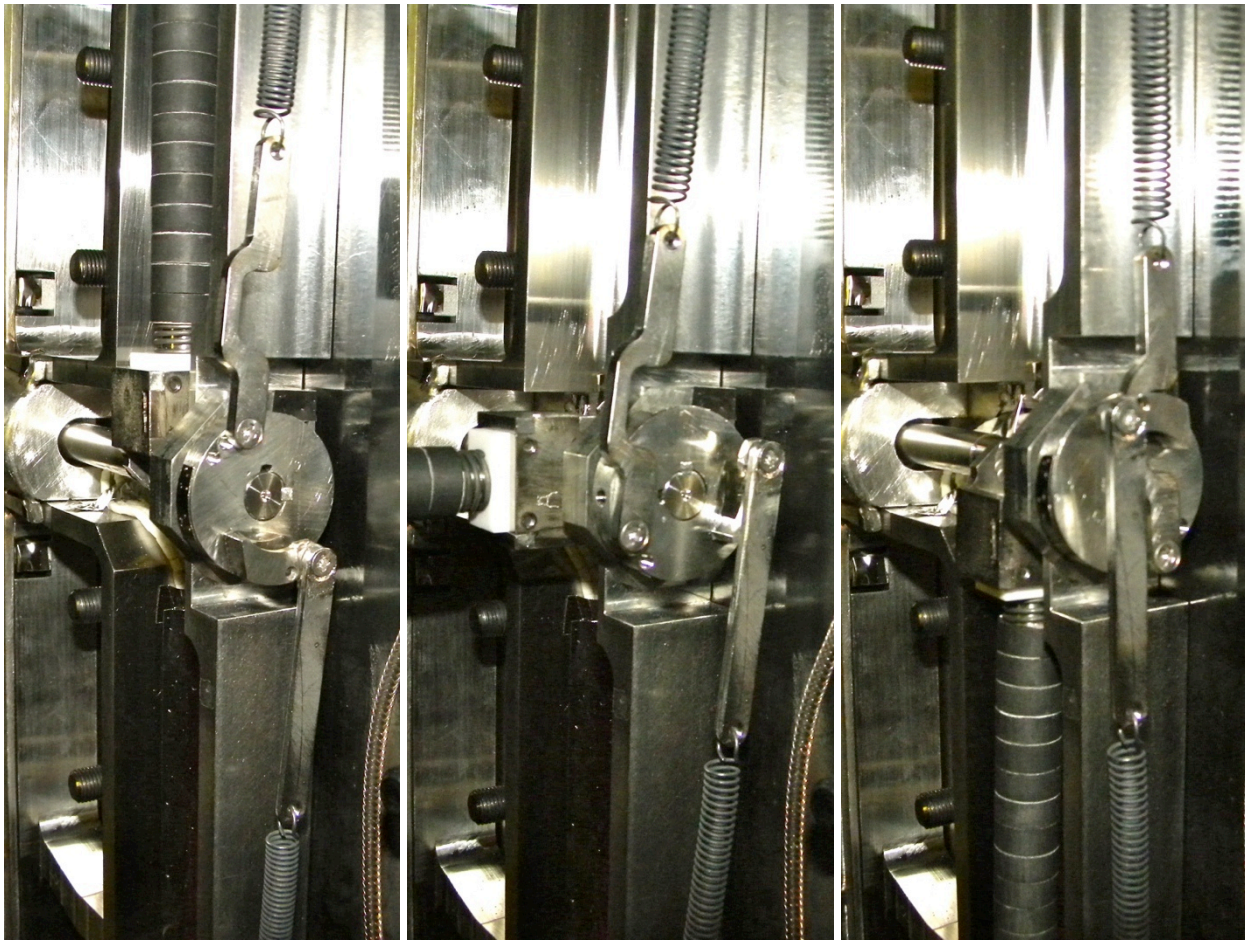


Fig 3-16 The spring assembly and the motion of the crank wheel.

The spring assemblies were designed by D Taussig.

[5] No axial or radial forces are applied to the drive shaft. These would need to be canceled out by the bearings, which would in turn increase the friction in the system.

3.7 Actuators

The actuator coils required an additional design review all on their own due to concerns that the shafts would not stay concentric and collinear after they were removed to wind the coils. Engineers Doug Taussig and Matt Hansink deserve a special acknowledgement for their excellent work on these very precise parts. We converged on a design that is milled out on a 5-axis C&C and then matched drilled in a jig with the shaft blocks. Each coil is 7.9'' long not including the shafts and 1.26'' wide. There are 42 windings with a total magnetic area of 0.186 m^2. The armature frame is made from 316 stainless and the shaft ends are made out of Nitronic-60. A 3D model of the armature frame is shown in Fig 3-17.

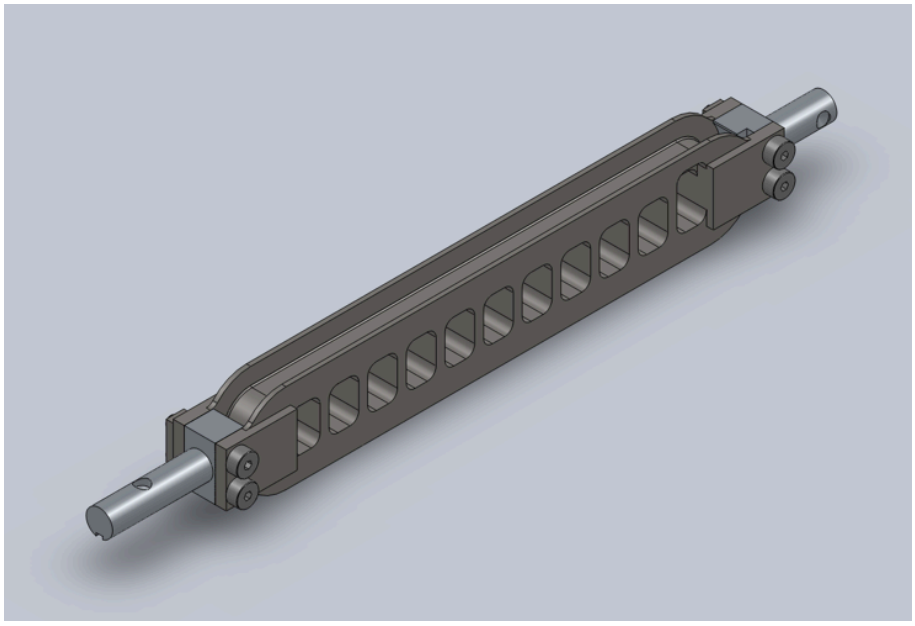


Fig 3-17 A schematic of the actuator coil design.

The coils were over engineered in both size and strength. The large size minimizes the magnetic perturbations to the plasma and provides a large safety factor. During normal operation, coil currents do not exceed 2A, but these actuators are capable of handling ten times that current for a period of up to 10s. This provides a method for forcefully retracting a probe stuck in the path of the plasma before more drastic measures such as venting the vessel are necessary.

3.7.1 Winding

The coil windings too were difficult to develop since most wire insulators such as polyamide, and Vespel are not permitted in this part of the vessel since it bakes at nearly 400° C. Ceramic insulator coatings passed Vacuum Committee tests but were then rejected due to health concerns over the high lead oxide content. We were finally able to develop a custom space-grade cable for our purposes. Copper was necessary in order to keep the power requirements low for these coils. Nextel, a glass fibre insulation developed for the space program, was the only insulator approved for this application. Nextel must be baked in air at high temperature in order to remove a lubricant that must be present during the winding process to avoid fraying. However, copper will oxidize when baked in air, so the wires are protected with a 2% nickel flashing. The wires pass through an oval hole in one of the shaft ends where they are crimped to stranded molybdenum wires that are insulated using ceramic beads since we found that the copper wire was susceptible to metal fatigue even from a 180° rotation. The oval hole holds the crimps in place forcing the thicker molybdenum wire to rotate instead of the weaker copper wire below the junction. The actuator coils were wound by hand and were secured using sheet steel and a spot-welder. One actuator coil as installed is shown in Fig 3-18.

The coil windings were designed by D Taussig with help from C Tsui.

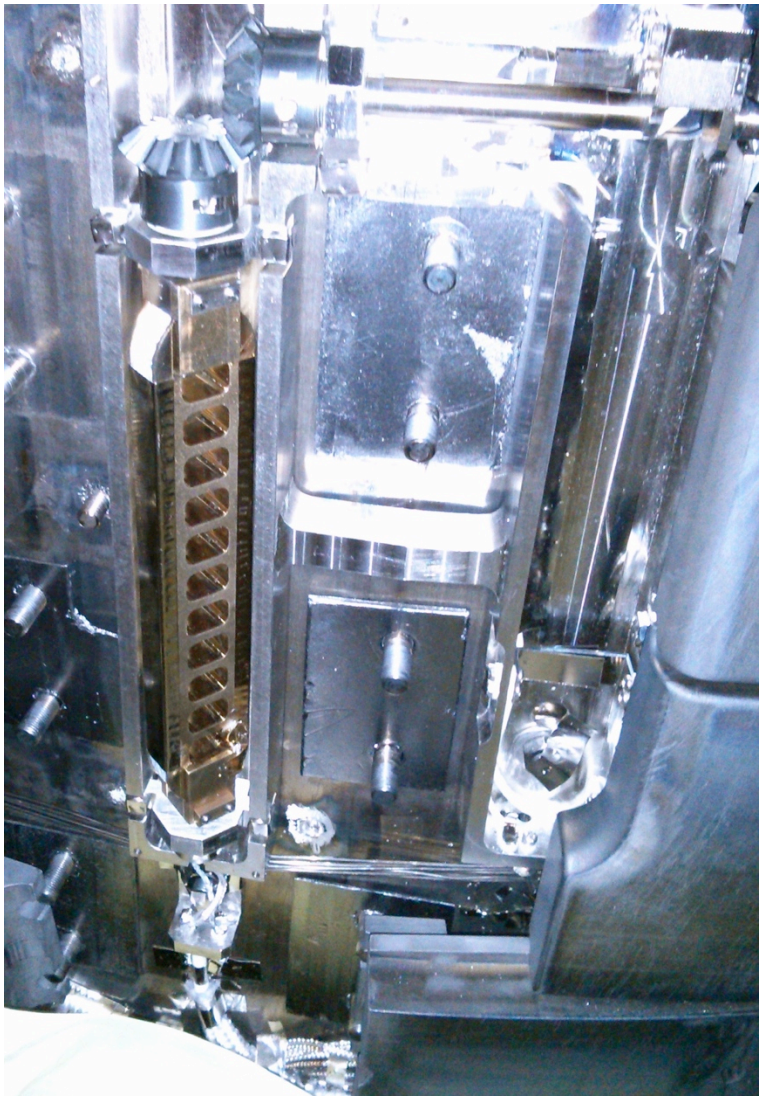


Fig 3-18 One of the actuator coils mounted in the Swing-Probe frame.

Chapter 4 Swing-Probe Control and Data Acquisition

The Swing-Probe uses two magnetic coils to power its motion. When current is run through the coils in the presence of the toroidal magnetic field, the $\vec{I} \times \vec{B}$ force causes the probe arm to swing through the plasma over ~200 ms. The coils are also used as encoders, providing information about the probe's position as a function of time. The toroidal field is typically ~3.5 T at the R of the Swing-Probes.

In this chapter, we describe the concepts and systems necessary to control the Swing-Probes including how to calculate the probe position from the coil back-EMF, the control theory for coil-driven in-vessel probes, and the electronics equipment that actually drives the system.

4.1 Coordinates

For this section, we follow the ITER coordinate conventions (see [27] for details) starting with the standard cylindrical coordinates $\left[\hat{R},\hat{\phi},\hat{Z}\right]$ where $\hat{\phi}$ is positive in the counter clockwise direction when viewed from above. See Fig 4-1. The actuator coils rotate about the $\hat{Z}$ axis, so we define the coil area vector $\vec{A}$ as the total area of all 42 windings pointing normal to the coil face and following the right hand rule for the direction of current flow, i.e. when a positive current is applied to the coil, the magnetic field produced is parallel to $\vec{A}$.

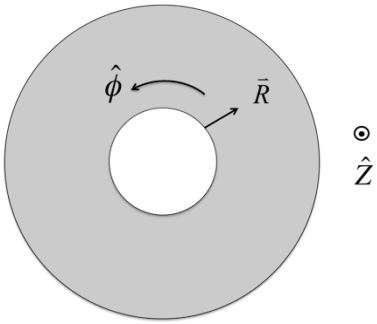


Fig 4-1: The toroidal coordinates for DIII-D. The smaller white circle represents the centerpost.

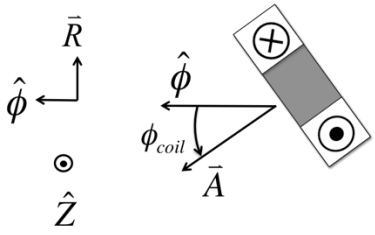


Fig 4-2: The coil angle ϕ_{coil} as viewed from above.

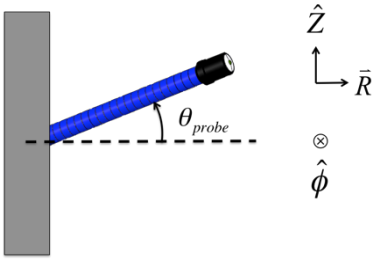


Fig 4-3: The probe angle θ_{probe}. The block on the left represents the armour tiles of the centerpost.

The angle of the coil $\phi_{coil} = 0$ when $\vec{A}$ is facing the toroidal direction $\hat{\phi}$ and increases when the coil rotates counter clockwise when viewed from above. See Fig 4-2. In this coordinate system, the torque delivered by a current flowing around a loop or coil is given by:

$$\tau = -IAB\sin\left(\phi_{coil}\right) \qquad \textbf{eq. 4-1}$$

Additionally, we define the probe angle θ_{probe} as the angle of the probe arm above horizontal. See Fig 4-3. We can convert between the probe angle and the upper coil angle using:

$$\theta_{probe} = \phi_{coil}^{upper} + \theta_0^{upper} \qquad \textbf{eq. 4-2}$$

And between the probe angle and the lower coil angle using:

$$\theta_{probe} = -\phi_{coil}^{lower} - \theta_0^{lower} \qquad \textbf{eq. 4-3}$$

where θ_0^{lower} and θ_0^{upper} are the probe angles at which the lower and upper coils respectively have $\phi_{coil} = 0$. This relationship is set by the orientation of the miter gears and must be measured before the probes are installed.

4.2 Coil Back-EMF

As a coil rotates with respect to the magnetic field, a back-EMF voltage is generated in the coil which is defined as:

$$EMF = -\frac{d\left(\vec{B}\cdot\vec{A}\right)}{dt} \qquad \textbf{eq. 4-4}$$

Assuming that $\vec{B} = B_{\hat{\phi}}$, then

$$EMF = AB\sin\left(\phi_{coil}\right)\frac{d\phi_{coil}}{dt} - A\cos\left(\phi_{coil}\right)\frac{dB}{dt} \qquad \textbf{eq. 4-5}$$

which is picked up in the potential difference across the actuator coil and its power supply. The voltage sensor therefore measures:

$$V = IR - EMF \qquad \textbf{eq. 4-6}$$

Note the standard sign convention here, which is often mistaken [28]. The back-EMF is treated as a voltage source and is therefore positive when it is on the source side of Ohm's law. This is important if we are using the $-A\cos\phi_{coil}\frac{dB}{dt}$ term from a ramping magnetic field to determine the polarity of the coils and what direction the torque will be applied in for a given direction of current. We use these equations to calculate the probe angle as a function of time.

4.2.1 Using the Back-EMF to Determine the Probe's Motion

To calculate the probe position as a function of time, we solve for $\frac{d\phi_{coil}}{dt}$ from eq. 4-5 by assuming that $\frac{d\vec{B}}{dt}$=0.

$$\frac{d\phi_{coil}}{dt} = \frac{EMF}{AB\sin\phi_{coil}} \qquad \textbf{eq. 4-7}$$

Knowing the coil's starting angle, we use a 4th order Runge-Kutta method to solve for ϕ_{coil} and $\frac{d\phi_{coil}}{dt}$ as a function of time. However, each coil crosses through ϕ_{coil} = 0° or 180° during the probe swing, causing a divide by zero error and predicting an infinite angular velocity. We therefore calculate the probe angle θ_{probe} by using a weighted average of the $\frac{d\phi_{coil}}{dt}$ values calculated from each coil. When a coil is aligned within 10° of the field, its contribution is neglected.

The computational routine can either use eq. 4-7 or a more general case where $\vec{B} = \left[B_{\hat{R}}, B_{\hat{\phi}}, B_{\hat{Z}}\right]$ (average through the actuator coil) and $\frac{d\vec{B}}{dt} \neq 0$. The derivation is omitted here for clarity, but is available in [29]. $\vec{B}$ and $\frac{d\vec{B}}{dt}$ are calculated for each coil independently by taking spline interpolates from the EFIT reconstructions.

$$\frac{d\phi_{coil}}{dt} = \frac{EMF + \frac{d\vec{B}}{dt} \bullet \vec{A}}{A\left(B_{\hat{R}} \cos\phi_{coil} + B_{\hat{\phi}} \sin\phi_{coil}\right)} \qquad \textbf{eq. 4-8}$$

When B_t is held constant, the probe positions calculated using eq. 4-7 and eq. 4-8 agree within 0.3° for the conditions tested thus far but the general case is useful when checking the arming of the Swing-Probes since this occurs while B_t is ramping. Including the uncertainties introduced from the backlash in the gears and in the starting angle defined by the spring catches, we assume that the total uncertainty in the probe position is on the order of +/- 1° which corresponds to 3-4mm along the arc of the swing.

The back-EMF work was done by C Tsui.

4.3 Physical Model

In addition to the ability to determine the probe position as a function of time, we need to be able to predict and control the probe's movement. This method, including the accounting for eddy currents closely follows [30].

4.3.1 Eddy Current Drag

The back-EMF detected in the coils is also driven around any structural loops within the frame of the coil. If the loop is conducting, the induced voltages generate currents, which in turn drive a torque that opposes the rotation, acting to slow the system. Assuming a constant $B=B_t$, the back-EMF voltage around a loop is:

$$EMF = AB\sin\left(\phi_{loop}\right)\frac{d\phi_{loop}}{dt} \qquad \textbf{eq. 4-9}$$

where ϕ_{loop} is the angle between the magnetic field direction and the area vector describing the current loop, i.e. the back-EMF goes to zero when the loop is facing directly into the field. If we can estimate the electrical resistance R of the conducting loop, in this case by using the resistivity of 316 stainless steel and the thickness of the mandrel, then we can estimate the magnitude of the current flowing around the loop. Substituting into eq. 4-1, the torque from one eddy current loop path can be described as:

$$\tau_{eddy} = -\kappa B^2 \sin^2\left(\phi_{loop}\right)\frac{d\phi_{loop}}{dt} \qquad \textbf{eq. 4-10}$$

where $\kappa = A^2/R$ is a constant describing the magnitude of the braking. There are two main eddy current paths in each coil frame. See Fig 4-4. The larger path describes currents flowing around the loop of the coil mandrel and is described with κ_{loop}. The smaller path describes currents flowing up one face, through the shaft interface block, and down the opposite face, and is described with κ_{face}. κ_{loop} and κ_{face} can be estimated from geometry to be 4.1 x10^{-3} m^4/Ω and 1.3x10^{-3} m^4/Ω, respectively.

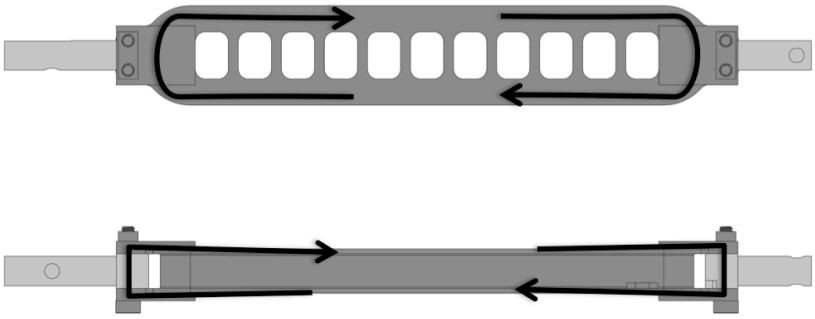


Fig 4-4 The eddy current paths in the actuator coil frames. κ_{loop} represents the current path in the top image and κ_{face} represents the one in the bottom image.

For κ_{loop}, the mandrel frame forms a conducting path with a cross section of about $6.8\times10^{-5}m^2$ with a length of 0.37 m. The resistivity of 316 Stainless is 7.496x10^{-7} Ωm, so the resistance of the current path $R \cong 4$ x10^{-3} Ω. The area of this loop is A≅ 4.0 x10^{-3} m^2, so $\kappa_{loop} \cong 4.1$ x10^{-3} m^4/Ω. Estimates are more complicated for sheets or blocks of metal, see [30]. In practice, these estimates need to be adjusted in the physical model to match the actual behaviour of the probes as observed using the coil back-EMF, but we find the values match within 15%. Eddy current braking from the other components such as the probe arm is negligible compared to that from the coil frames.

4.3.2 Assembling the Physical Model

We can describe the total torque from one coil by combining the torque from the coil current (eq. 4-1) with the eddy current braking (eq. 4-10) for both current paths. The 'loop' eddy current path is aligned with the coil, so we substitute ϕ_{coil} for ϕ_{loop}. For the 'face' path, we use $\phi_{coil}+90°$.

$$\tau_{coil} = -IAB\sin(\phi_{coil}) - \left(\kappa_{loop}\sin^2(\phi_{coil}) + \kappa_{face}\sin^2(\phi_{coil}+90°)\right)B^2\frac{d\phi_{coil}}{dt} \qquad \textbf{eq. 4-11}$$

The actuators are aligned such that they are ~45° from the field when the probe arm is stowed and counter-rotate such that when one is parallel to $\hat{\phi}$, the other is perpendicular within 10°. The motion of the probe assembly can be expressed as:

$$I_m \frac{d^2\theta_{probe}}{dt^2} = \left(\begin{array}{c} \tau_{coil}^{upper}\left(\phi_{coil}, \frac{d\phi_{coil}}{dt}\right) - \tau_{coil}^{lower}\left(\phi_{coil}, \frac{d\phi_{coil}}{dt}\right) + \\ \tau_{springs}\left(\theta_{probe}\right) + \tau_{gravity}\left(\theta_{probe}\right) + \tau_{friction}\left(\frac{d\theta_{probe}}{dt}\right) \end{array} \right) \qquad \textbf{eq. 4-12}$$

where …

I_m	is the effective moment of inertia of the assembly taken from the 3D model.
$\frac{d^2\theta_{probe}}{dt^2}$	is the angular acceleration of the probe.
$\tau_{coil}^{upper}\left(\phi_{coil}, \frac{d\phi_{coil}}{dt}\right)$	is the total torque delivered by the upper coil as a function of ϕ_{coil}, and $\frac{d\phi_{coil}}{dt}$ calculated using eq. 4-11.
$-\tau_{coil}^{lower}\left(\phi_{coil}, \frac{d\phi_{coil}}{dt}\right)$	is the total torque delivered by the lower coil. This term is negative because the lower coil rotates in the negative direction when the probe rotates in the positive direction.
$\tau_{springs}\left(\theta_{probe}\right)$	is the torque delivered by the spring assembly calculated using Hooke's law and the geometry of the crank wheel.
$\tau_{gravity}\left(\theta_{probe}\right)$	is the torque from the force of gravity pulling down on the probe arm, and is highest when the arm is horizontal and θ_{probe}=0.
$\tau_{friction}\left(\frac{d\theta_{probe}}{dt}\right)$	is set to a constant dynamic value opposing the angular velocity of the system, and is adjusted to match the back-EMF data.

κ_{loop}, κ_{face} and $\tau_{friction}$ are adjusted such that the back-EMF from the model matches that of the actual probe. This model is an ordinary differential equation, which we solve using standard numerical methods. When we are developing the current control waveforms, it is easier to use a spreadsheet version of the physical model. This version solves for θ using the derivatives from the last time step.

We program the actuator power supplies (two Kepco BOP 36-12 units) using the current controls since using the voltage controls would cause the currents to increase to the power supply limits whenever the coil reached its null point at ϕ_{coil} = 0,180°. This would cause unnecessary heating in the coils and create a larger than necessary error field. Since the currents must change polarity partway through the swing, there's no obvious way to use a feedback control system. We therefore use open loop control waveforms.

The results from the probe angle calculations can be seen in Fig 4-5 to Fig 4-8 in comparison to the physical model after κ_{loop}, κ_{face} and $\tau_{friction}$ have been carefully tuned, i.e. the model has been fit to the data. κ_{loop} = 3.6 x10^{-3} m^4/Ω, κ_{face} = 0.9 x10^{-3} m^4/Ω and $\tau_{friction}$ = 0.02 N-m.

For t < 100 ms, a small negative current causes both actuator coils to hold the probe against the armed limit switch. At t = 100 ms, a positive current through both coils pushes the probe away from the limit switch and into the plasma. The upper coil becomes parallel with $\vec{B}$ at t = 170 ms, at which time this coil has zero torque and its back-EMF reaches zero; See Fig 4-6 in black/blue. This is the appropriate time to change the polarity of the current flowing to this coil; See the black line in Fig 4-5. The coil geometry is setup such that when one coil is parallel to $\vec{B}$, the other coil is perpendicular to $\vec{B}$. At t = 260 ms, the lower coil is now parallel to $\vec{B}$ and the upper coil is perpendicular. Also at t = 260 ms, the magnitude of the currents flowing to both coils is decreased, though still torqueing the probe towards its destination. This decrease in current allows the eddy current braking to slow the probe down for a softer landing. The probe hits the stowed limit switch at t = 300 ms, and bounces several times

which can be seen in the angle calculated from the back-EMF (red points Fig 4-7 which is represented by the zig-zags seen in Fig 4-6.

There is some variability in the probe's motion from shot to shot such that the landing time varies by up to 5ms. We find that once the physical model has been setup for any given shot, it accurately predicts the probe behaviour within the same margin of error.

The physical models are predominately the work of C Tsui.

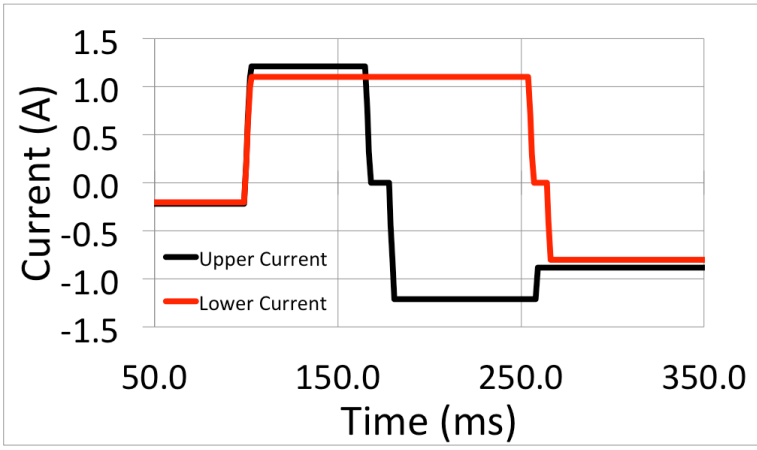


Fig 4-5 A typical current control waveform for the fireing sequence of the lower Swing-Probe. The coils hold the probe against the armed limit switch until 100ms, drive the probe forward from 100ms to 260ms, changing polarity as necessary. After 260ms the forward torque is decreased to allow the probe to decelerate.

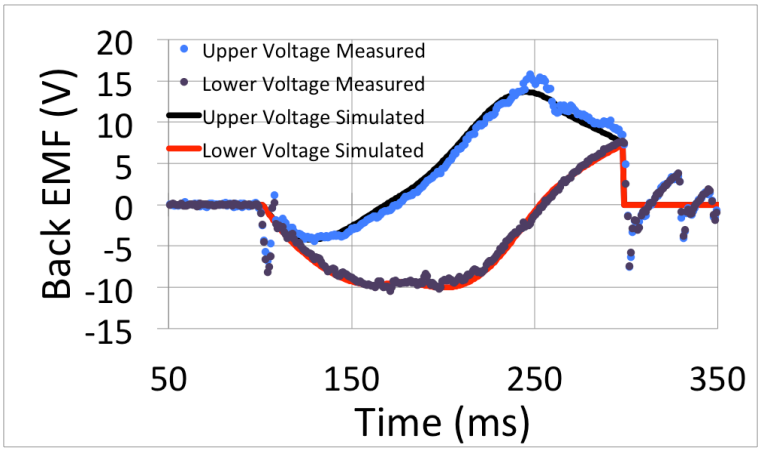


Fig 4-6 The simulated and measured back EMF from a typical swing compared to simulated values from a well-tuned physical model. Differences are due to the backlash in the gears, e.g. the upper actuator transitions from accelerating to decelerating at 250ms, and the shift within the gears causes a jump in the back-EMF. After the probe lands at 300ms it bounces several times causing the back-EMF to flip polarity rapidly.

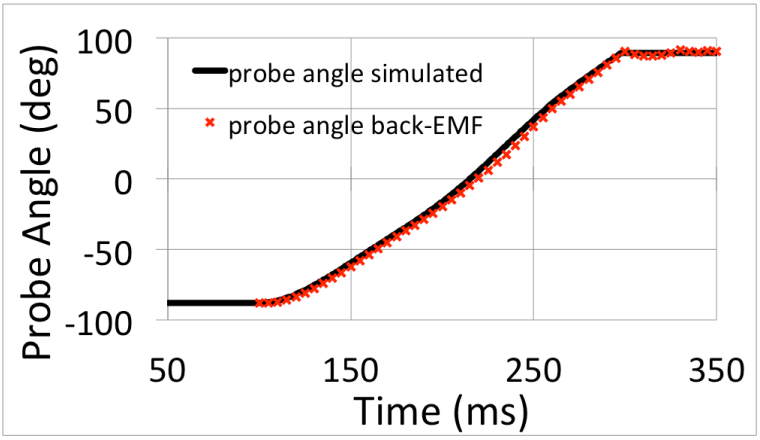


Fig 4-7 The probe angle simulated in the model and calculated from the back EMF.

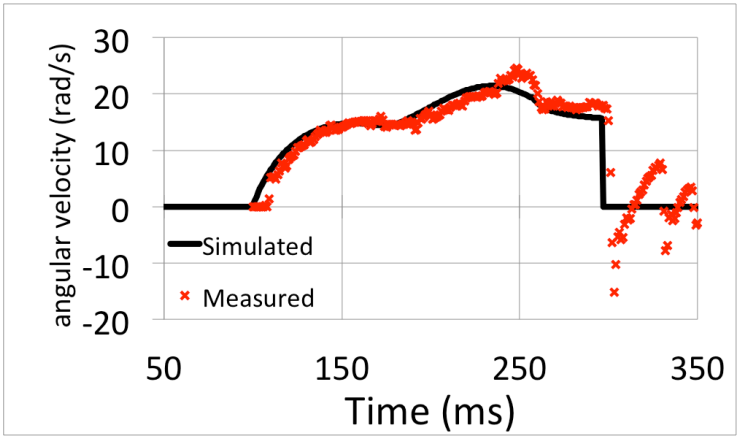


Fig 4-8 The probe's angular velocity simulated in the model and calculated from the back EMF.

4.4 Electronics and Data Acquisition

The Swing-Probes require an intricate control system. The electronics setup is detailed in Fig 4-9. Diagnostics engineer Matt Watkins deserves a special acknowledgement for the Swing-Probe electronics, which he designed to meet some very stringent requirements. For example, the limit switch control circuits float such that they can continue to operate even if there is a short somewhere in the system. This was an important level of redundancy since many of the insulators used in vessel were untested.

The electronics system is only able to control one probe at a time and requires a patch panel change to swap between them. Each actuator coil is powered by a Kepco BOP 36V-12A. The electrodes are powered by three Kepco BOP 100V-4A connected in series to provide +/- 300V. A signal generator controls all 5 power supplies and is remotely programmed via fibre optic USB. The control signals pass through isolation relays on the circuit board that open in the event of a fault. These components are kept in a rack in the machine pit that is held at machine potential, which can float several thousand volts away from earth ground. The entire rack, including the electronic grounds,

requires 6kV electrical isolation to earth ground as well as any components that a person may come in contact with.

The control system, interface computer, and the data acquisition system are therefore separated from the power supplies and are located outside of the shield wall in the Annex. All communication between the control systems in the Annex and the power supplies in the machine pit are relayed via fibre optics, providing the necessary level of electrical isolation.

Unfortunately, fibre optic transmitters are typically unable to transmit a DC signal, which is necessary for Langmuir probe analysis. Most analog fibre transmitters are designed to transmit video signals and therefor have a low frequency cut-off ~10Hz. The only transmitters that can operate below 10Hz are quite limited in terms of dynamic range and ease of use. We used the Analog Modules 732 fibre optic links, which are the same links used by all of the Langmuir probe systems in DIII-D that have been upgraded from the old CAMAC digitizers which are capable of operating in the machine pit.

The analog fibre links have major drawbacks. The gain and offset drift widely unless the fibre optic connectors are glued in place. Even then, manual calibrations were needed almost daily to keep the control signals properly zeroed. Shot-by-shot calibrations correct for the offset drift, but not for the drift in the gain. Complex signal conditioning circuit boards were needed at both ends to make the most of the limited dynamic range. The boards also provide buffering, anti-aliasing, control the limit switches and interface with the RESET interlock.

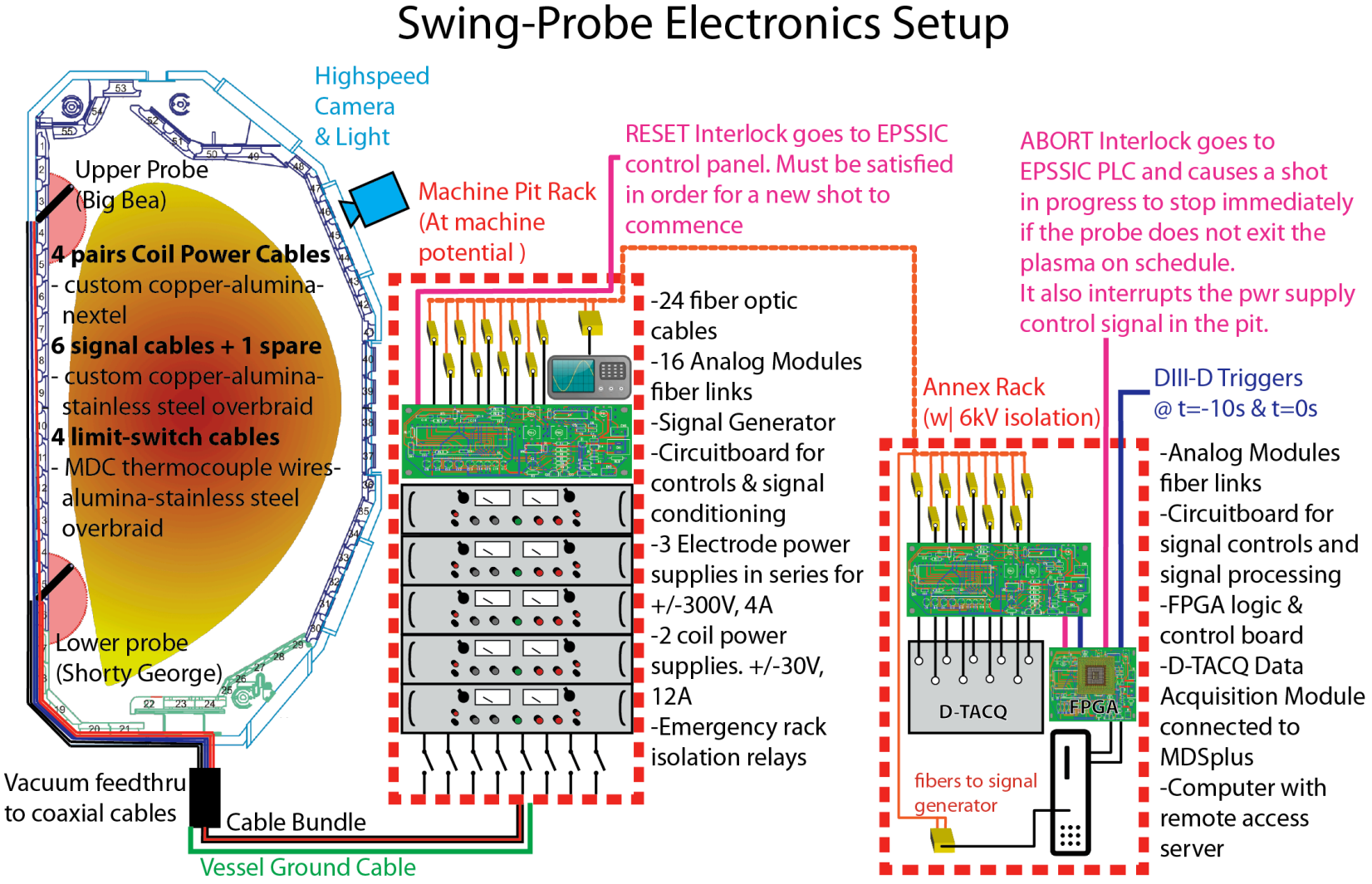


Fig 4-9 The electronics setup for the DIII-D Swing-Probes

The Analog Modules 732 fibre transmitters have a signal to noise ratio of 52dB, meaning that it introduces noise which is ~1/400th its maximum voltage range. Including the fact that the offsets drifts actually decreases the useful maximum voltage range, we find the true value is closer to 1/300. For comparison, our 14-bit digitizers have signal to noise ratio of 84dB = 1/16,000. This signal to noise ratio is not sufficient for the interpretation of Langmuir probe data, particularly for the current signal for a swept probe where we are interested in a dynamic range spanning nearly 3 orders of magnitude.

There was no good solution to this problem. Reliable analog fibre links suitable for this application are under development, and will be replacing the 732s at DIII-D in the years to come. As a stopgap measure, the electrode current and voltage signals are transmitted twice with two separate gain levels. The voltage channels are limited at +/- 300V and at +/- 60V giving a minimum voltage resolution of 200mV. The current channels are limited at +/- 1A and at +/- 50mA giving a minimum current resolution of 0.2mA. These signals are digitized separately and must be knit together as part of the data analysis.

Digitization is performed using a D-TACQ data acquisition module operating at 500ksamples/s on 13 channels at 14 bits. The D-TACQ uploads the data to DIII-D's data servers running MDSplus.

The timing control, control relays, and the ABORT interlock are implemented on an Altium Field Programmable Gate Array (FPGA) development board. Initially, it was planned to use the digital to analogue converters on the board to output the control waveforms from onboard storage, in which case the FPGA board would act as a fully programmable and fully integrated control system for the Swing-Probes. Unfortunately, the unreliability of the fibre optic links made controlling the power supplies from the annex too unpredictable, and the FPGA board cannot survive the radiation load in the pit. Instead, a stand-alone signal generator is used in the pit and can be programmed by USB via a fibre optic patch box. Most electronics do operate reliably in the machine pit with the exception of those with microprocessors or FPGA chips.

The electronics systems were predominately designed by M Watkins.

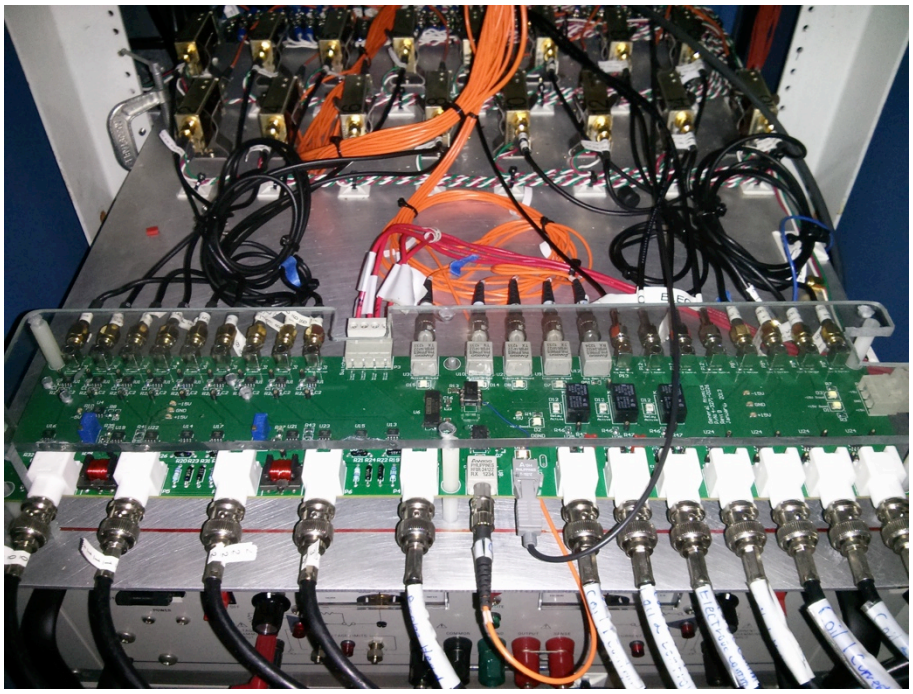


Fig 4-10 A photo of the machine pit rack's communication panel. The yellow boxes at the top are the fibre optic links that transmit data to the Annex from the data processing and control circuit board in the center of the photo. At the bottom, the first of five Kepco power supplies is visible.

4.5 Control systems and Interlock Integration with DIII-D

Before the start of any DIII-D shot, E Power Supply System Integrated Control (EPSSIC) checks that Swing-Probe RESET interlock is good. The RESET interlock is a simple circuit located in the machine pit and it reads 'good' if both probes are in contact with their stowed limit switches. This insures that a probe is not caught in the path of the plasma. A shot cannot commence until this interlock is satisfied even if the Swing-Probes are not being used.

To operate a probe, the appropriate control waveforms generated using the physical models are uploaded to the signal generator in the machine pit, the signal generator is armed to output on trigger, and the FPGA board is enabled. The t = -10 s synchronous trigger activates the FPGA board which starts its own clock and triggers the signal

generator. The digitizer begins collecting data at t = -6 s. The control relays remain open until t = -5 s providing a true zero calibration reference. The toroidal field begins is ramp up around t = -2 s. The probe swings from its stowed limit switch to its armed limit switch between t = -500 ms and t = -300 ms when the toroidal field is still ramping. At t = -100 ms, the FPGA board checks if the probe has successfully armed. If it has not, the control relays in the pit are opened, allowing the springs to pull the probe out of the vessel without interference from the actuators which are likely malfunctioning. The FPGA board also trips the ABORT interlock, which causes DIII-D's relays to open under load, stopping the shot before plasma breakdown.

If the probe has successfully armed, the shot proceeds while the actuators hold the probe against its armed limit switch. At the designated time, the actuators and the springs work together to pull the probe through the plasma and back to its stowed limit switch while the electrode voltage is alternated. At a pre-set time, the FPGA board checks if the probe has returned to its stowed position and will trip the ABORT interlock if it has not. In this case, the opening DIII-D's relays causes a disruption, but this is preferable to leaving the probe exposed to a heated plasma.

The Swing-Probe electronics are still experimental and additional development would be ideal. Integrating the voltage sensor at the probe head and the limit switch sensors was more difficult than anticipated, resulting in a noisy and occasionally unreliable limit-switch system. The system will sometimes trigger the ABORT interlock even when the system operated correctly, which is frustrating and expensive for the DIII-D program. Currently, the control system is very prone to human error. The Swing-Probe operator must connect the right probe, manually setup two channels on the signal generator with the correct timing, and setup the correct timing in the FPGA board. A small mistake in any of these steps will cause the probe to misfire and abort a shot. A relatively modest engineering investment could improve reliability, reduce human error and potentially expand Swing-Probe capability to 2 swings per shot.

The control systems were predominately designed by M Watkins.

Chapter 5 Langmuir Probe Data Analysis

The raw data extracted from the Swing-Probe system (or any Langmuir probe system) are temporal voltage and current signal traces. In order to extract useful plasma parameters from these data, one requires a least squares fitting of an I-V (current-voltage) characteristic function. Although the concept is simple, I-V fitters are prone to converging to local minima in the reduced Chi-squared instead of the global minimum. As a result, these fitters can be very finicky and require a great deal of intervention and inspection in order to produce trustworthy results. Langmuir probe data were typically unavailable to DIII-D physicists until one or more days following the experiment due to the long processing time required.

As part of this thesis, a new I-V fitter was developed that successfully addresses these problems. This software is reliable enough to be run automatically immediately following a plasma shot, and returns values of the plasma parameters quickly enough that the experimental session leader can use it to make informed adjustments to the following shot. Furthermore, the software is flexible enough to accommodate complex I-V fitting functions. It was therefore possible to develop and use an I-V fitter that accounts for sheath expansion in a theoretically consistent way. The novel aspects of this approach are discussed in sections 5.2-5.4.

5.1 Reliable Langmuir Probe Fitter

This code uses the MPFIT library [31] which is an IDL port of the MINPACK library [32], and based on the Levenberg-Marquardt algorithm. MPFIT allows simple boundary constraints to be imposed on each parameter. Testing for and constraining I_{sat} and V_f to reasonable ranges speeds up the fitting, and stops it from converging to a local minimum outside of the permitted values. The resulting program converges to

appropriate plasma parameters a large majority of the time and (more importantly) will consistently report the error condition when it does not.

MPFIT also allows model parameters to be prescribed and held constant during the fitting procedure, making it straightforward to include into the fitting function semi-constant parameters such as the projected surface area. This added flexibility makes it possible to use some of the more complex fitting functions described in the following sections.

5.2 Sheath Expansion

The standard Langmuir I-V fitting function can be expressed as:

$$I = A_{coll} J_{sat} \left[1 - EXP\left[\frac{e\left(V - V_f\right)}{kT_e} \right] \right] \qquad \textbf{eq. 5-1}$$

where A_{coll} is the collection surface area, which is equal to the cross-field area of the magnetic flux-tube subtended by the electrode. This area is increased by the presence of the plasma sheath that grows thicker as the potential difference across it increases. We define the "Standard I-V Fitter" to be that which uses eq. 5-1 with $A_{coll} = A_0$, see Fig 5-1.

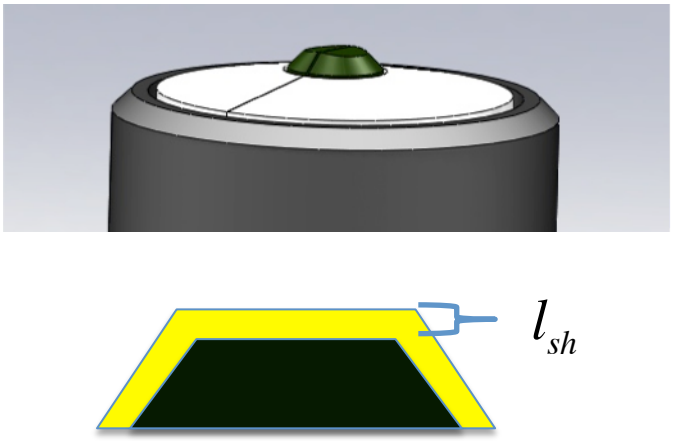


Fig 5-1 (top) a schematic of the Swing-Probe head and electrodes. (bottom) The collection surface area of the Swing-Probe electrode. The black area represents the projected geometrical surface area A_0 while the yellow area represents the additional collection area due to the sheath.

At voltages sufficiently below the local $V_{plasma,}$ all of the electrons are repelled from the electrode and all of the ions are collected from the Debye sheath as quickly as they can enter it. The region characterized by this behavior is called the 'ion branch' and eq. 5-1 says that the ion current should remain constant even as the voltage is decreased further. i.e. the ion current is said to be 'saturated.' However, the thickness of the sheath does increase as the potential difference across it increases and thus, the ion current can continue to increase with further decrease in voltage.

While this effect can be ignored for large Langmuir probes, smaller ones, such as those used in all of the reciprocating probes in DIII-D [33,34] have I-V characteristics where the 'saturated' current is far from constant. See example in Fig 5-2.

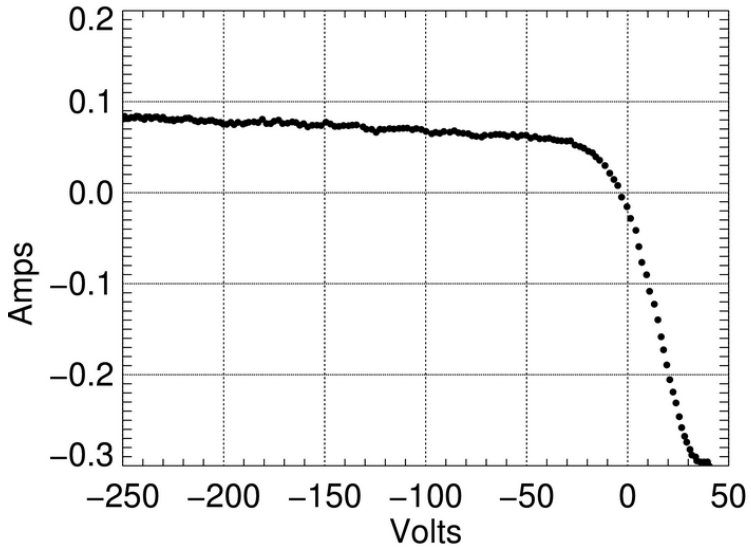


Fig 5-2 An example of an I-V characteristic where the ion saturation branch does not actually saturate due to sheath expansion. The data are digitized at 500 kHz. It takes < 500 μs to collect the data shown here.

Strongly non-saturating examples such as the one in Fig 5-2 can cause the Standard I-V Fitter to report inaccurately high values for I_{sat} and T_e.

While sheath expansion has been long studied, see e.g. [35], methods for dealing with its effects on Langmuir probes interpretation are relatively limited. One method [36] is to include a sloped line to the fit by adding a fourth fitting parameter, see eq. 5-2. This is an empirical modification to match the non-saturating current and is not derived from theory

$$I = I_{sat}\left[1 - EXP\left[\frac{e\left(V - V_f\right)}{kT_e}\right]\right] + \left(\Delta I / \Delta V\right)\left(V - V_f\right) \qquad \textbf{eq. 5-2}$$

where the parameter $\left(\Delta I / \Delta V\right)$ is varied, along with the values of the 3 parameters used in the Standard I-V Fitter: I_{sat}, T_e and V_f. This 4th parameter deals with the non-saturating ion collection. The extracted values of $\left(\Delta I / \Delta V\right)$ change from I-V characteristic to I-V

characteristic. The addition of a 4th totally free parameter is a concern as it may result in over-interpretation of the probe data. It is difficult to determine whether the improved statistics of the fit, e.g. reduction in Chi-squared, is significant or is just caused by the over-fitting of noise. While it is often possible to make this judgment call by eye on a case-by-case basis, this becomes impractical if one has to fit large numbers of I-V characteristics. In this thesis, I propose a theoretically constrained treatment of sheath expansion as an alternative; this results in just the same 3 free fitting parameters of the Standard I-V function.

The goal here is not to develop or evaluate a refined model of the sheath or sheath expansion. These models have been developed and studied elsewhere, see e.g. [35]. The goal here is to create a reliable robust fitter that can extract T_e, J_{sat} and V_f automatically in low n_e, T_e conditions where the ion current does not saturate as well as in high n_e, T_e conditions where sheath expansion effects are negligible.

This appears to be the first analysis tool for protruding probes that accounts for sheath-expansion using a theoretically constrained function. These tools have been developed extensively for flush-mounted probes [37–39], but the flush geometry adds complications from magnetic pre-sheath effects and neighbouring sheath geometries such that these models generally are not useful for protruding probes.

5.3 Child-Langmuir sheath expansion function

We will consider an approach to fitting Langmuir I-V characteristics that uses the Child-Langmuir law to account for sheath expansion in an approximate way. We start with the I-V characteristic eq. 5-1. The ion collection surface area A_{coll} is simply assumed to increase with the voltage drop across the sheath according to eq. 5-3

$$A_{coll} = A_0 + l_p l_{sh}\left(\Delta V_{sh}\right) \qquad \textbf{eq. 5-3}$$

where l_{sh} is the thickness of the sheath and l_p is the perimeter of the electrodes. For the Swing-Probe electrodes, A_0 = 4.4mm^2 and l_p = 8.6mm when considering the current collected in both directions along the field.

Using the Child-Langmuir Law, we can obtain that to a first approximation [40]

$$l_{sh} \approx \lambda_D \left(e\Delta V / kT_e \right)^{3/4} \qquad \textbf{eq. 5-4}$$

where the Debye length λ_D is given as [41]

$$\lambda_D = \sqrt{\frac{\varepsilon_0 kT_e}{e^2 n}} = 7434 \times \sqrt{\frac{T\left[eV\right]}{n\left[m^{-3}\right]}} \quad \left[m\right] \qquad \textbf{eq. 5-5}$$

We define the Simple Sheath Expansion Fitter by assuming that $\Delta V_{sh} = \left| V - V_f \right|$

$$l_{sh} = c\lambda_D \left(e\left| V - V_f \right| / kT_e \right)^{3/4} \qquad \textbf{eq. 5-6}$$

where we have included a numerical factor c since eq. 5-4 is only approximate, see e.g. Fig. 1 of [35], Fig. 2.6 of [40], etc. The value of *c* is evaluated below.

5.3.1 Evaluating c

The Simple Sheath Expansion Fitter was applied to several sets of I-V data with low-noise and strong sheath expansion over a range of conditions (B_t, I_p, P_{inj}) in order to find an optimal value for c to match the shape of the non-saturating ion branch. We found that a value of c = 1.8 produced a good match to the data. Some examples are shown in Fig 5-3 to Fig 5-8. The Simple Sheath Expansion Fitter (solid lines) is applied to the datapoints in green (only) with c=1.8. The dotted lines represent the same parameters plotted with no sheath expansion, c=0.

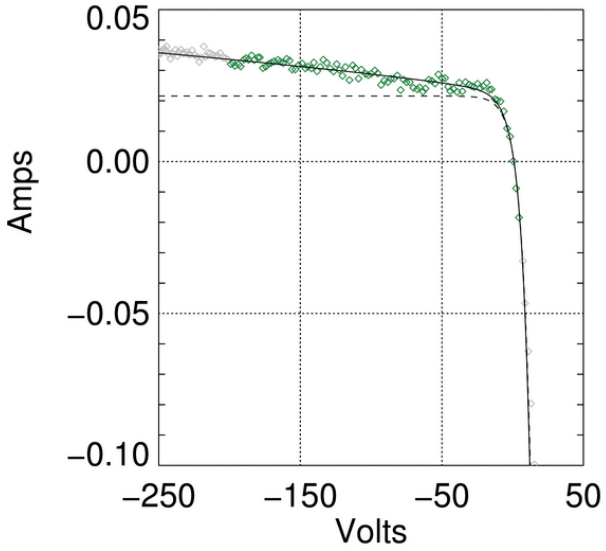


Fig 5-3 Sheath expansion example from typical ohmic shot 152838 sweep#90.
B_t = -2.05 T I_p = 1.3 MA P_{tot} = 946 kW
I_{sat} = 0.0216 A V_f = 0.923 V T_e = 6.67 eV

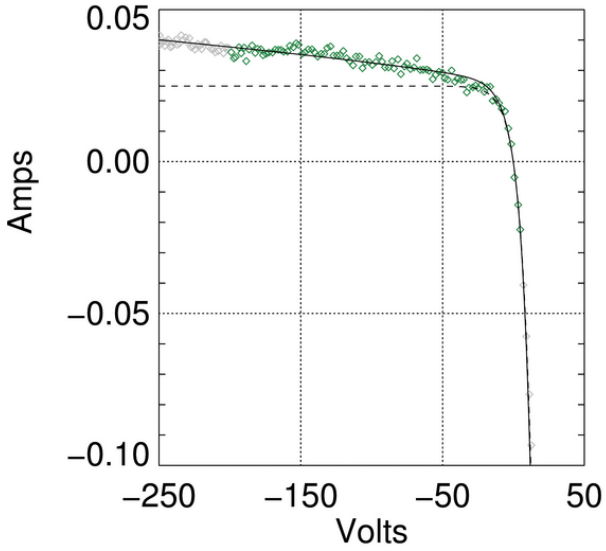


Fig 5-4 Sheath expansion example from typical ohmic shot 152838 sweep #98.
B_t = -2.05 T I_p = 1.3 MA P_{tot} = 946 kW
I_{sat} = 0.0248 A V_f = -0.164 V T_e = 7.66 eV

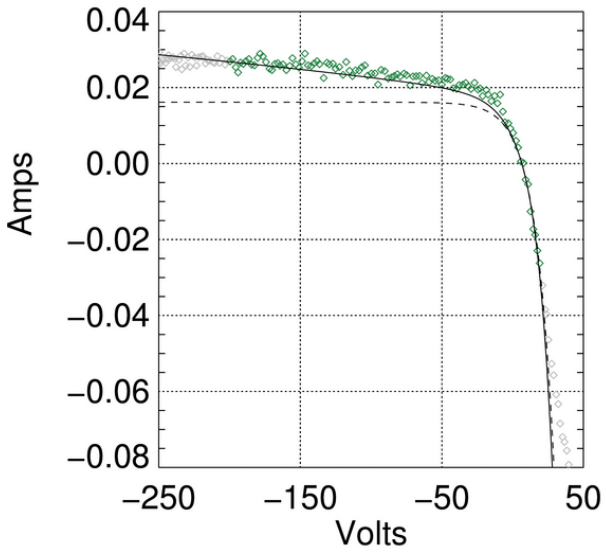


Fig 5-5 Sheath expansion example from low I_p plasma shot 155650 sweep#87.
B_t = -2.05 T I_p = 610 kA P_{tot} = 355 kW
I_{sat} = 0.0162 A V_f = 6.48 V T_e = 12.7 eV

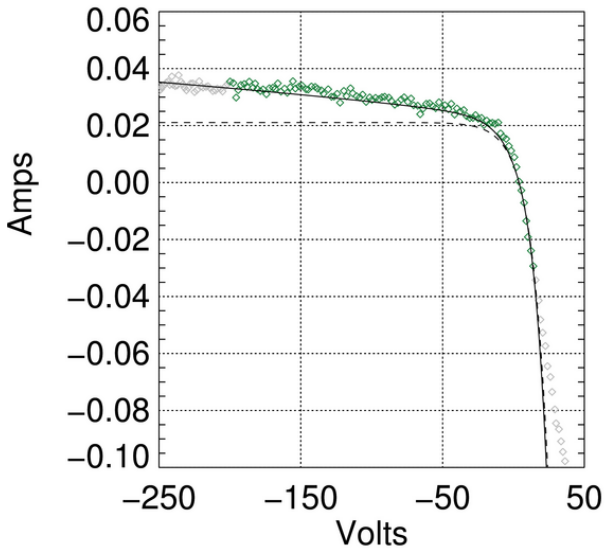


Fig 5-6 Sheath expansion example from low I_p plasma shot 155650 sweep#91.
B_t = -2.05 T I_p = 610 kA P_{tot} = 355 kW
I_{sat} = 0.0211 A V_f = 3.88 V T_e = 11.6 eV

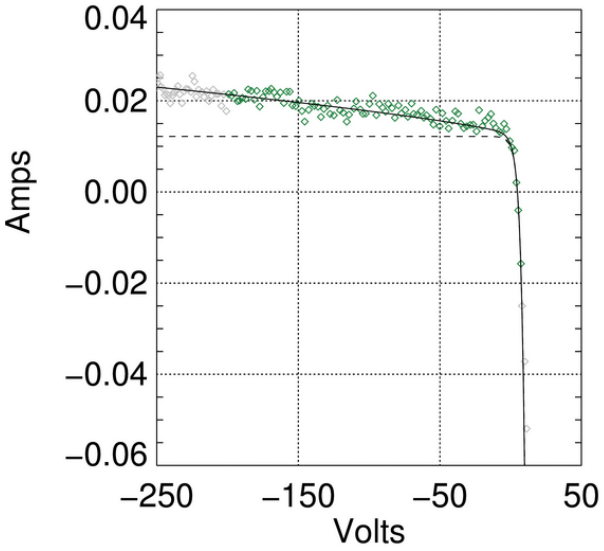


Fig 5-7 Sheath expansion example from high power low B_t shot 153836 sweep#360.
B_t = -1.95 T I_p = 1.1 MA P_{tot} = 9.78 MW
I_{sat} = 0.0122 A V_f = 4.78 V T_e = 2.9 eV

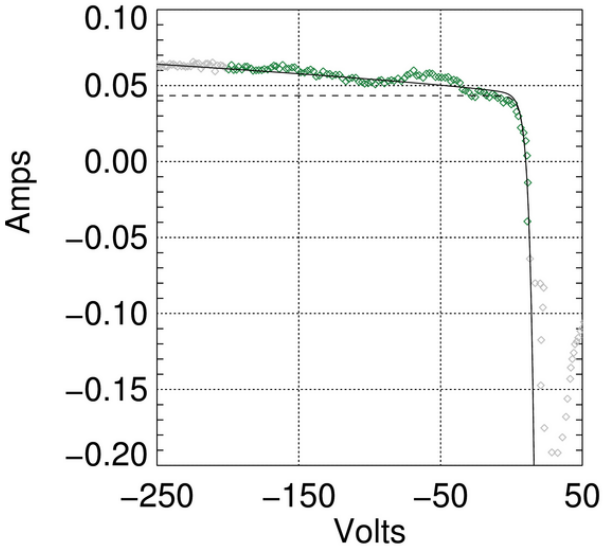


Fig 5-8 Sheath expansion example from high power low B_t shot 153836 sweep#85.
B_t = -1.95 T I_p = 1.1 MA P_{tot} = 9.78 MW
I_{sat} = 0.0434 A V_f = 10.1 V T_e = 3.37 eV

We find that a value of c = 1.8 appears to match the shape of the non-saturating ion branch well in all the above conditions. That is, once the value of c has been determined for a particular electrode, this fitter is able to reliably calculate the magnitude of the sheath expansion effect for all I-V characteristics measured by this electrode.

We note that these examples are not typical Swing-Probe I-V characteristics and were measured within several millimeters of the centerpost wall. The low-density conditions here provide strongly non-saturating I-V characteristics for testing sheath expansion fitters. The magnitude of the sheath-expansion effect at $V-V_f$ = -100V is listed in Table 5-1

n_e [m^{-3}]	T_e [eV]	λ_D [m]	l_{sh} (-100V) [mm]	Sheath 'area' $l_p l_{sh}(-100V)$ [mm^2]	Increase in Collection Area (-100V)
5E+18	10	1.1E-5	1.1E-1	1.7	37%
5E+18	25	1.7E-5	8.5E-2	1.3	30%
5E+18	50	2.4E-5	7.1E-2	1.1	25%
1E+19	10	7.4E-6	7.5E-2	1.2	26%
1E+19	25	1.2E-5	6.0E-2	0.9	21%
1E+19	50	1.7E-5	5.0E-2	0.8	18%
5E+19	10	3.3E-6	3.4E-2	0.5	12%
5E+19	25	5.3E-6	2.7E-2	0.4	9%
5E+19	50	7.4E-6	2.3E-2	0.3	8%

Table 5-1 The magnitude of the sheath expansion effect tabulated for the condition where $V-V_f = -100$ V. Sheath expansion is affected most strongly by the density.

5.3.2 Comparing I-V Fitting Models

To evaluate the performance of the Simple Sheath Expansion Fitter, we will need to consider a wider range of conditions including those where the sheath expansion effects are smaller. We will compare the Standard IV Fitter, the Simple Sheath Expansion Fitter with c = 1.8, and the Simple Sheath Expansion Fitter using c as a freely fitted (4th) parameter. For clarity, we will refer to the latter as the 'Four-Parameter Fitter.' We will also compare the 'Linear Sheath Expansion Fitter' described in eq. 5-2.

We will compare these fitters using reduced-chi-squared, which is defined as:

$$\chi^2_{red} = \frac{1}{\nu}\Sigma_{-200V}^{\sim V_f}\frac{(O-E)^2}{\sigma^2} \qquad \textbf{eq. 5-7}$$

where O is observed data, E is the theoretical data and ν is the number of degrees of freedom = N-n-1 where N is the number of datapoints being fit, n is the number of fitting parameters. σ^2 is the known variance of the data, which will be estimated from datapoints between -250V and -200V. Since the σ^2 estimate is rough, we loosely expect that $\chi^2_{red} \gg 1$ implies a poor fit, $\chi^2_{red} \cong 1$ implies an optimal fit, and $\chi^2_{red} \ll 1$ implies over fitting. Of course, we expect that the fitters with 4 free parameters will produce lower

values of χ^2_{red} than those with only 3 free parameters. We aim to show that the closer fits from the 4 parameter fitters often impedes the reliability with which we can extract useful values for I_{sat}, T_e, and V_f.

In Fig 5-9-Fig 5-12, the fitters will be applied over the data (black) with voltages from -200V to just above the floating potential. Data outside of this range (grey) are not used in the fitting.

The Standard I-V Fitter will be fitted over a smaller voltage range for conditions with strong sheath expansion effects in order to mitigate the over-estimation of I_{sat}. This is a level of active management that would be performed by any responsible diagnostician. The above χ^2_{red} definition is therefore meaningless for the standard I-V fitter, so values will not be reported.

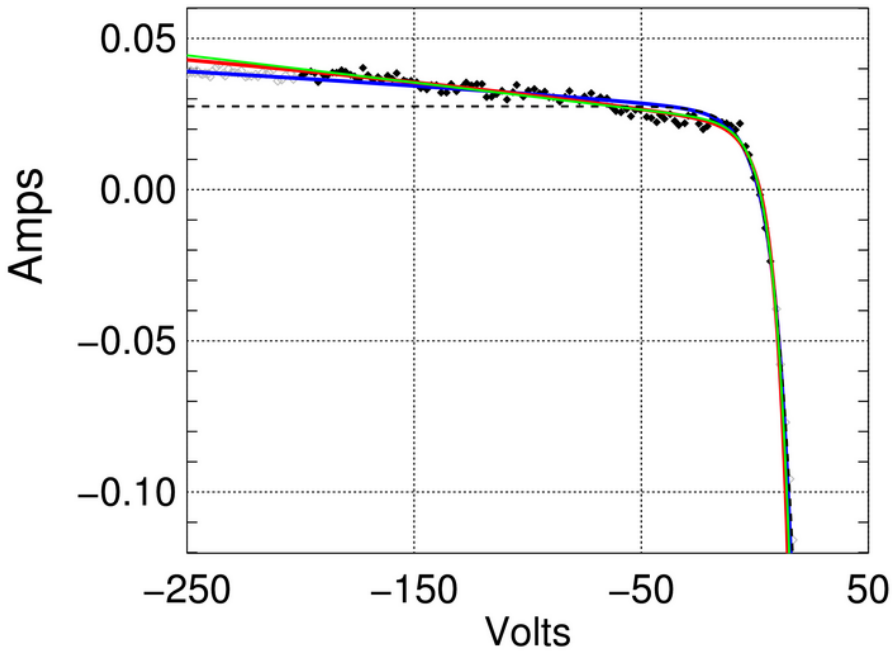


Fig 5-9 An I-V characteristic example with a significantly non-saturating ion branch. (shot 152838 sweep#100)

Standard I-V Fitter (dashed)	Simple Sheath Expansion Fitter (blue)	Four-Param Fitter (red)	Linear Sheath Expansion Fitter (green)
I_{sat} = 0.013 A	I_{sat} = 0.024 A	I_{sat} = 0.019 A	I_{sat} = 0.022 A
V_f = 6.5 V	V_f = 1.6 V	V_f = 2.2 V	V_f = 2.0 V
T_e = 9.3 eV	T_e = 8.2	T_e = 6.6 eV	T_e = 7.0 eV
	c = 1.8 (fixed)	c = 3.1	ΔI/ΔV = -9.0E-05
	χ^2_{red} = 4.99	χ^2_{red} = 2.81	χ^2_{red} = 2.73

See Fig 5-9. Unsurprisingly, the Four-Param Fitter and the Linear Sheath Expansion fitter have closely agreeing curves as they both adjust to minimize the residual in the ion saturation branch. Notice that the Simple Sheath Expansion Fitter and the Four-Parameter Fitter report significantly different parameters despite the fact that they have similar curves. The Four-Parameter Fitter has decreased χ^2_{red} by 40% compared to the Simple Sheath Expansion Fitter. It's difficult to tell from this example whether the Four-Parameter Fitter is doing a better job, or if it is over fitting the small fluctuation between -75V and -25V. However, the fact that I_{sat} and T_e can change so dramatically for a small and reasonable adjustment to c argues against accounting for sheath expansion by using

a fourth totally free parameter. Here, the Linear Sheath Expansion Fitter achieves the smallest χ^2_{red}.

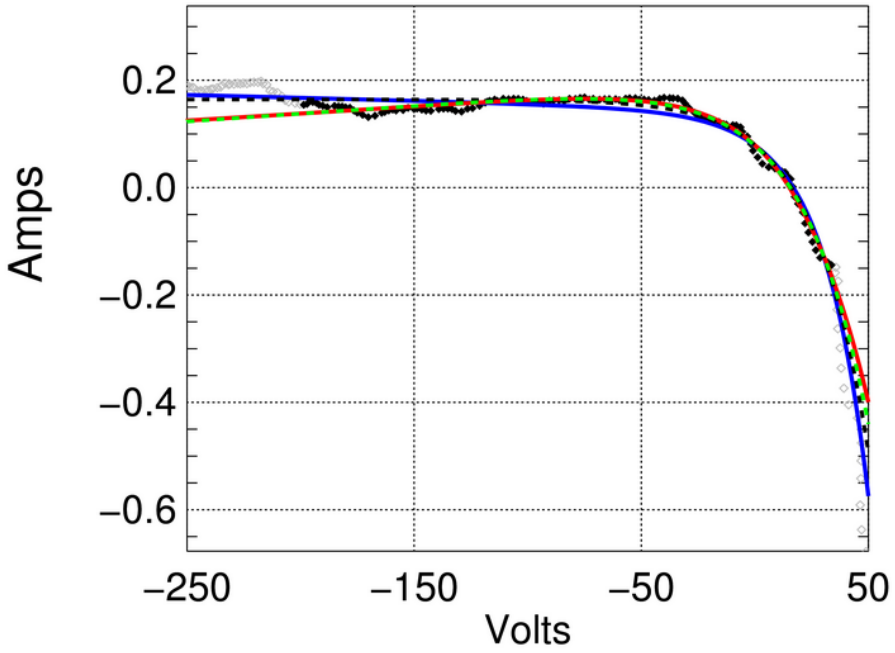


Fig 5-10 A moderately noisy I-V characteristic example.

(shot 152838 sweep#227)

Standard I-V Fitter (dashed)	Simple Sheath Expansion Fitter (blue)	Four-Param Fitter (red)	Linear Sheath Expansion Fitter (green)
I_{sat} = 0.164 A	I_{sat} = 0.135 A	I_{sat} = 0.216 A	I_{sat} = 0.201 A
V_f = 16.2 V	V_f = 16.9 V	V_f = 15.3 V	V_f = 15.6 V
T_e = 24.4 eV	T_e = 20.5 eV	T_e = 31.2 eV	T_e = 29.3 eV
	c = 1.8 (fixed)	c = -3.5	ΔI/ΔV = 0.000294
	χ^2_{red} = 2.42	χ^2_{red} = 0.764	χ^2_{red} = 0.84

Fig 5-10 shows a moderately noisy case where we still expect to extract useful data from the I-V characteristic. Sheath expansion is small for these conditions, and the Standard I-V fitter performs well as expected. However, the Four-Parameter Fitter and the Linear Sheath Expansion Fitter are over fitting the noise with χ^2_{red} substantially smaller than that of the Simple Sheath Expansion Fitter and less than 1. The negative value of c=-3.5

is not physical, nor is the positive ΔI/ΔV. It is possible to constrain c and ΔI/ΔV to reasonable ranges, but constraining c to a single well-tuned value provides good results.

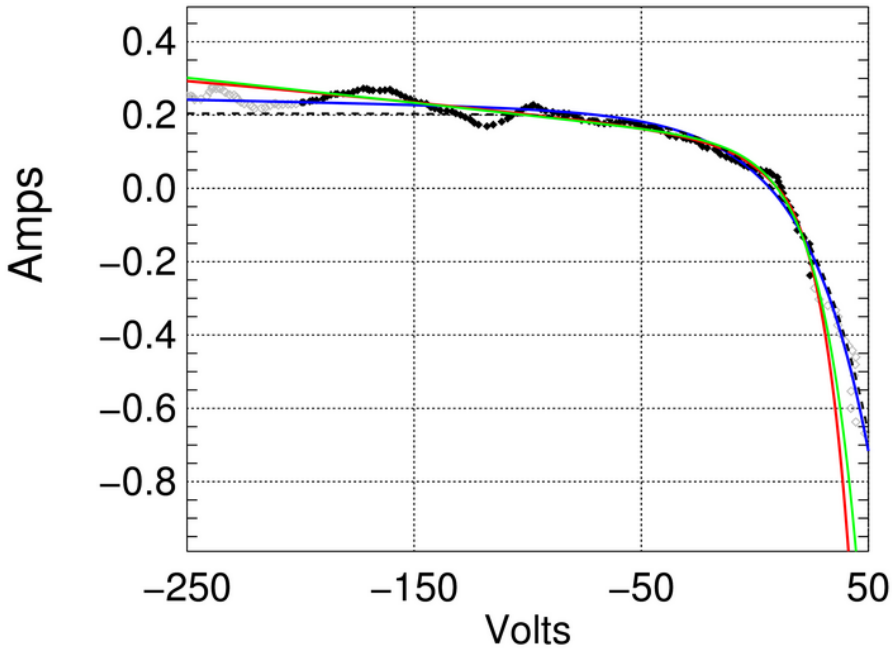


Fig 5-11 An I-V characteristic example with significant plasma fluctuations.

(shot 152838 sweep#253)

Standard I-V Fitter (dashed)	Simple Sheath Expansion Fitter (blue)	Four-Param Fitter (red)	Linear Sheath Expansion Fitter (green)
I_{sat} = 0.204 A	I_{sat} = 0.198 A	I_{sat} = 0.101 A	I_{sat} = 0.124 A
V_f = 7.77 V	V_f = 6.77 V	V_f = 10.1 V	V_f = 9.9 V
T_e = 29.1 eV	T_e = 29.1 eV	T_e = 14.9 eV	T_e = 15.9 eV
	c = 1.8 (fixed)	c = 11	ΔI/ΔV = -0.000684
	χ^2_{red} = 1.78	χ^2_{red} = 0.933	χ^2_{red} = 1.03

Fig 5-11 shows a very noisy example. In this case, the Simple Sheath Expansion Fitter rejects the fit as the parametric error exceed accepted thresholds. Specifically for this case, T_e can be varied significantly without a significant change in χ^2_{red}, so the fitter rejects the data over lack of confidence over the reported parameters. The fitted line does however appear to represent the data quite well. Once again, the Four-Parameter Fitter and the Linear Sheath Expansion Fitter over-fit the noise, registering an

unphysically large sheath expansion effect for this example, though we can only tell here by comparing against the unfitted data shown in grey.

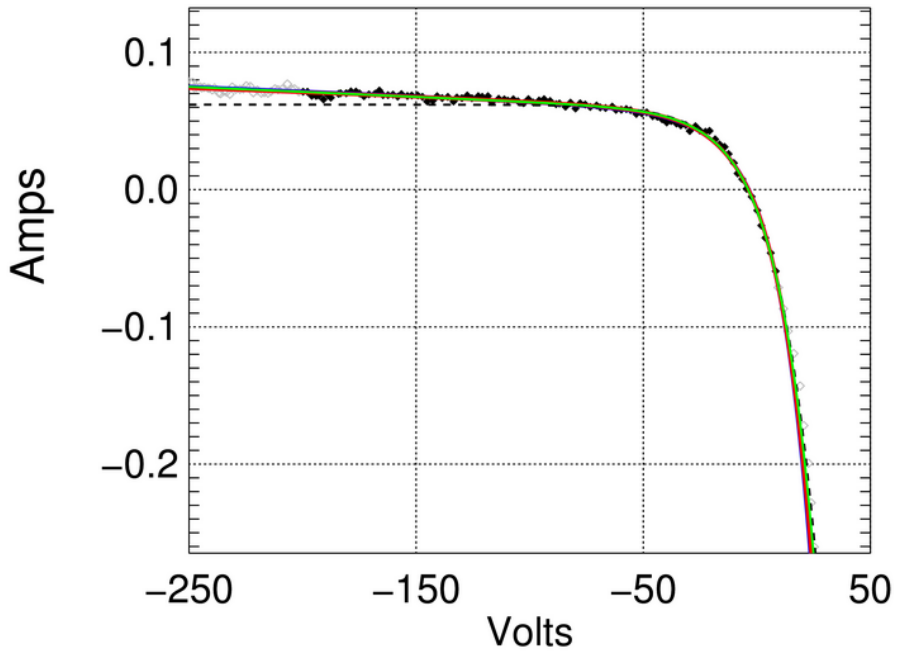


Fig 5-12 A very low-noise I-V characteristic example.

(shot 152838 sweep#304)

Standard I-V Fitter (dashed)	Simple Sheath Expansion Fitter (blue)	Four-Param Fitter (red)	Linear Sheath Expansion Fitter (green)
I_{sat} = 0.062 A	I_{sat} = 0.053 A	I_{sat} =0.054 A	I_{sat} = 0.057 A
V_f = -4.0 V	V_f = -3.6 V	V_f = -3.7 V	V_f = -3.8 V
T_e = 17.9 eV	T_e = 15.7 eV	T_e = 16.1 eV	T_e = 16.6 eV
	c = 1.8 (fixed)	c = 1.6	ΔI/ΔV = -7.35E-05
	χ^2_{red} = 0.779	χ^2_{red} = 0.766	χ^2_{red} = 0.770

Fig 5-12 shows an especially low-noise I-V characteristic. Sheath expansion effects are small but visible. We see that in ideal conditions, all four fitters agree within a reasonable margin. Furthermore, we see that using a fixed value of c=1.8 does an excellent job predicting the small magnitude of the sheath expansion effect for these conditions. Note that the Four-Param Fitter finds c = 1.6 which is very close to our fixed

value of 1.8. This is typical for low-noise I-V characteristics with significant sheath expansion to ± 30%

5.3.3 Motivating the ¾ power scaling

Since the sheath-expansion fitter invokes the Child-Langmuir law instead of an empirical linear fit, it is interesting to ask whether the ¾ths power is actually representative of the ion branch. This cannot be determined by any individual I-V characteristic, but it is possible that independent power-law fits to a large database could motivate either the $|V - V_f|^{3/4}$ or the $|V - V_f|$ scaling.

A database of I-V characteristics were collected from an ohmic shot where the Swing-Probe data shows low levels of plasma fluctuations. The magnitude of the sheath expansion effects were evaluated using:

$$S_{sh} \equiv n_e^{1/2} (kT_e)^{3/4} \qquad \textbf{eq. 5-8}$$

and I-V characteristics with S_{sh} values < 5.5e4 were selected, yielding a database of ~100 fits. These were then fit with an independent power-law of the form:

$$I = I_{sat}^{+} + a|V - V_f|^b \qquad \textbf{eq. 5-9}$$

where I_{sat}^{+} is taken from the 3-param fit for the same I-V characteristic. Data were fitted over for voltages less than -100V to ensure that the probe was in ion-saturation for all fitted conditions. An example of this fit is shown in Fig 5-13.

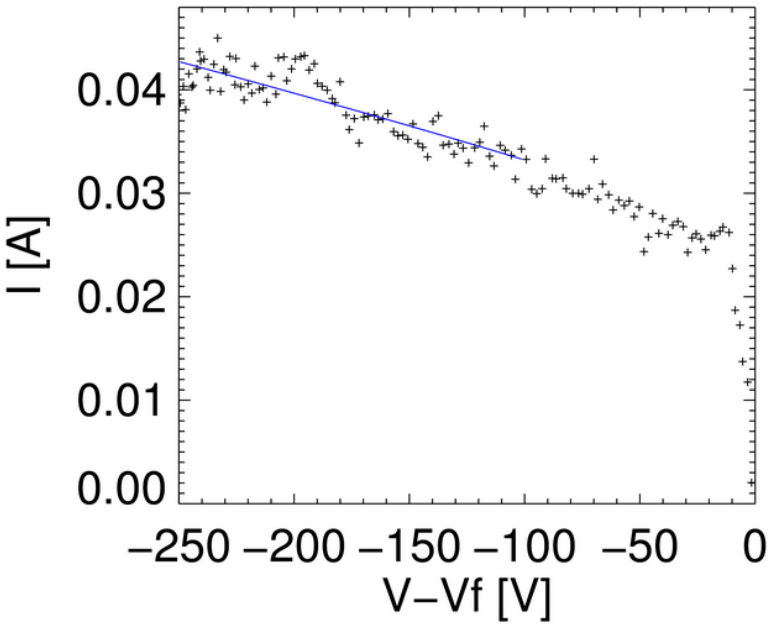


Fig 5-13 An example of an independent power-fit to the ion branch from -100V to -250V. For this example,the fit (blue line) has the values I_{sat}^{+} = 0.025 A, a = 1.4e-4, and b = 0.87, see eq. 5-9

The average value of 'b' taken over the database was 0.82 with a standard deviation of 0.38. The standard deviation is larger than the difference we wish to resolve. We cannot therefore differentiate whether the saturation current varies as a function of $\left|V - V_f\right|^{0.75}$ or $\left|V - V_f\right|^{1.0}$.

5.3.4 Conclusion

In conclusion, using a totally free 4th parameter to account for sheath expansion in real I-V characteristic data was found to result in over-fitting problems. The sheath expansion effect is on the same magnitude as the noise in the data, and so, it is not possible to separate out the two effects on a sweep-by-sweep basis. While the fitters with more freely fitted parameters will always produce a closer fit to the datapoints,

they do so in a way that is not physically meaningful. The sheath expansion parameters c or $\Delta I/\Delta V$ have no physical meaning, and allowing them to change affects the I_{sat} and T_e values significantly.

However, if the sheath expansion effects are relatively small, calculating a precise value for c is not needed for each I-V characteristic. We have shown that using a constant ballpark value for c=1.8 appears to account for sheath expansion across all conditions for the Swing-Probe electrodes. Compared to the Simple Sheath Expansion Fitter, the Standard IV fitter systematically overestimates I_{sat} and T_e. Properly managing the Standard IV fitter by truncating the data partway through the ion-saturation branch helps to mitigate these problems, but this needs to be checked manually to ensure good results, a laborious procedure. This type of management is not necessary with the Simple Sheath Expansion Fitter making it practical to automate data analysis once any given electrode has been analyzed to obtain the value of c.

5.4 Alternate Sheath expansion Models

It might be objected that the simple sheath expansion function is too simple and that overlooking certain details could cause problems. We consider a number of refinements to this model to see how large of an effect they have.

The simple sheath expansion function has zero sheath thickness when V = V_f, instead of a thickness of λ_D. The function also has the deficiency that the electrons have the same collection area as the ions, but it does not make sense that the electron collection area should grow with decreasing V since the sheath becomes increasingly repellent to electrons as V decreases. We therefore specify and test more refined sheath expansion fitting functions.

5.4.1 I-V Fitter #2 uses separate ion and electron collection areas

Here we specify different collection areas for the ions, A^i_{coll}, and electrons, A^e_{coll}:

$$I = J_{sat}\left(A^{i}_{coll} - A^{e}_{coll}EXP\left[\frac{e\left(V - V_f\right)}{kT_e}\right]\right) \qquad \textbf{eq. 5-10}$$

$$A^{e}_{coll} = A_0 \qquad \textbf{eq. 5-11}$$

$$A^{i}_{coll} = A_0 + l_p l_{sh}\left(V - V_f\right) \qquad \textbf{eq. 5-12}$$

$$l_{sh} = \left\{ \begin{array}{cc} c\lambda_D\left[1 + \left(e\left(V - V_f\right)/kT_e\right)^{3/4}\right] & \text{for V} < \text{V}_f \\ c\lambda_D & \text{for V} > \text{V}_f \end{array} \right\} \qquad \textbf{eq. 5-13}$$

Here, l_{sh} reaches its minimum thickness λ_D when V=V_f and stays constant for the region where V>V_f.

5.4.2 I-V Fitter #3 includes sheath contraction for V > V_f

As a further refinement we might expect the sheath to contract when V is increased above V_f and the ion collection area would decrease to A_0. Here we retain eq. 5-10 through eq. 5-12 but replace eq. 5-13 with eq. 14:

$$l_{sh} = \left\{ \begin{array}{cc} c\lambda_D\left[1 + \left(e\left(V - V_f\right)/kT_e\right)^{3/4}\right] & \text{for V} < \text{V}_f \\ 0 & \text{for V} > \text{V}_f \end{array} \right\} \qquad \textbf{eq. 14}$$

Of course this creates a discontinuity in A^{i}_{coll} at V=V_f.

5.4.3 Comparing the three sheath expansion fitters

Fig 5-14 shows the differences between the ion collection areas for the three models. We compare the three models starting with an example with a significant sheath expansion effect, see Fig 5-15.

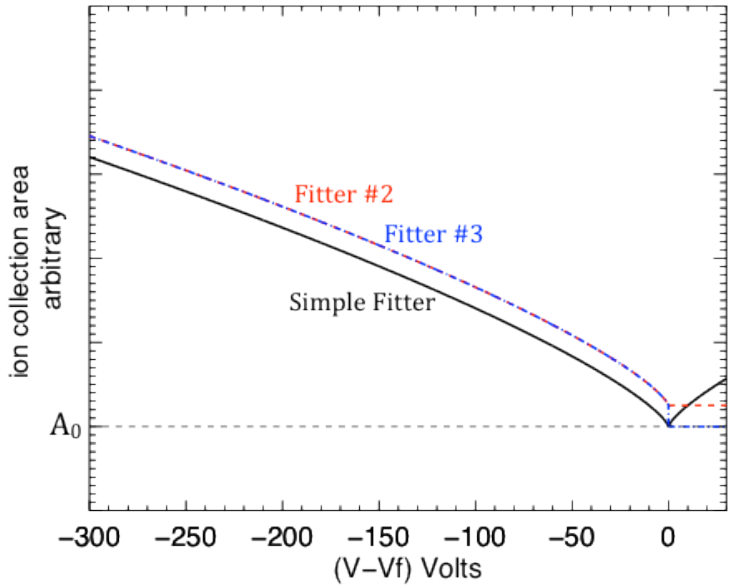


Fig 5-14 An exaggerated schematic showing the ion collection area increasing with the 3/4th power of ($V - V_f$). Simple Sheath Expansion Fitter (solid black), I-V Fitter #2 (dashed red), I-V Fitter #3 (dashed blue).

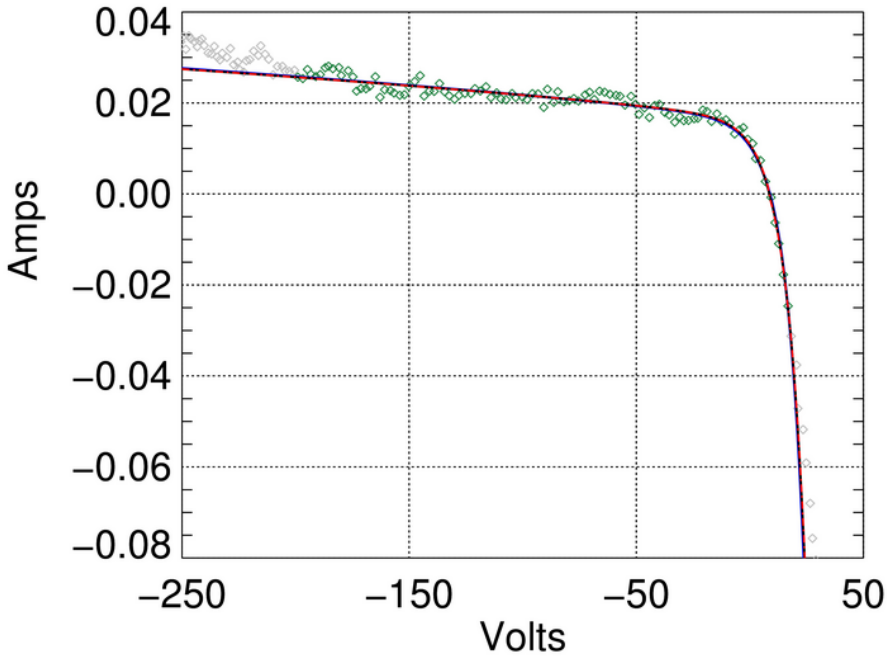


Fig 5-15 An I-V characteristic example with a significant sheath expansion effect. (shot 155646 sweep#86)

Simple Sheath expansion Fitter (blue)	Fitter #2 (black)	Fitter #3 (red)
I_{sat} = 0.015 A	I_{sat} = 0.015 A	I_{sat} = 0.015 A
V_f = 8.8 V	V_f = 8.1 V	V_f = 8.3 V
T_e = 8.5 eV	T_e = 8.5 eV	T_e = 8.5 eV
χ^2_{red} = 0.338	χ^2_{red} = 0.329	χ^2_{red} = 0.325

See Fig 5-15. In this example, all three curves lay overtop one another and it is difficult to tell them apart. We note that the plasma parameters are also in agreement except for a small variation in V_f.

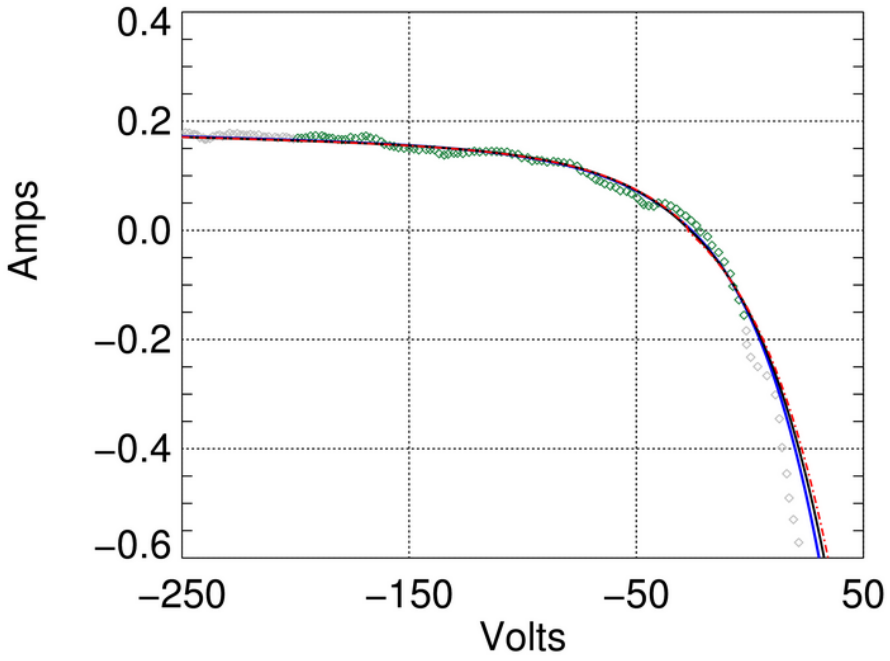


Fig 5-16 A high temperature noisy I-V characteristic example.

(shot 155646 sweep#206)

Simple Sheath expansion Fitter (blue)	Fitter #2 (black)	Fitter #3 (red)
I_{sat} = 0.139 A	I_{sat} = 0.131 A	I_{sat} = 0.131 A
V_f = -25.8 V	V_f = -28.4 V	V_f = -27.6 V
T_e = 35 eV	T_e = 35 eV	T_e = 36 eV
χ^2_{red} = 9.54	χ^2_{red} = 11.0	χ^2_{red} = 12.2

See Fig 5-16. Here, in this noisier example, we see a larger difference between the three fitters with a significant difference in χ^2_{red} and see that the functions diverge slightly in the electron branch. Despite this, the plasma parameters are in good agreement for all three fitters.

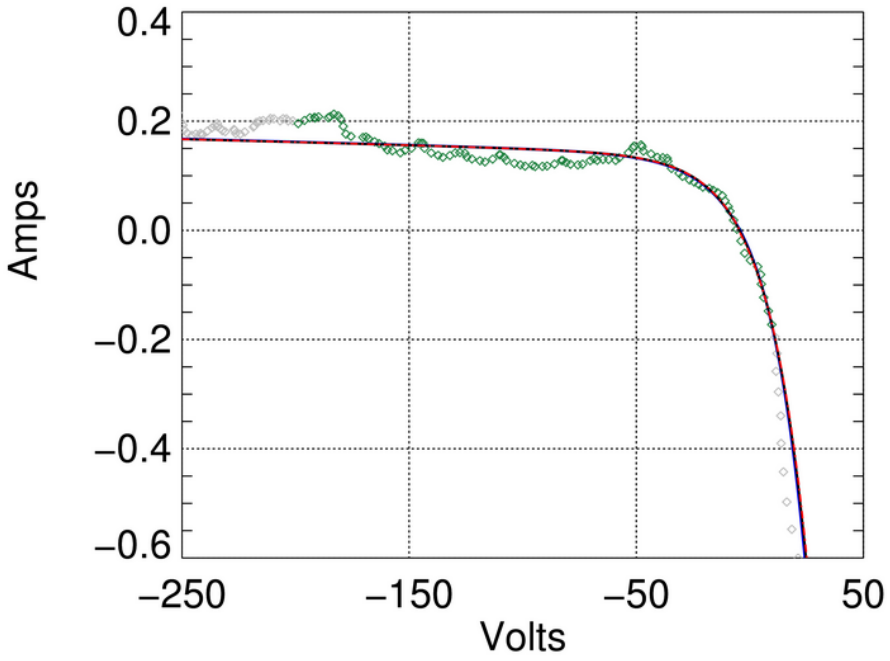


Fig 5-17 A noisy I-V characteristic example.

(shot 155646 sweep#337)

Simple Sheath expansion Fitter (blue)	Fitter #2 (black)	Fitter #3 (red)
I_{sat} = 0.133 A	I_{sat} = 0.128 A	I_{sat} = 0.128 A
V_f = -4.3 V	V_f = -5.2 V	V_f = -5.0 V
T_e = 17.1 eV	T_e = 17.1 eV	T_e = 17.3 eV
χ^2_{red} = 4.24	χ^2_{red} = 4.24	χ^2_{red} = 4.22

In this noisy example, Fig 5-17, we again note that the three fitters are in agreement.

It appears that the refinements to the Simple Sheath Expansion Fitter used in fitter #2 and fitter #3 have a minimal effect on the operation of the software and on the extracted plasma parameters. Furthermore, the magnitudes of the changes to A_0 are on the order of its measurement uncertainty. Therefore we conclude that for the general data analysis of the Swing-Probe I-V characteristics, such refinements add complexity without improving the fitter. Unless otherwise noted, all of the I-V characteristic data analyses in this thesis are done using the Simple Sheath Expansion Fitter.

5.4.4 Conclusion

In conclusion, we have developed a new Langmuir probe I-V characteristic fitter that is robust, reliable, and that needs almost no intervention to produce good quality fits. This fitter accounts for sheath expansion using a theoretically constrained Child-Langmuir function. Once the fitter is setup for any given electrode, it has proven to be able to calculate the magnitude of the sheath expansion effect for all conditions measured thus far. The model takes the projected surface area A_0, the electrode perimeter l_p, and an empirical sheath expansion coefficient 'c' as constant inputs, then freely fits I_{sat}, T_e, and V_f for each I-V characteristic. This fitter is a significant improvement compared to those previously used at DIII-D.

5.5 Swing-Probe Specific Data Analysis

The digitized signals are converted into the electrode voltages and currents, and calibrated to mitigate the drifts introduced by the fibre optic links. This includes knitting together the low-gain and high-gain datasets and correcting for systematic errors such as line capacitance and the current flowing across the voltage sensor.

To calculate J_{sat}, T_e and V_f, the two opposite-facing electrodes are treated as a "single probe," i.e. the currents from each half are summed together before being handed off to the sheath-expansion IV fitter. Plasma density n is calculated using:

$$\frac{n_e}{2} = n_{se} = \frac{J_{sat}^{+}}{c_s \cdot e} \qquad \textbf{eq. 5-15}$$

where the ion sound speed c_s is calculated from the local temperature using:

$$c_s = \sqrt{\frac{ZkT_e}{m_i}} \qquad \textbf{eq. 5-16}$$

where Z is the charge state and m_i is the ion mass.

To calculate Mach number $M_{||}$, we run the sheath-expansion IV fitter on the current and voltage signals from the two electrodes individually to extract a value of J_{sat} for each electrode and each IV characteristic. $M_{||}$ is calculated assuming [42]:

$$M_{\parallel} = 0.43 \ln\left(\frac{J_{sat+}^{left}}{J_{sat+}^{right}}\right) \qquad \textbf{eq. 5-17}$$

We define $M_{||}$ as being positive when the plasma is flowing in the $+\hat{\phi}$ direction, which is counter-clockwise when looking down at the torus.

Chapter 6 Validation of Swing-Probe Measurements

While the unambiguous I-V characteristics lend good confidence in the Swing-Probe measurements, validating the new data against other diagnostics is also an important check.

6.1 Validation Against the Divertor Target Langmuir Probes

When the lower Swing-Probe passes the tiles on its way towards the plasma, it crosses the same R,Z coordinates as one of the fixed probes mounted on the centerpost, see Fig 6-1. These two probe can therefore be compared directly for validation.

The lower Swing-Probe and this intersecting fixed probe agree for most conditions. In strongly detached conditions, where the neutral density increases in the inner divertor, the lower Swing-Probe and this fixed probe can disagree very strongly. However, in these regimes, the fixed probes also disagree with each other confirming that Langmuir probe measurements become unreliable in detached conditions.

The Swing-Probe also shares flux surfaces with many more of the fixed Langmuir probes, and for strongly attached conditions, they should produce similar measurements.

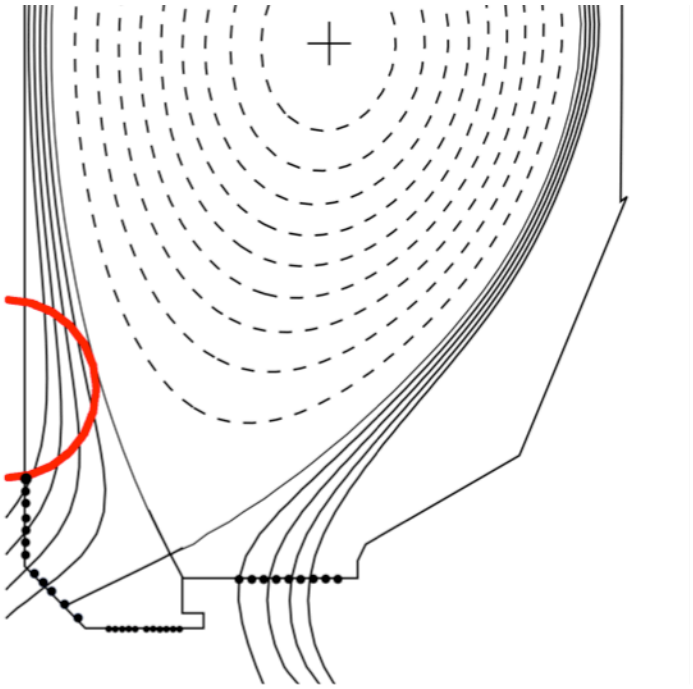


Fig 6-1: The path of the lower Swing-Probe (red) and the locations of the fixed Langmuir probes in the lower divertor (black). Note that the Swing-Probe crosses the poloidal location of the highest fixed probe.

Fig 6-2 shows lower Swing-Probe and fixed probe T_e and J_{sat} data in a low density ohmic shot. B_t = 2.1 T, I_p = 1.3 MA, $\bar{n}_e$ = 1.6e19 m^{-3}, P_total = 1 MW. The inner target is as strongly attached as possible without auxiliary heating. Unfortunately, the actual degree of detachment is not known since there are no DTS measurements in the inner divertor and Langmuir probes do not provide reliable measurements of T_e in detached conditions.

The x-axis reports the distance outside of the LCFS normalized on the outboard side at the height of the magnetic axis. The 1cm ticks in Fig 6-2 correspond to the 1cm flux surfaces shown in the SOL in the EFITs such as the one shown in Fig 6-1. We will follow the standard practice and refer to this transformation as being 'normalized to the Outer MidPlane' (OMP). Though technically, the magnetics group defines the midplane to be where Z = 0, while the flux surface are actually normalized at the outside SOL at the

height of the magnetic axis. The magnetic axis is marked by the large '+' symbol in the center of the EFIT. We will also report the coordinates by the normalized Psi coordinate Ψ_n.

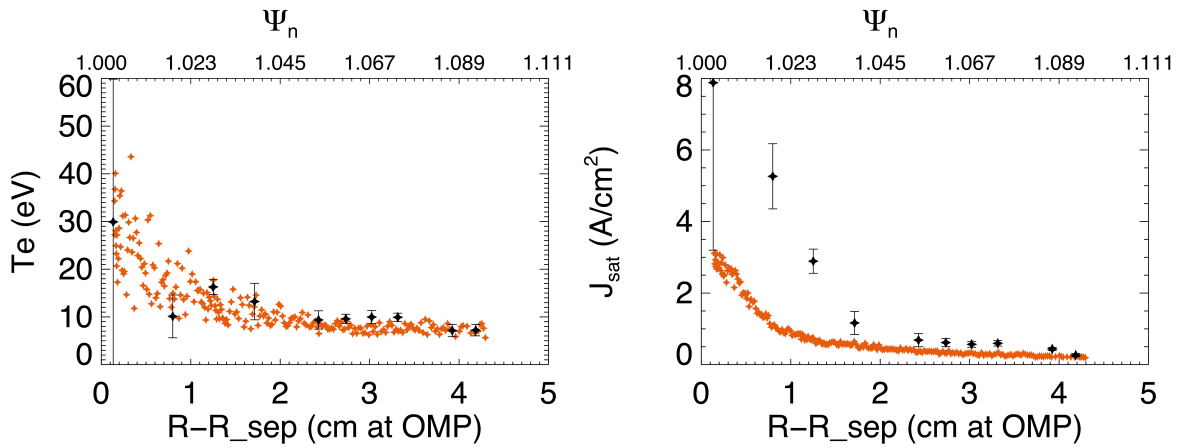


Fig 6-2: Agreement is shown between the Swing-Probe data (orange) and the fixed Langmuir probes in the target (black). The rightmost fixed probe datapoint at R-R_{sep} = 4.2 cm shares the same R,Z coordinates as the Swing-Probe when it passes. The error-bars for the fixed-probe data represent one standard deviation.

In Fig 6-2, excellent agreement is seen between the intersecting fixed probe (at R-$R_{se=}$ = 4.2 cm) as well as the fixed probe on the centerpost (all data for R-R_{sep} > 2 cm). We hypothesize that the plasma is attached in this region away from the separatrix. Closer to the separatrix, the measurements begin to strongly disagree, and at the strikepoint, the fixed probe data become very noisy; note that some of the error bars are off-scale. We may conjecture that the plasma is detached in this near-separatrix region, in which case the apparent agreement between the target-probe and Swing-Probe T_e values would appear to be coincidental. The difference in the J_{sat} values, if real, is not understood: if the plasma is detached, one would expect the target-probes to measure lower values of J_{sat} than the Swing-Probes, not higher. We therefore adopt the working hypothesis that these specific target-probe data are not reliable and that the excellent agreement between the Swing-Probe and the fixed-probe in the far-SOL, which we assume is attached, constitutes a confirmatory test of the Swing-Probe measurements.

6.2 Validation Against the Divertor Thomson Scattering

The Swing-Probe data have also been compared with the Divertor Thomson system in conditions where they are expected to agree. We consider a low-density ohmic discharge that's limited on the centerpost and where the divertor Thomson measurement chords have been raised to intersect with the plasma. See Fig 6-3. For such a plasma it is theoretically expected that there will be little variation of the plasma along any given flux surface.

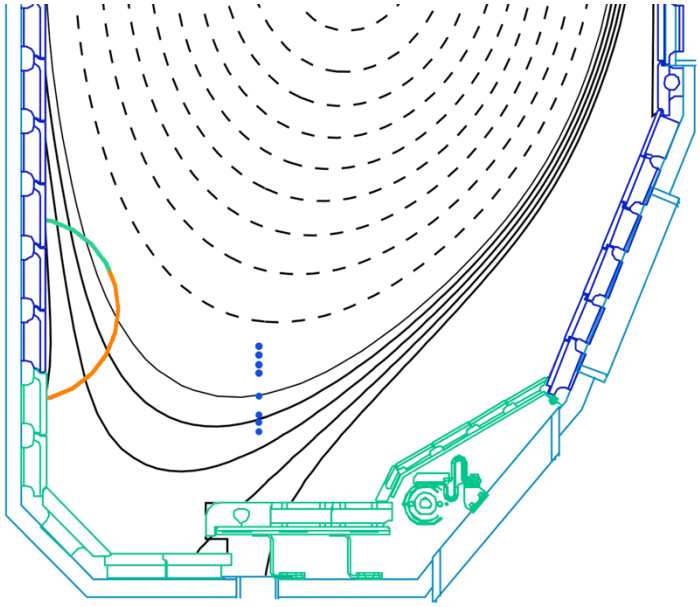


Fig 6-3: The plasma EFIT overlaid with the lower Swing-Probe path (green and orange) and the divertor Thomson measurement locations (blue). Note that the Swing-Probe and the TS share measurements on many flux surfaces both inside and outside the LCFS.

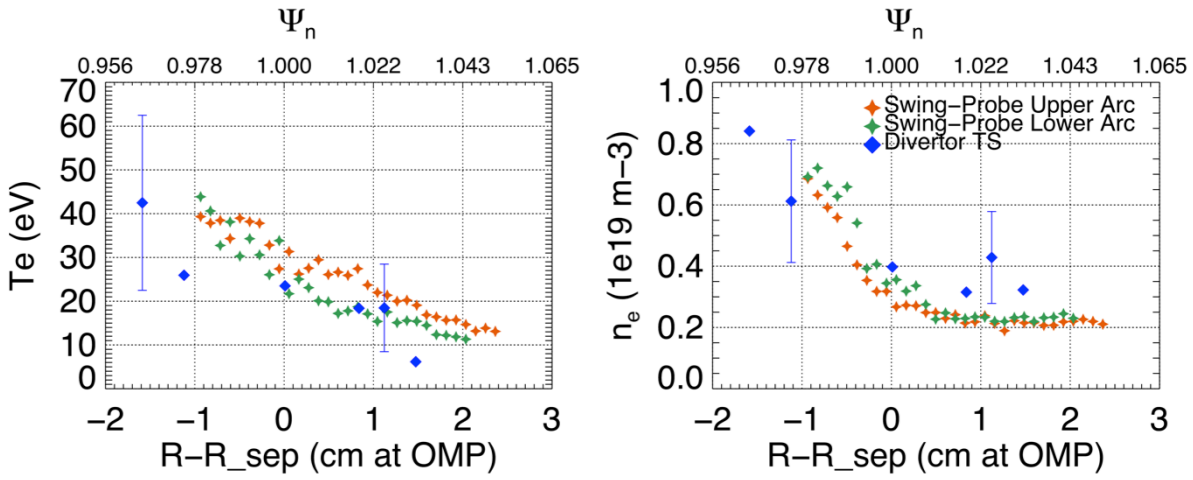


Fig 6-4: The averaged values for the Divertor TS (one value per channel) and for the Swing-Probe (1 mm bin averaged). We see good agreement between the electron temperature and the density measurements made by the two diagnostics.

There is reasonable agreement in the T_e measurements between the lower and upper arcs of the Swing-Probe and those measured by the divertor Thomson scattering. For n_e, the agreement is not quite as strong. In particular, the divertor Thomson appears to have one channel at $R\text{-}R_{sep}$ = 1.2cm that reads higher than its neighbours and the Swing-Probe. Even so, this appears to be an acceptable level of agreement between two very different diagnostics.

Note that the wide scatter in the Divertor TS data is not due to electronic noise or to any deficiency in the diagnostic. The TS makes measurements on the nanosecond timescale and is therefore able to detect high-speed plasma fluctuations. Langmuir probe measurements are made on the 100 μs timescale (the Swing-Probe takes one measurement every 500 μs) so these measurements average over the high-speed plasma fluctuations. There may also be longer time-scale variations, e.g. over a period of a second, due to lack of full control of the principal discharge parameters such as the line-averaged density; however, data collected over such variations are not usually combined but are treated as representing different plasma conditions.

Chapter 7 2013 Campaign Experiments

Swing-probe operations in the 2013 campaign were very successful. Both probes performed as designed and collected data over a wide range of plasma conditions. Tasks and experiments for the campaign are listed in Table 7-1.

Unlike the 2012 campaign, the Swing-Probe control systems were up and ready for the month of start-up tasks which includes several days worth of tests with toroidal field but no plasma allowing us to test the Swing-Probes with the interlocks disabled for troubleshooting. Both probes armed and fired successfully on the first attempt before the physical model was re-tuned to accommodate for the new friction coefficients, which stayed low and consistent ≤ 0.05 N-m. The resistance of the coils fluctuated with the vessel bake, and one of the limit switches shorted to ground due to shifting of the ceramic beads in the in-vessel cable. Fortunately, the electronics systems were able to accommodate for this short so long as a second did not develop in the same circuit.

The swing-probes were carefully tested during plasma experiments in the start-up phase, and the plasma was moved toward the probe very gradually. We found that the very long cables both inside the vessel and from the vessel to the electronics rack contributes a high capacitance of 8.3μF for each electrode in the lower probe and ~9μF for each electrode in the upper probe. The capacitance is corrected for during data analysis, but we find that it is preferable to drive the electrodes with a sinusoidal voltage waveform instead of a triangle or sawtooth waveform since the high frequency components of these waveforms are strongly affected by the impedance in the cables.

Date	Tasks & Experiments
Mar 6 - 8	Lower probe actuator tests. Arms and fires for all toroidal field strengths. Data acquisition bugs found and corrected. Friction is negligible.
Mar 13 – Apr 4	Upper probe actuator tests. Arms and fires successfully for all toroidal field strengths. Limit switches do not work consistently. Friction ≤ 0.05N-m.
Apr 5	First plasma measurements taken
Apr 9	Piggyback into Inner Wall Limited plasmas. Lower probe reaches the 1.5cm flux surface.
Apr 15	Lower probe swings 250ms after an USN MARFE. First data verification against Divertor Thomson Scattering
Apr 26	Dedicated startup ½ day. Simple as possible LSN plasma. 7 shots, probe reached the 0.8cm flux surface. Tangential TV, floor probes and Divertor Thomson scattering running.
Apr 29	First probe actuation following boronization. Probe operation not affected and no change in friction was measured.
May 1	Divertor Detachment Experiment ½ day Density ramp in LSN ohmic plasmas. All diagnostics running.
June 25	Divertor Detachment Experiment ½ day Low Power H-mode
July 16-18	Divertor Detachment Experiment full day Medium Power H-mode High Power H-mode (10MW with a line average density ≅7e19m^{-3}, probe took useful measurements up to the LCFS)
Aug 29	Dimes Experiment ½ day. Piggyback data comparison against Coherent Imaging Flow Camera
Oct 17	Inner Wall Limited Heat Flux Experiment ½ day. Plunge depth record: -1.5cm flux surface (inside the LCFS)

Table 7-1: A list of Swing-Probe participation in the 2013 campaign. Useful data was collected in the tasks highlighted in green.

The true test for the Swing-Probes was the first boronization, which we were afraid would somehow affect the friction in the drive trains once again. The lower probe was

exercised on the very first shot following the boronization – a low power test shot that starts every day – and no change in the friction was measurable in the probe's performance.

The probes continued to perform reliably throughout the 2013 campaign, taking data in the boundary group's divertor detachment experiment. This experiment was split over 4 half days studying LSN plasmas in L-mode, low power H-mode, medium power H-mode, and high power H-mode respectively. A piggyback experiment provided useful comparison data between the upper Swing-Probe and the Coherent Imaging Flow Camera observing that region. Finally, an inner wall limited heat flux experiment was performed at the very end of the campaign.

In-vessel maintenance showed no damage to any of the probe systems, and no visible erosion to the electrodes. An electrical short to ground in the upper limit-switch for the upper probe was isolated and repaired, and a loose terminal in the upper actuator coil in the lower probe was tightened, but otherwise, no other maintenance was necessary. There was no observable carbon deposition on any of the ball bearings, so we expect the system should continue to operate with little to no in-vessel maintenance for several years.

Chapter 8 L-mode Divertor Detachment Experiment

While an attached divertor is characterized by plasma freely streaming towards the target, a detached divertor has sufficient interaction with recycled neutrals that the plasma flow towards the target becomes more diffusive. Effects such as volumetric momentum loss, power loss, and recombination act to decrease the peak heat flux and spread out the heat over a larger area. As fusion devices grow larger and more powerful, detachment will become necessary to protect the target surfaces.

On May 1st 2013, an L-mode divertor detachment experiment was performed with the goal of building on past work with the improved datasets provided by the lower Swing-Probe, the Coherent Imaging Flow Camera, and the upgraded Divertor Thomson Scattering system. Additionally, data were collected using the Tangential TV cameras, the filterscopes, the divertor infrared camera, the X-point reciprocating probe, and the fixed Langmuir probes. We used a lower single null plasma with ohmic heating and diagnostic beam blips from the neutral beams. The strikepoint positions were adjusted to control the Swing-Probe penetration depth in the first part of the shot (see the EFIT shape in Fig 8-1) and then they were swept to enhance diagnostic coverage in the latter part.

The goal was to begin with a shot where both the inner and outer targets were attached, then increase the density gradually in each subsequent shot until the outer target was observed to detach. Unfortunately, at such a low power (~1MW), we were unable to lower the density sufficiently to attach the inner target because of the locked mode limit. Swing-Probe data were taken over a 12 shot density scan.

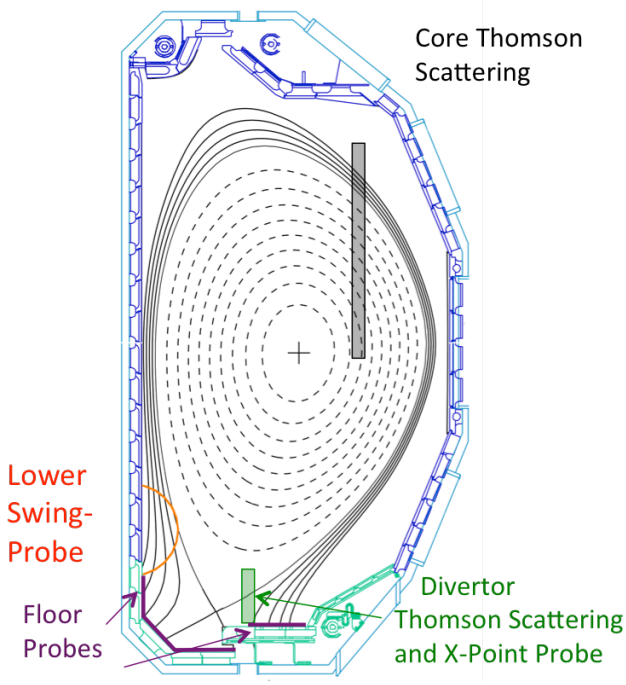


Fig 8-1: The EFIT for the divertor detachment experiments where the lower Swing-Probe reached the LCFS. The locations of some of the spatially localized diagnostics used in this experiment are shown.

8.1 The Simplest Shot

The first shot of the sequence with a line average density of 1.6×10^{19} m^{-3} represents our simplest case, being the most strongly attached. Comparing the Swing-Probe data against the core Thomson scattering upstream, we note that the Swing-Probe T_e changes very gradually along the SOL. However, we see an increase in the density measured by the Swing-Probe relative to that by the TS close to the LCFS. See Fig 8-2.

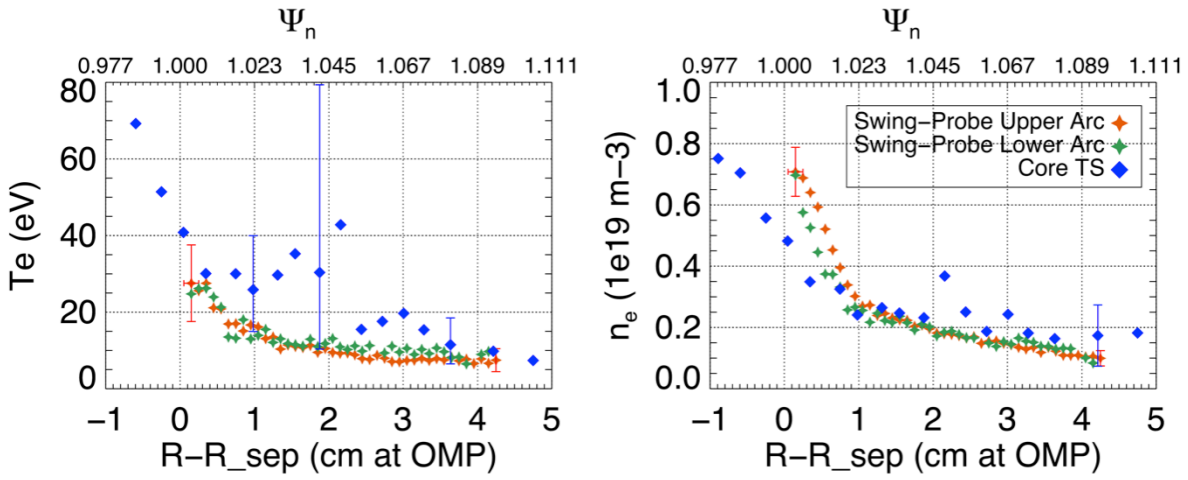


Fig 8-2: A comparison between the core Thomson scattering (1 average value per channel) and the lower Swing-Probe (1mm bin averages) for the lowest density shot of the L-mode detachment study. The uncertainty-bars show the scatter in the data. The Thomson T_e data shows significant scatter between 1 and 3 cm due to fluctuations in the plasma.

The TS data are averaged using an error-weighted mean. In this case, very hot data points $\gg$ 100 eV between 1 cm < R-R_{sep} < 3 cm add considerable scatter to the averaged data, and a normal unweighted mean produces a value that can differ considerably from the average one would estimate by eye. The colder and more tightly clustered datapoints tend to have smaller reported errors, so the weighted average produces more reasonable looking results. The uncertainty-bars in Fig 8-2 represent the scatter in the data and are set by eye.

The Swing-Probe measured parallel flow velocity and floating potential for this shot are shown in Fig 8-3.

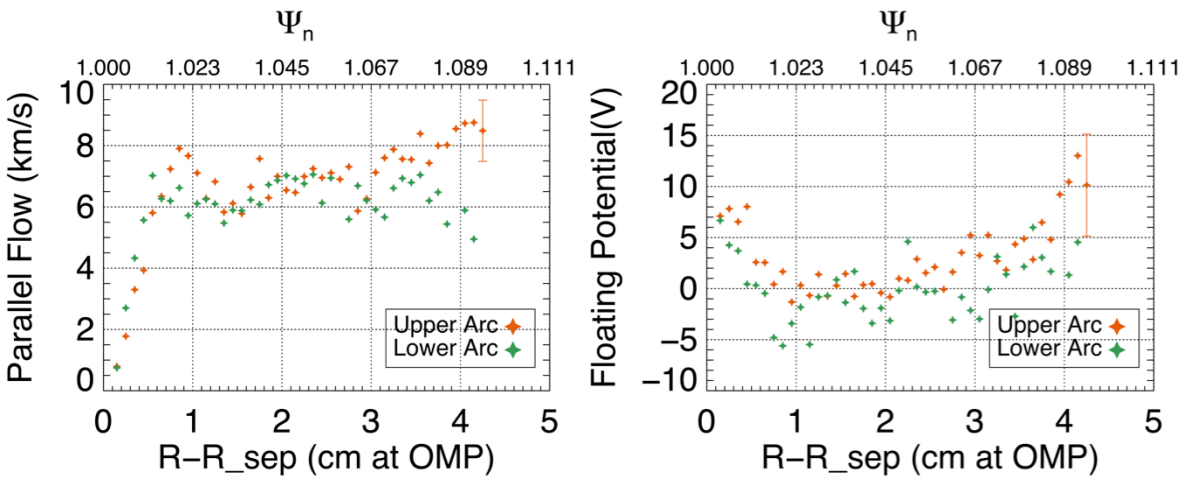


Fig 8-3: 1mm bin averages of the parallel flow velocity (left) and the floating potential (right) measured by the swing probe in the lowest density shot of the L-mode detachment study. Positive flow is in the $+\hat{\phi}$ direction and towards the inner target.

8.2 EFIT Uncertainties

Unfortunately, EFIT uncertainties are quite large at the core TS measurement locations because they are far from the magnetic sensors at the vessel wall and far from the X-point. It is normal and often necessary to shift the TS data. The Porter method uses a tanh fit to the core Thomson T_e profiles to estimate the location of the LCFS [43]. This method shifts the LCFS inward by 7.5mm OMP for this shot [44]. While this method works well for the steep gradients of H-mode plasmas, it is less useful for L-mode plasmas such as this one. Preliminary OEDGE modeling has been performed on this shot [45], and shows that the Core TS, Divertor TS and Swing-Probe measurements are mutually consistent if the Core TS data is shifted outward by $\Psi_n = 0.005$, which is equivalent to shifting the LCFS inward by 2.1 mm OMP. We must assume for these conditions that the upstream to downstream alignment uncertainty is on the order ~7.5 mm.

We require an accurate method of finding the location of the LCFS that can be quickly applied all of the shots in this density scan. One such method is to balance the input power P_{SOL}^{input} with the electron parallel conducted heat flux out of the SOL $P_{SOL}^{exhaust}$. P_{SOL}^{input} is obtained by subtracting the power radiated inside the separatrix from the total heating power. The kinetically-corrected-Spitzer expression for the electron conducted power $q_{\parallel e}^{kcSp}(R)$ is calculated from T_e and n_e core Thomson profiles. $P_{SOL}^{exhaust}$ can then be obtained by integrating $q_{\parallel e}^{kcSp}(R)$ across the SOL. The limit of that integral and thus the location of the corrected LCFS is defined such that $P_{SOL}^{exhaust} = P_{SOL}^{input}$. This method is described in detail in [46].

The effects of shifting the location of the LCFS on T_e and n_e values at the LCFS are shown in Table 8-1.

LCFS locator	Δ(R-R$_{sep}$) mm OMP	$\Delta\Psi_n$	T_{e_sep} [eV]	n_{e_sep} [1e19 m^{-3}]
EFIT	-	-	41 +/- 22	0.48 +/- 0.17
Porter	7.5	0.0175	77 +/- 17	0.75 +/- 0.11
OEDGE	2.1	0.005	51 +/- 17	0.55 +/- 0.15
Power Balance	2.4	0.006	52 +/- 17	0.55 +/- 0.15

Table 8-1: Shifts to the location of the LCFS for shot 152835 and the resulting T_e and n_e values for the LCFS.

The calculated shifts in the LCFS from Power Balance are listed in Table 8-2 for all the shots in the set.

shot #	Line Average Density [e19/m^3]	P_heat [MW]	P_rad total [MW]	P_rad core [MW]	$\Delta(R-R_{sep})$ mm OMP	T_{e_sep} [eV]	n_{e_sep} [1e19 m^{-3}]
152835	1.57	1.01	0.31	0.10	2.4	52	0.55
152836	1.60	0.91	0.42	0.11	1.5	51	0.57
152837	2.18	1.01	0.42	0.10	3.7	48	0.77
152838	2.02	0.94	0.36	0.11	1.6	51	0.73
152839	2.02	0.97	0.38	0.12	3.2	49	0.70
152840	2.39	1.00	0.46	0.11	4.7	48	0.93
152841	2.39	1.01	0.45	0.12	4.1	48	0.95
152842	2.81	0.97	0.38	0.12	4.2	48	1.12
152843	3.24	1.15	0.63	0.14	5.5	52	1.52
152844	3.60	1.26	0.75	0.15	6.6	47	1.68
152845	4.46	1.29	0.90	0.19	8.4	44	2.12
152846	4.80	1.40	1.15	0.25	9.9	44	2.49

Table 8-2: The power values used to calculate input power to the SOL, the shifts to the location of the LCFS for the shots in the density scan calculated using power balance and the resulting T_e and n_e values for the LCFS.

We note that the shift in the location of the LCFS increases systematically with increasing line average density. The reason for this is not known, but the EFIT reconstructions depend on assumptions made about the edge currents, and edge currents are suppressed with increasing density. The power radiated from the core is calculated from the automatic tomographic inversion performed on the bolometer array data. We note that the power radiated from the confined plasma is roughly 25% of the total radiated power, which is comparable to the fraction found using the validated bolometer inversions [47].

8.3 Trends Across the Density Scan

We compare the measurements between the core Thomson scattering and the Swing-Probe averaged over two regions: just outside the LCFS within 0.15 cm ≤ $R-R_{sep}$ ≤ 0.4 cm (see Fig 8-4) and in the far SOL within 3.0 cm ≤ $R-R_{sep}$ ≤ 3.5 cm (see Fig 8-5). The near LCFS region is selected based on the closest approach of the Swing-Probe on the first

shot of this experiment, while the far SOL region represents a part of the SOL where the core TS measurements have relatively low noise across all of the shots. The core TS data have been shifted using power balance by the distances listed in Table 8-2.

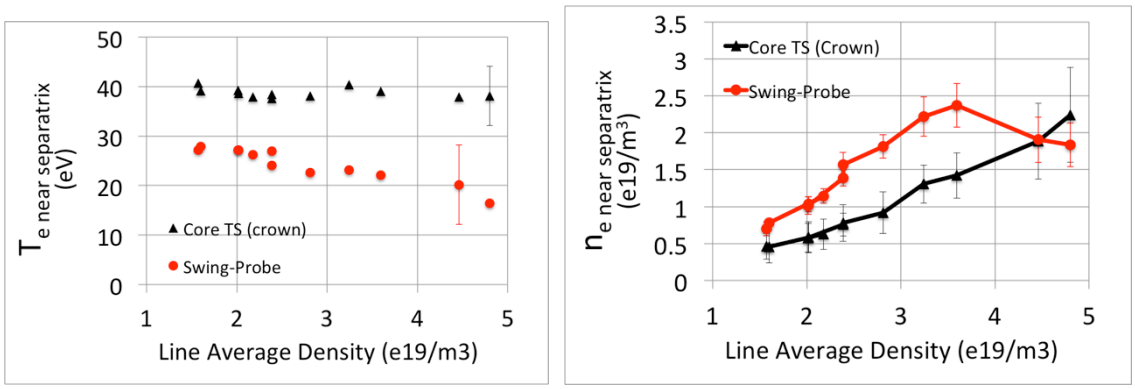


Fig 8-4: Comparison of T_e and n_e between the core Thomson scattering (black) and the lower Swing-probe (red) close to the LCFS. The error bars represent the scatter in the data. Averages were taken over the range 0.15 cm ≤ R-Rsep ≤ 0.4 cm. TS data have been shifted using power balance . Each datapoint represents one shot.

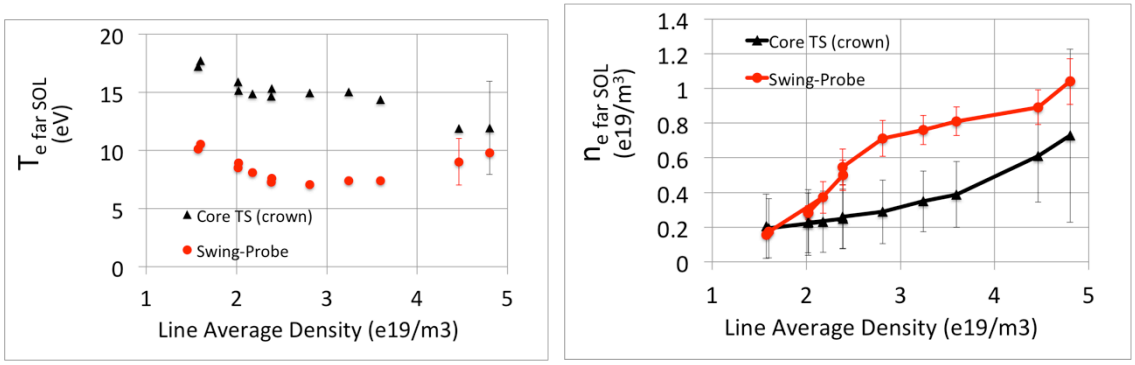


Fig 8-5: Comparison of T_e and n_e between the core Thomson scattering (black) and the lower Swing-probe (red) in the far SOL. The error bars represent the scatter in the data. Averages were taken over the range 3.0 cm ≤ R-Rsep ≤ 3.5 cm. TS data have been shifted using power balance.

We see that there is a drop in T_e along the field lines of up to 50%. The Swing-Probe almost always measures a significantly higher density than the core TS with the exception of the two highest density conditions near the LCFS. We find that for most

conditions, the electron pressure (p_e) is conserved along the field line between the two measurements. See Fig 8-6.

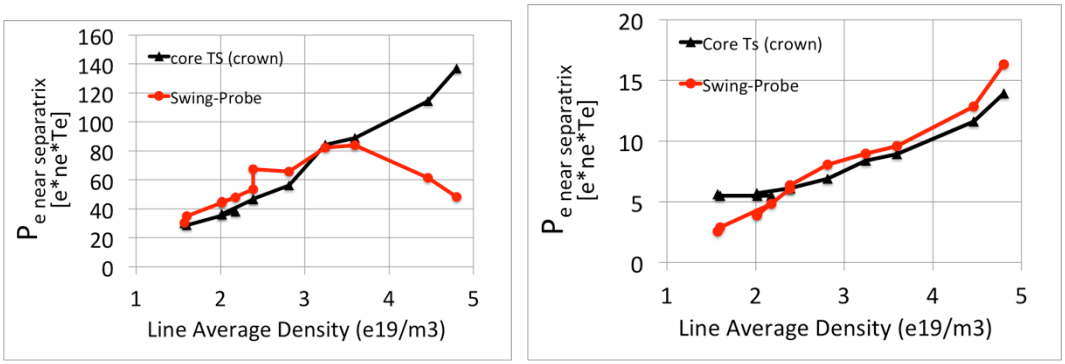


Fig 8-6: Comparison of p_e between the core Thomson scattering (black) and the lower Swing-probe (red) near the LCFS (left) and in the far SOL (right). TS data have been shifted using power balance..

Close to the LCFS, the Swing-Probe and core TS p_e match and increase linearly with the line average density until the line average density $\bar{n}_e$ exceeds 4e19 m^{-3} at which point the Swing-Probe density drops ~20%. We do not see a corresponding density drop in the core TS, which continues to increase linearly. We see a similar linear response and match in the p_e values measured in the far SOL, but the values here do not diverge at higher $\bar{n}_e$.

For the shots where $\bar{n}_e$ > 4e19 m^{-3}, the divertor Thomson scattering (DTS) shows the target temperature at the *outer* strikepoint is 1-2 eV, indicating that the outer strikepoint is beginning to detach. There are no DTS data for the inner target.

This behaviour is typical of the divertor detachment behaviour observed at the target. We find that for $\bar{n}_e$ < 4e19 m^{-3}, the inner SOL is in the conduction-limited regime where substantial parallel T_e gradients exist, but the p_e is conserved along the field lines. When $\bar{n}_e$ increases above 4e19 m^{-3}, we hypothesize that detachment occurs at the inner target, and the density and pressure near the LCFS in the inner divertor begin to roll over and decrease. It is perhaps surprising to see this trend in the Swing-Probe data which is taken a significant distance above the target.

Chapter 9 Inner Wall Limiter Heat Load Experiment

Of the experiments listed in Table 7-1, we have decided to make the Inner Wall Limiter (IWL) Heat Load experiment as the main physics focus for this thesis. The results from this experiment are highly relevant to ITER[6] and have the smallest dependence on results from other diagnostics or analysis from computational codes. There is concern at ITER that the present design of the inner wall limiter is based on a partly incorrect physics basis, so the 2013 DIII-D IWL experiment was undertaken at ITER's request to attempt to address this problem. The Swing-Probe played a central role in this experiment and turned out, in fact, to be the single most important diagnostic for the experiment.

9.1 Background

In ITER, heat fluxes are sufficiently high and time scales are sufficiently long that even the ramp-up and ramp-down limiter plasmas have the potential to cause damage to the plasma facing surfaces. Although ITER is considered to be a divertor tokamak, it will operate in a limiter configuration during plasma ramp up and ramp down. The high-field-side limited ramp-up plasma is projected to have a peak parallel heat flux of 25MW/m^2 with P_{sol} as high as 5MW [52]. The large heat fluxes, combined with the fact that ITER cannot transition from the limiter configuration to a diverted one until 9.5s into the shot [53] leaves the plasma-facing tiles on the centerpost with a very marginal safety factor for melting or delaminating if any of the design assumptions are even somewhat incorrect.

---

[6] ITER will be the world's largest tokamak. It is currently under construction in France as an international collaboration. The significant increase in plasma volume and power compared to current tokamaks will push ITER's first wall modules to their limits.

At the time of the DIII-D IWL experiment, measurements in JET [54] and COMPASS [55,56] indicated that there could in fact be a significant mistake in the assumed shape of the heat flux profile. It was assumed that the parallel heat flux increases exponentially with approach to the LCFS with a single exponential decay length λ_q that was extrapolated from a multi-machine scaling database; see [57] for further details. The plasma facing modules on ITER's centerpost have accordingly been designed with a logarithmic shape that spreads the heat out evenly over the whole surface. However, infrared camera measurements of deposited power flux density on the inner walls of the above-mentioned tokamaks indicate that the prescribed exponential may not sufficiently describe the parallel heat flux for inner wall limited plasmas. See Fig 9-1.

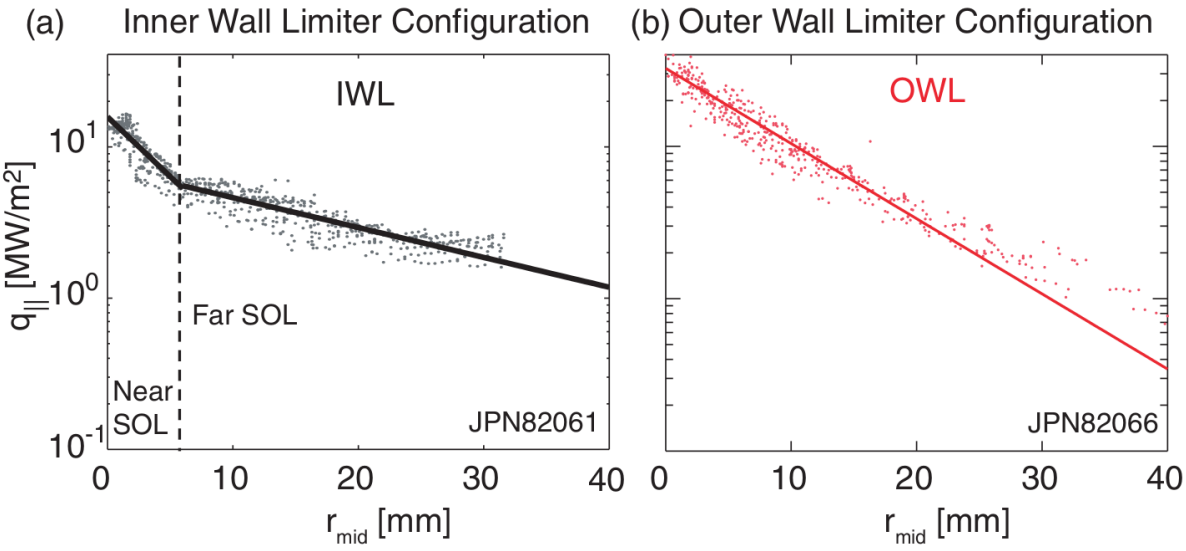


Fig 9-1: Measurements of the parallel heat flux of (a) an inner wall limited and (b) an outer wall limited plasma in JET mapped to the outer-midplane, inferred from infrared thermography Taken from [54]**.**

For the IWL configuration, the parallel heat flux ($q_{||}$) appears to increase significantly faster in the near SOL than in the far SOL. These heat flux data can be reasonably well fit using a double exponential function of the form:

$$q_{\parallel} = q_{\parallel 0}^{long} \exp\left[-\frac{R - R_{sep}}{\lambda_{long}}\right] + q_{\parallel 0}^{short} \exp\left[-\frac{R - R_{sep}}{\lambda_{short}}\right] \qquad \textbf{eq. 9-1}$$

where $q_{\parallel 0}^{short}$ has been reported to be roughly the same magnitude as $q_{\parallel 0}^{long}$ for cases where the inner walls that are approximately toroidal.

At the time of the DIII-D experiment this local enhanced heating had only been observed in JET and COMPASS, and then only using infrared cameras to measure q_{dep} (deposited power flux density). Outboard side reciprocating probe measurements had failed to detect this feature in both machines. If ITER engineers are going to use this information to improve the design of the inner limiters, greater confidence is needed in the measurements. It is necessary to replicate these results in more machines and with different diagnostics.

It has been hypothesized that the enhanced heat flux feature would be easier to observe on the inboard side because of the wider spread of the flux-surfaces and the lower turbulence levels in the inboard SOL plasma compared with the outboard side, so an inboard-side probe can achieve much finer spatial resolution than a similar probe on the outboard-side. The Swing-Probes therefore turned out to be the ideal diagnostic for investigating the parallel heat flux profile of IWL plasmas in DIII-D. We present here the first Langmuir probe measurements showing the enhanced heat flux feature for an IWL plasma.

9.2 The DIII-D IWL Experiment

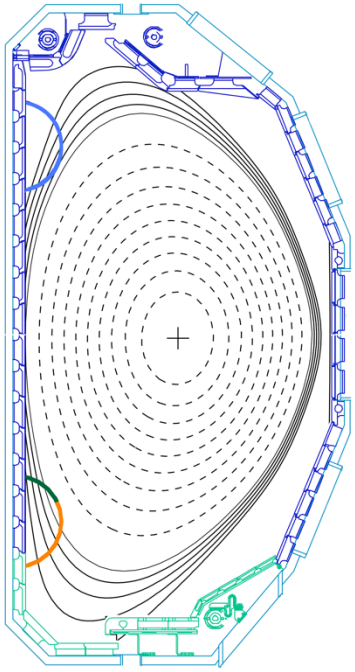


Fig 9-2: The elongated (κ ≅ 1.6) plasma shape.

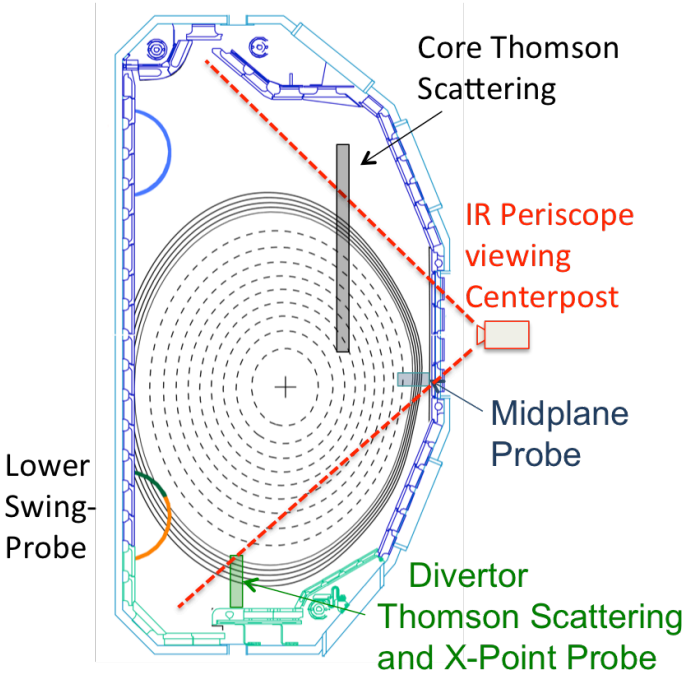


Fig 9-3: The round (κ ≅ 1.25) plasma shape (vertical location varied from shot to shot). The diagnostics used in this experiment are shown.

Data were collected over 8 shots with varying plasma current I_p (see Table 9-1). An elongated plasma (κ ≅ 1.6, see Fig 9-2) was used for the first part of each shot and a rounder plasma (κ ≅ 1.25, see Fig 9-3) was used for the second part of each shot. Measurements were made using the bottom Swing-Probe, the outboard side midplane probe, the x-point probe, the core and divertor Thomson scattering systems, and an IR camera viewing the inner wall.

Shot	Time [ms]	Ip [A]	κ	top gap [cm]	bottom gap [cm]	P_{tot} [MW]
155643	2000	1.19	1.59	3	3	1.05
155643	3500	1.19	1.26	v large	6	1.05
155645	**2000**	**1.19**	**1.58**	**3**	**3**	**1.06**
155645	3500	1.19	1.25	v large	v large	1.01
155646	**2000**	**1.2**	**1.59**	**3**	**3**	**1.05**
155646	3500	1.2	1.26	v large	v large	0.99
155647	2000	1.19	1.57	2	6	1.05
155647	3500	1.19	1.26	v large	6	1.06
155648	2000	1.19	1.57	6	2	1.05
155648	**3500**	**1.19**	**1.26**	**v large**	**6**	**1.06**
155649	2000	0.61	1.57	4	2	0.37
155649	**3500**	**0.61**	**1.24**	**v large**	**6**	**0.37**
155650	**2000**	**0.61**	**1.6**	**3**	**2.5**	**0.39**
155650	3500	0.61	1.23	v large	v large	0.36
155651	**2000**	**0.61**	**1.66**	**1.5**	**2.2**	**0.37**
155651	3500	0.61	1.23	v large	v large	0.36

Table 9-1: The shots from the 2013 DIII-D IWL experiment. Shots and times where the Swing-Probe operated are shown in bold.

Fig 9-4 shows the parallel heat flux $q_{||}$ as a function of R-R_{sep} for the SOL data where $q_{||}$ is calculated from the T_e and J_{sat} values of each I-V characteristic using:

$$q_{||} = \gamma k T_e J_{sat} / e \qquad \textbf{eq. 9-2}$$

where here we assume that the sheath heat transmission coefficient $\gamma = 7$.

Here, we consider data with R-R_{sep} > 0 cm from shot 155646, which is a typical example with neither the strongest nor the weakest signs of the enhanced heat flux feature. For additional clarity, we will only consider the data from the bottom-half arc (marked in orange in Fig 9-2). The top-half arc agrees with the bottom-half within 50%, but combining the data from both arcs adds considerable scatter to the plots. J_{sat} and T_e drop quickly for values of R-R_{sep} > 3 cm where the flux surfaces are interrupted by the shelf in the lower divertor. These data have been omitted for clarity.

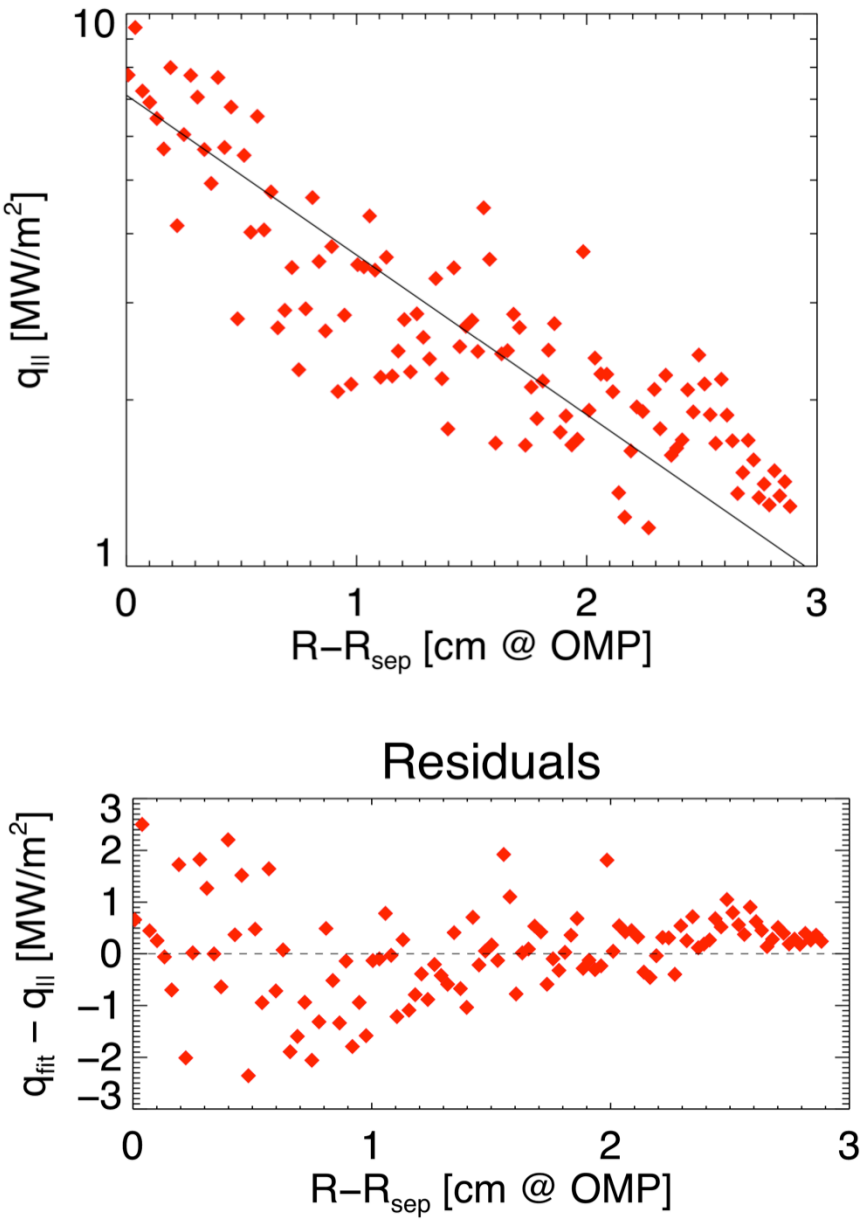


Fig 9-4 Data from the lower-half arc of 155646

Top: The parallel heat flux as a function of the distance from the LCFS normalized at the outboard midplane. The trend line is from a single-exponential best-fit: $q_{||}$ =(5.9 MW/m^2) EXP[(R-R$_{sep}$)/(1.91 cm)]

Bottom: The residual of the exponential trend line from the top.

We note that λ_q = 1.91 cm for this trend line, which is similar to the values obtained in earlier measurements made using the outboard side midplane probe [57], adding

confidence to these data. Furthermore, we note that within the scatter of the data, there is no strongly convincing indication that a single exponential function does not adequately describe the data. This helps to explain why this feature may have gone undetected until now. The residual plot however does suggest some deviation from the trend line. For example, the majority of the datapoints for R-R_{sep} < 5 mm have positive residuals suggesting that $q_{\parallel}$(R=0) could be 2-3 MW/m^2 higher than estimated by the trend line.

The residuals along with measurements in other machines and from the DIII-D IR camera motivate us to take a closer look. We find that the scatter in the $q_{\parallel}$ measurements is dominated by the large scatter in the T_e values, and yet the average value of T_e changes very little (a factor of 2 or less) across the whole SOL. See Fig 9-5.

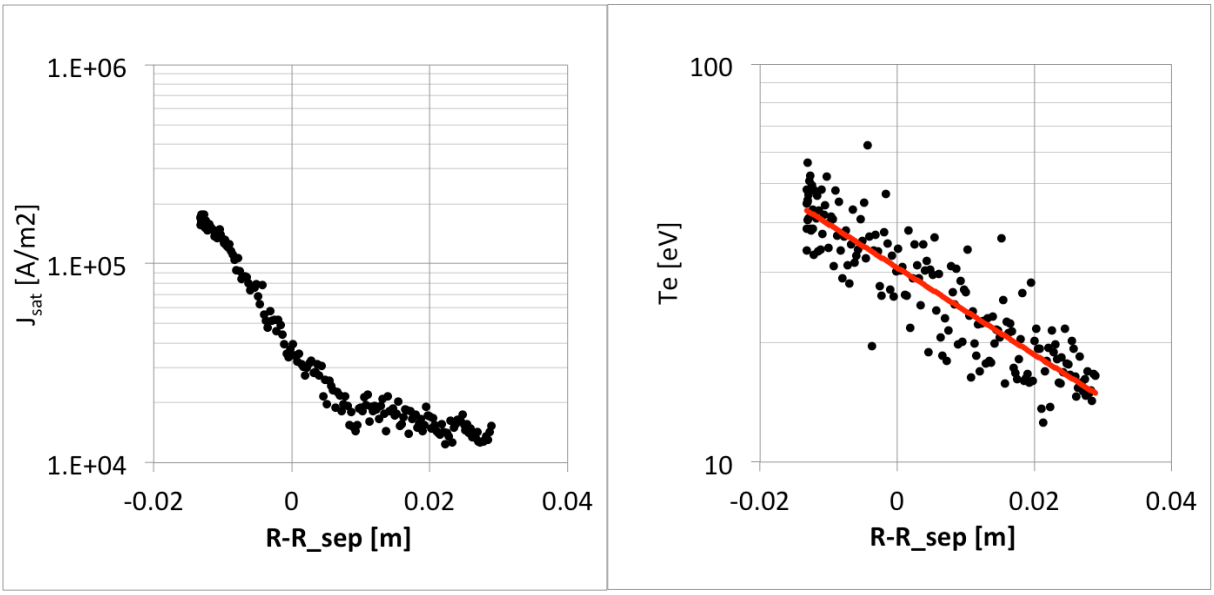


Fig 9-5: The J_{sat} (left) and T_e (right) data for the lower-half swing of 155646. The trend line is from an exponential best fit T_e = 30.785 EXP[-25.14(R-Rsep)]

We therefore apply an exponential fit to the T_e values and recalculate $q_{\parallel}$ using the values from the trend line. We further reduce the scatter by taking 1mm bin averages of the data. See Fig 9-6.

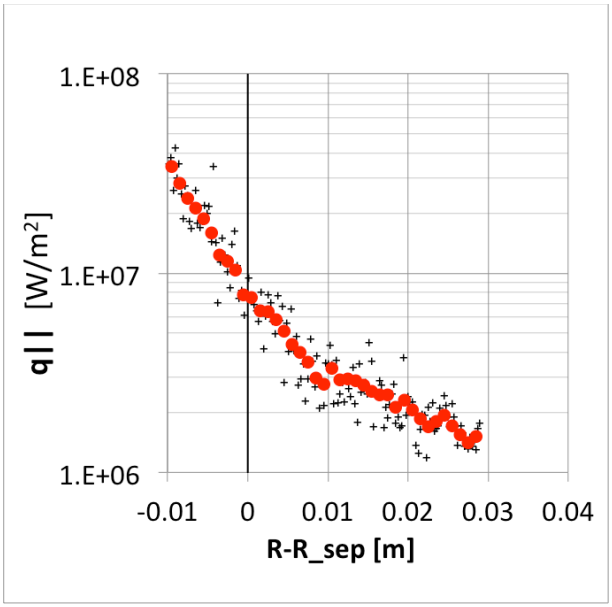


Fig 9-6: $q_{||}$ calculated from the smooth T_e values and averaged into 1mm bins are shown in red. For comparison, the unbinned data are shown in grey.

The trend in Fig 9-6 seems to be clearer than the one in Fig 9-4. To quantify the magnitudes of the long and short heat flux features, we fit eq. 9-1 to calculate $q_{||0}^{long}$, $q_{||0}^{short}$, λ_{long}, and λ_{short}. We report the relative magnitude of the short heat flux feature using the ratio

$$r_q = \frac{q_{||0}^{short}}{q_{||0}^{long}} \qquad \textbf{eq. 9-3}$$

Regardless of the method used for fitting, we find that uncertainties for r_q are large (up to ± 70%) since $q_{||0}^{long}$ and $q_{||0}^{short}$ can be inversely varied while maintaining a reasonable-looking fit.

In Fig 9-7, the $q_{||}$ data are fit using an apparent least-squares method. Note that the long heat flux component shown by the dotted line does not fit the data in the far-SOL. Unfortunately, in most of the shots the gaps to the wall are 3 cm or less, which provides

insufficient data to fully constrain the long component λ_{long} which can be as long as 4 cm.

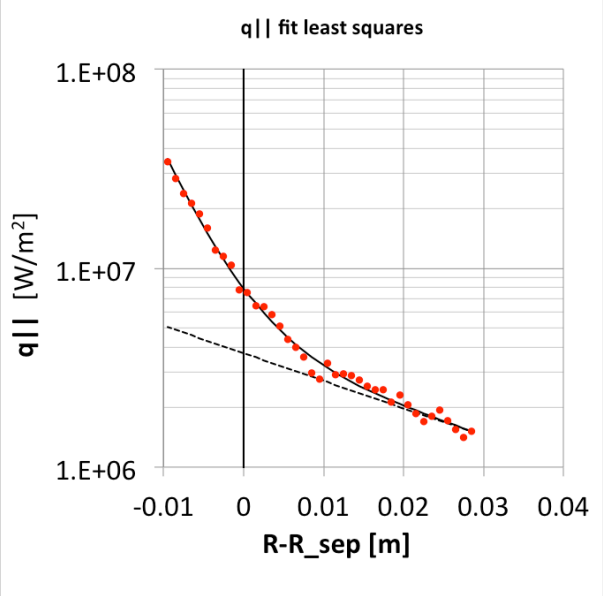


Fig 9-7: Example Least Squares Fit

$q_{\parallel 0}^{long}$ **= 3.74MW/m², λ_{long} = 31mm**

$q_{\parallel 0}^{short}$ **= 4.22MW/m², λ_{short} = 4.8mm**

r_q = 1.13

r_{double}=1.31

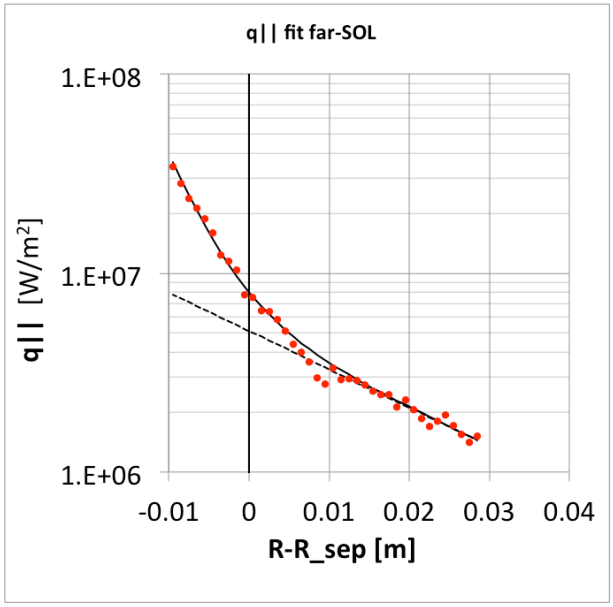


Fig 9-8: Example Far-SOL Fit

$q_{\parallel 0}^{long}$ **= 5.12MW/m², λ_{long} = 23mm**

$q_{\parallel 0}^{short}$ **= 2.93MW/m², λ_{short} = 4.2mm**

r_q = 0.57

r_{double}=1.32

A double exponential (solid line) and its long exponential component (dotted line) fitted using a least squares method (left) and by fitting the long exponential component to the far-SOL before fitting for the short exponential component (right).

While it does not decrease the uncertainties, we find better results by constraining the long and short exponential components separately. The long exponential is fit to the data for R-R_{sep} > 1 cm, though we shorten the region if any outliers are encountered. Then the short exponential component is fit on top of the long component to the data starting at R-R_{sep} = -1 cm over one order of magnitude in the values of $q_{\parallel}$, encompassing the LCFS. See Fig 9-8. Note that in this case, the long exponential (dotted line) does fit the data in the far-SOL. The two r_q values differ considerably for these fits.

The far-SOL fitting ignores the outliers near $R-R_{sep} = 1$ cm. The remaining double-exponential fittings in this thesis are made using this procedure.

As the uncertainties in r_q are large and depend strongly on how the double exponential is fit to the data, we will also report a second ratio to represent the relative magnitude of the short heat flux feature.

$$r_{double} = \frac{q_{\parallel 0}^{short} + q_{\parallel 0}^{long}}{q_{\parallel 0}^{\text{single}}} \qquad \textbf{eq. 9-4}$$

where $q_{\parallel 0}^{long}$ and $q_{\parallel 0}^{short}$ are from the double-exponential fit and $q_{\parallel 0}^{\text{single}}$ is the intercept value from a single exponential fit to the bin-averaged $q_{\parallel}$ data in the SOL. r_{double} represents how much one would underestimate $q_{\parallel 0}$ at the LCFS by assuming that $q_{\parallel}$ increases following a single exponential across the SOL. In this case, $q_{\parallel 0}^{\text{single}} = 6.08 MW/m^2$, so $r_{double} = 1.31$ and 1.32 for the least-squares fit and the far-SOL fit respectively. The exceptionally close agreement here is coincidental. The uncertainty for r_{double} is typically +/- 15% and is much smaller than that for r_q because the intercept value of the double exponential $q_{\parallel 0}^{long} + q_{\parallel 0}^{short}$ remains nearly constant regardless of how one fits the $q_{\parallel}$ data. For subsequent experiments, it is critical to have wider gaps between the plasma and secondary limiters to provide data for 4-6cm into the far-SOL. With 6cm of data, all double exponential fitting procedures tested agree within ~2%.

9.3 Swing-Probe Results

Here we present the lower Swing-probe data from the 2013 DIII-D IWL experiment. For each shot, we present the data from the lower half swing and the upper half swing (data taken from the orange arc and the green arc respectively as shown in figures Fig 9-2 and Fig 9-3).

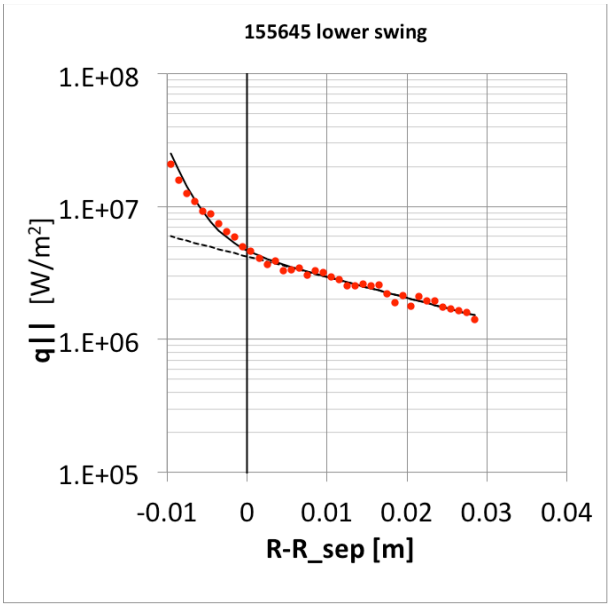


Fig 9-9: Swing-probe $q_{||}$ fit from shot 155645 @ 2000ms lower swing

$q_{||0}^{long}$ **= 4.23MW/m², λ_{long} = 28mm**

$q_{||0}^{short}$ **= 0.46MW/m², λ_{short} = 2.6mm**

r_q = 0.11

r_{double}=1.12

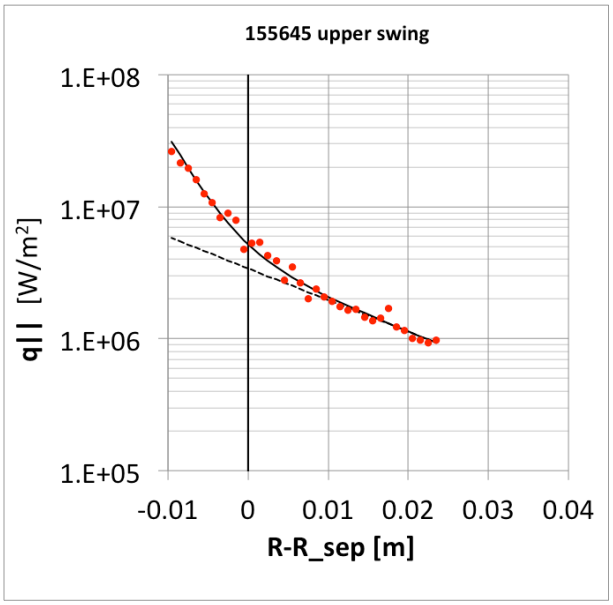


Fig 9-10: Swing-probe $q_{||}$ fit from shot 155645 @ 2000ms upper swing

$q_{||0}^{long}$ **= 3.42MW/m², λ_{long} = 18mm**

$q_{||0}^{short}$ **= 1.81MW/m², λ_{short} = 3.6mm**

r_q = 0.53

r_{double}=1.22

See Fig 9-9 and Fig 9-10. We note that there are significant differences in the $q_{||}$ data between the upper and lower swings, particularly in the shape of the data. The differences are not consistent with a systematic error in the location of the LCFS or with the probe position as a function of time. Outgassing from the probe head could be affecting the measurements due to a very high-density experiment immediately preceding the IWL experiment. For the upper swing, the probe crosses fewer flux surfaces as some have already intercepted the centerpost. Note that there is less data available in Fig 9-10.

For shot 155645 I_p = 1.19 MA and we probe the elongated plasma with κ = 1.58. For the lower swing (left) a short component within the SOL itself is not very evident, and the

transition could occur at the LCFS. For the upper swing (right), it appears there is a short component within the SOL.

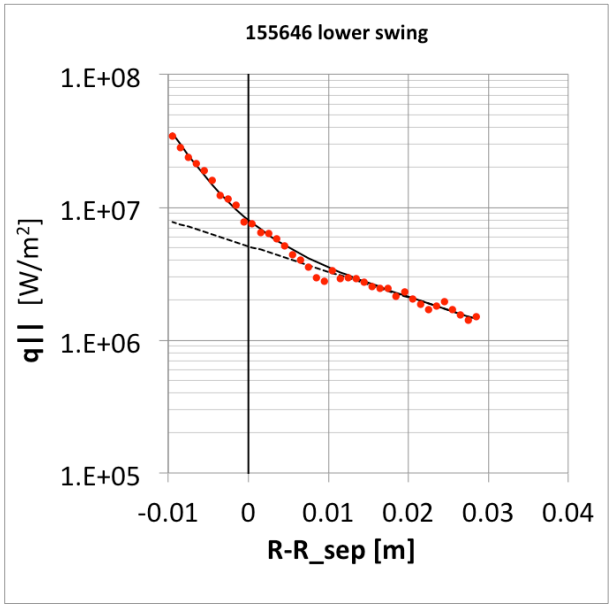


Fig 9-11: Swing-probe $q_{||}$ fit from shot 155646 @ 2000ms lower swing

$q_{||0}^{long}$ **= 5.12MW/m², λ_{long} = 23mm**

$q_{||0}^{short}$ **= 2.93MW/m², λ_{short} = 4.2mm**

r_q = 0.57

r_{double}=1.38

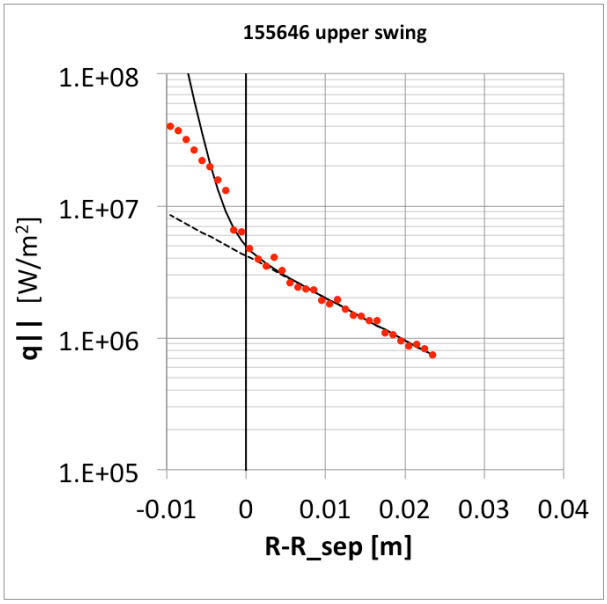


Fig 9-12: Swing-probe $q_{||}$ fit from shot 155646 @ 2000ms upper swing

$q_{||0}^{long}$ **= 4.20MW/m², λ_{long} = 14mm**

$q_{||0}^{short}$ **= 0.77MW/m², λ_{short} = 1.5mm**

r_q = 0.18

r_{double}= 1.26

Shot 155646 has the same conditions as the previous shot with I_p = 1.20 MA and high elongation κ = 1.59. In the lower swing (Fig 9-11), there appears to be a short component within the SOL, but the short component is not very evident in the upper swing (Fig 9-12).

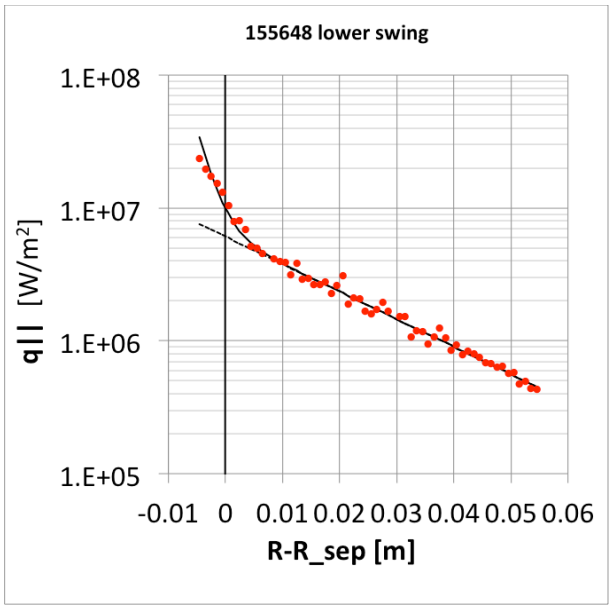


Fig 9-13: Swing-probe $q_{||}$ fit from shot 155648 @ 3500ms lower swing

$q_{||0}^{long}$ **= 6.1MW/m², λ_{long} = 21mm**

$q_{||0}^{short}$ **= 3.82MW/m², λ_{short} = 2.3mm**

r_q = 0.63

r_{double}= 1.43

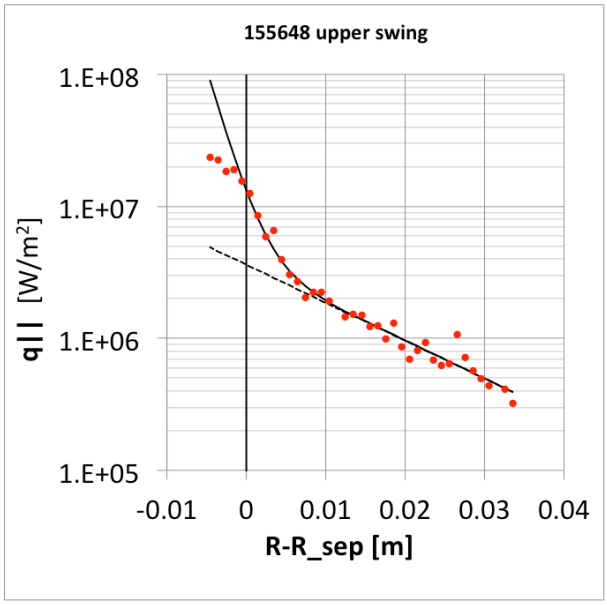


Fig 9-14: Swing-probe $q_{||}$ fit from shot 155648 @ 3500ms upper swing

$q_{||0}^{long}$ **= 3.63MW/m², λ_{long} = 15mm**

$q_{||0}^{short}$ **= 9.93MW/m², λ_{short} = 2.1mm**

r_q = 2.74

r_{double}= 2.27

In shot 155648, measurements were taken in the round plasma κ = 1.26 with I_p = 1.19 MA. This is the only shot where data were taken at these conditions. There are clear short components in both swings (Figs Fig 9-13 and Fig 9-14) and the short component in the upper swing is the largest observed in the experiment. With this plasma shape, the secondary gaps are large and useful data was taken well into the far-SOL. Again, there are less data for the upper swing because the outermost flux surfaces have already intercepted with the centerpost.

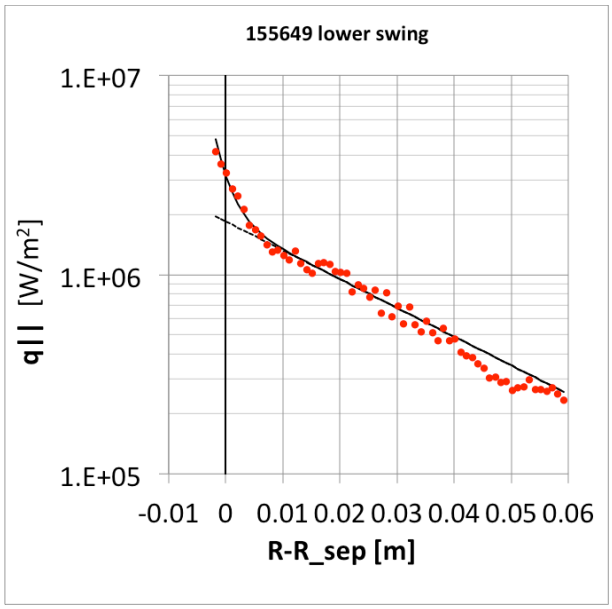


Fig 9-15: Swing-probe $q_{||}$ fit from shot 155649 @ 3500ms lower swing

$q_{||0}^{long}$ **= 1.85MW/m², λ_{long} =30mm**

$q_{||0}^{short}$ **= 1.32MW/m², λ_{short} = 2.4mm**

r_q = 0.71

r_{double}= 1.47

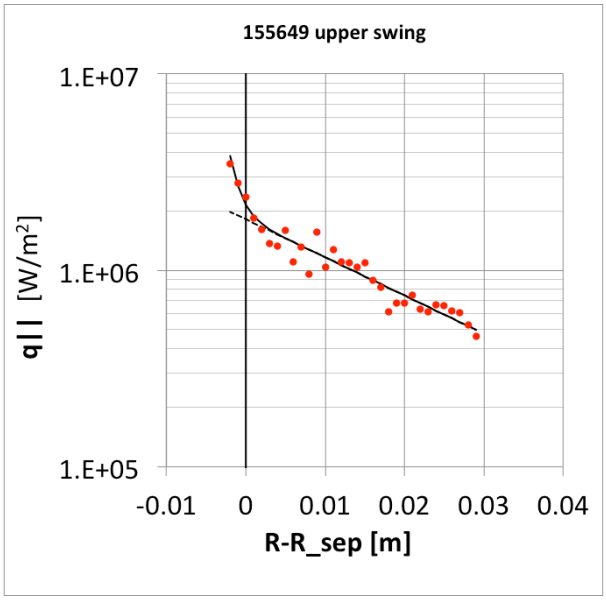


Fig 9-16: Swing-probe $q_{||}$ fit from shot 155649 @ 3500ms upper swing

$q_{||0}^{long}$ **= 1.82MW/m², λ_{long} =22mm**

$q_{||0}^{short}$ **= 0.34MW/m², λ_{short} = 1.2mm**

r_q = 0.19

r_{double}= 1.25

For shot 155649 and subsequent shots, the plasma current was decreased from 1.19 MA to 0.61 MA with a corresponding drop in power. Note the corresponding change in the $q_{||}$ scales. Data was taken in the round plasma shape with κ = 1.24. These are the only data taken for these conditions. There is a clear short component in the lower swing (Fig 9-15), but in the upper swing (Fig 9-16) $q_{||0}^{short}$ and λ_{short} are very small.

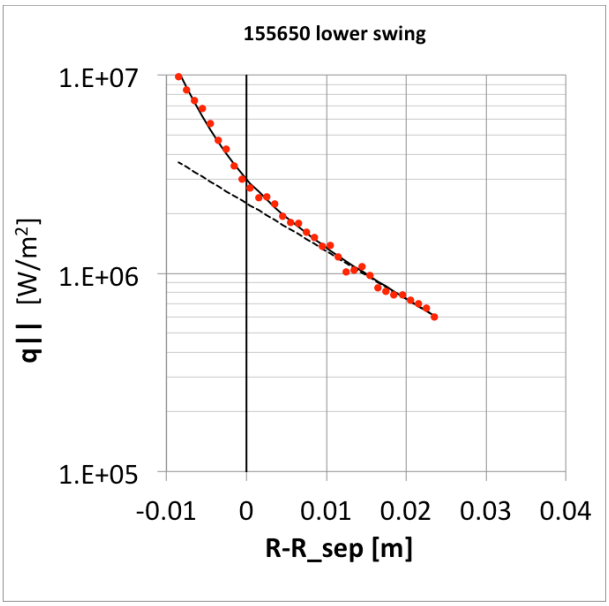


Fig 9-17: Swing-probe $q_{||}$ fit from shot 155650 @ 2000ms lower swing

$q_{||0}^{long}$ **= 2.26MW/m², λ_{long} = 18mm**

$q_{||0}^{short}$ **= 0.74MW/m², λ_{short} = 3.8mm**

r_q = 0.33

r_{double} = 1.14

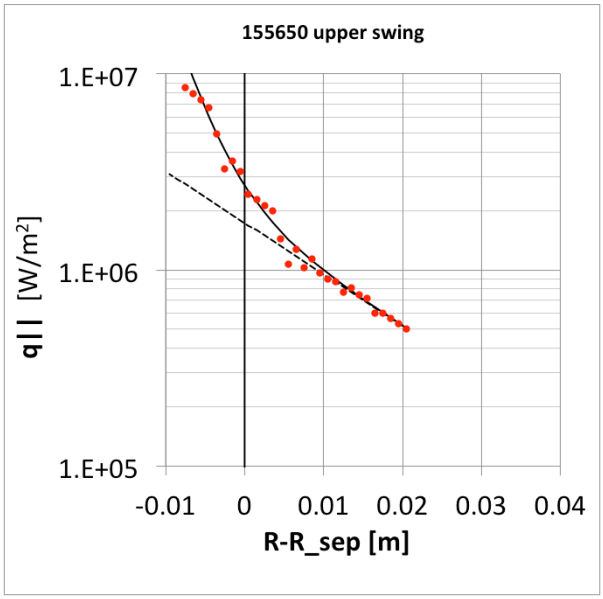


Fig 9-18: Swing-probe $q_{||}$ fit from shot 155650 @ 2000ms upper swing

$q_{||0}^{long}$ **= 1.74MW/m², λ_{long} = 17mm**

$q_{||0}^{short}$ **= 0.97MW/m², λ_{short} = 3.3mm**

r_q = 0.56

r_{double} = 1.23

For shot 155650, Ip = 0.61 MA and data were taken in the elongated plasma with $\kappa = 1.60$. Positioning was shifted such that the gaps to the secondary limiters are smaller, so we have less useful data to fit. There does appear to be a short component within the SOL for both the lower (Fig 9-17) and upper shot (Fig 9-18).

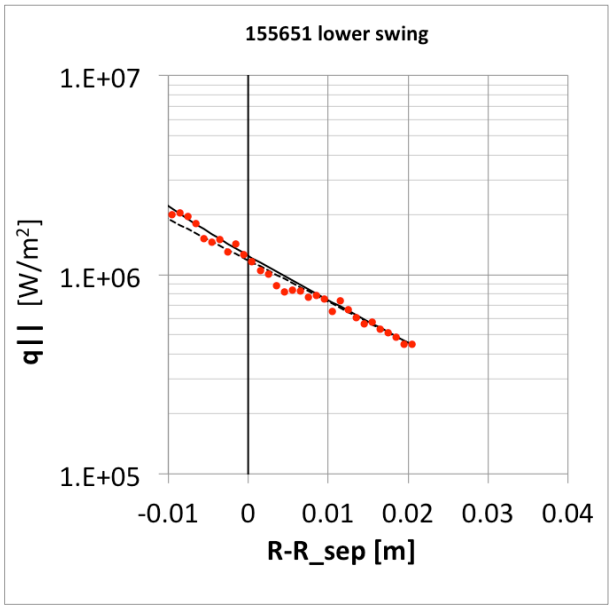


Fig 9-19: Swing-probe $q_{||}$ fit from shot 155651 @ 2000ms lower swing

$q_{||0}^{long}$ = 1.18MW/m², λ_{long} = 21mm

$q_{||0}^{short}$ = 0.07MW/m², λ_{short} = 6.7mm

r_q = 0.06

r_{double}= 1.13

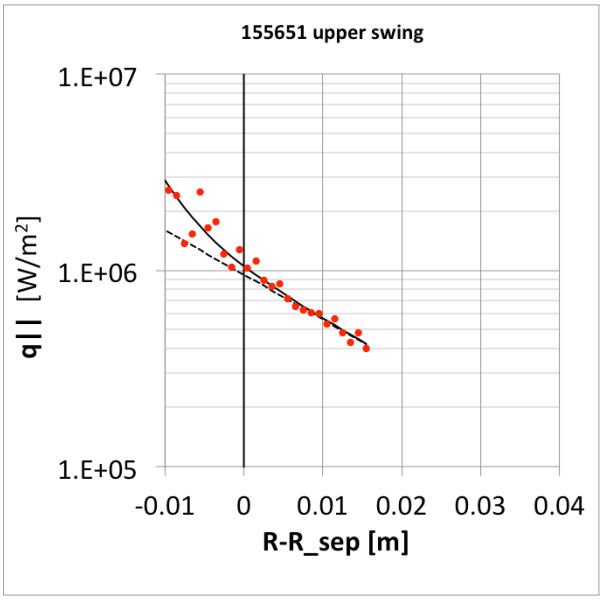


Fig 9-20: Swing-probe $q_{||}$ fit from shot 155651 @ 2000ms upper swing

$q_{||0}^{long}$ = 0.95MW/m², λ_{long} = 19mm

$q_{||0}^{short}$ = 0.10MW/m², λ_{short} = 4.0mm

r_q = 0.11

r_{double}= 0.98

For shot 155651, conditions are the same as in the previous shot with I_p = 0.61 MA and κ = 1.66. However, the positioning was shifted again, further limiting the useful data range. There are no discernable short components in either half swing (Fig 9-19 or Fig 9-20), but there are not enough data in the far-SOL to be conclusive.

The values thus obtained are averaged for the upper and lower swings of each shot and reported in Table 9-2.

Shot	Arc	I_p (MA)	κ	$q_{\|\|0}^{long}$ (MW/m^2)	$q_{\|\|0}^{short}$ (MW/m^2)	r_q	λ_q^{long} (mm)	λ_q^{short} (mm)	r_{double}
155645	Lower	1.2	1.60	4.23	0.46	0.11	28	2.6	1.12
155645	Upper	1.2	1.60	3.42	1.81	0.53	18	3.6	1.22
155646	Lower	1.2	1.60	5.12	2.93	0.57	23	4.2	1.38
155646	Upper	1.2	1.60	4.20	0.77	0.18	14	1.5	1.26
155648	Lower	1.2	1.25	6.10	3.82	0.63	21	2.3	1.43
155648	Upper	1.2	1.25	3.63	9.93	2.74	15	2.1	2.27
155649	Lower	0.6	1.25	1.85	1.32	0.71	30	2.4	1.47
155649	Upper	0.6	1.25	1.82	0.34	0.19	22	1.2	1.25
155650	Lower	0.6	1.60	2.26	0.74	0.33	18	3.8	1.14
155650	Upper	0.6	1.60	1.74	0.97	0.56	17	3.3	1.23
155651	Lower	0.6	1.66	1.18	0.068	0.06	21	6.7	1.13
155651	Upper	0.6	1.66	0.95	0.10	0.11	19	4.0	0.98
Average						0.56 ± 1.45			1.30 ± 0.67

Table 9-2: Parameters for best double-exponential fits to the Swing-Probe $q_{||}$ profiles calculated using individual values of J_{sat} and values from the fit to T_e. The uncertainty values in the averages are two standard deviations from the column.

Of the shots six listed above, four show short features with r_q~0.4, and one shows a much stronger q_{short} component with an average r_q=1.69. The remaining shot (155651) shows no sign of a short feature, but the gaps to the secondary limiters are so small that a short component could be obscured. The r_q column tells us that $q_{||0}^{short}$ is on the same order as $q_{||0}^{long}$. r_q = 0.56 ± 1.45. The r_{double} column tells us that the single-exponential description of the $q_{||}$ profile underestimates the heat flux at the LCFS by 30% ± 67%.

While the uncertainty in the EFIT and the location of the LCFS varies from experiment to experiment, we expect that it is small for this case since the LCFS is defined by the limiter, not by an X-point. The EFIT reconstruction is therefore less sensitive to the assumptions made regarding the distribution of the toroidal current density in the edge of the plasma. Furthermore, the accumulated errors should be especially small for the Swing-Probe measurements since it is close to the target and close to the magnetic sensors at the wall. The small uncertainties at the wall are amplified exponentially as the EFIT solution propagates inward [42]. The largest uncertainty in the LCFS position may be due to the radial alignment of the centerpost tiles, which is of the order +/-

2 mm. The uncertainty in the angle of the Swing-Probe is on the order of 1°- which translates to 3-4mm in real coordinates. When mapped to the outer mid plane, the flux expansion affects the magnitude of the uncertainty, in this case, decreasing it by a factor of 2-4 for the Swing-Probe (<2mm) and a factor of 1.5 for the limiter (<2mm). If we consider that the short exponential is measurable beyond the LCFS for 2-3 λ_{short}, then the features are on average 3 times larger than the total errors associated with the EFITs, the probe position, and the electrode dimensions.

9.4 IR Camera Results

The IR camera data are challenging to interpret in DIII-D because the centerpost is armoured by 48 columns and 18 rows of flat tiles and the tile alignment, while exceptionally good by normal standards, is nevertheless of order +- 2 mm (radial) which is enough to effect the power deposition pattern as seen by the IR. Each tile creates a very shallow cavity that contains a short-connection SOL, but readily interpretable data can be taken near the edges of each tile. The cleanest example comes from shot 155648, which has been processed by CJ Lasnier and JD Elder. See Fig 9-21. Unfortunately this is the only example with usable IR data for this experiment due to complications with the shape of the centerpost and the positioning of the LCFS contact point on the inner column.

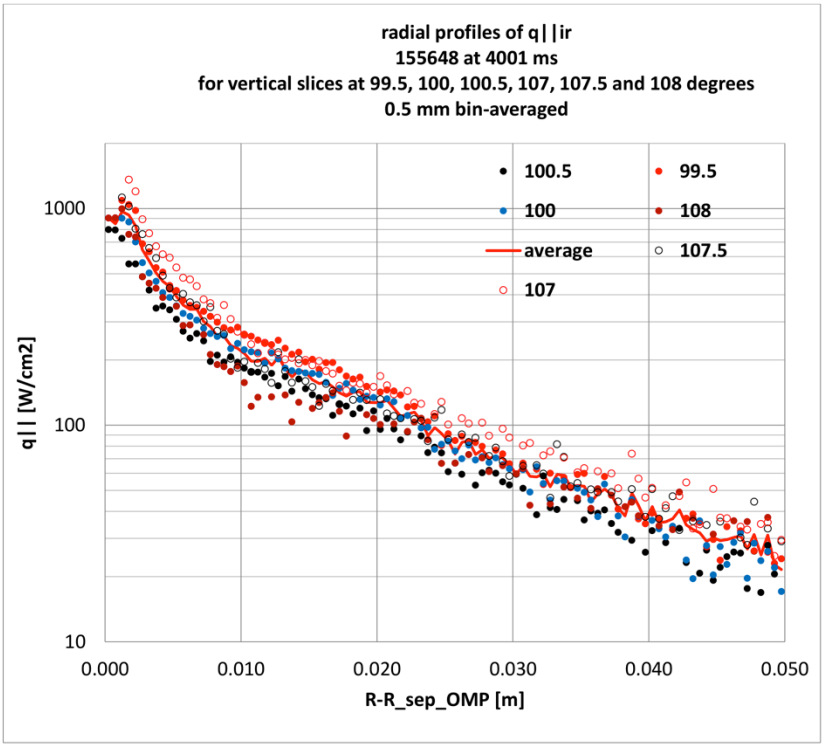


Fig 9-21: $q_{||}$ profiles inferred from IR camera data from the DIII-D centerpost from the toroidal angles with the clearest signals (close to the limiter apexes) as a function of R-R_{sep}.

This also happens to be the shot and time where the Swing-Probe shows the strongest q_{short} feature. Since the IR camera data were taken at the same time as the Swing-Probe, it provides a useful test on the consistency between the two diagnostics with regard to the shape of the $q_{||}$ profile. We do not make a comparison here of magnitudes as that requires knowledge of the sheath heat transmission coefficient. Both diagnostics show similar short features, see Fig 9-22.

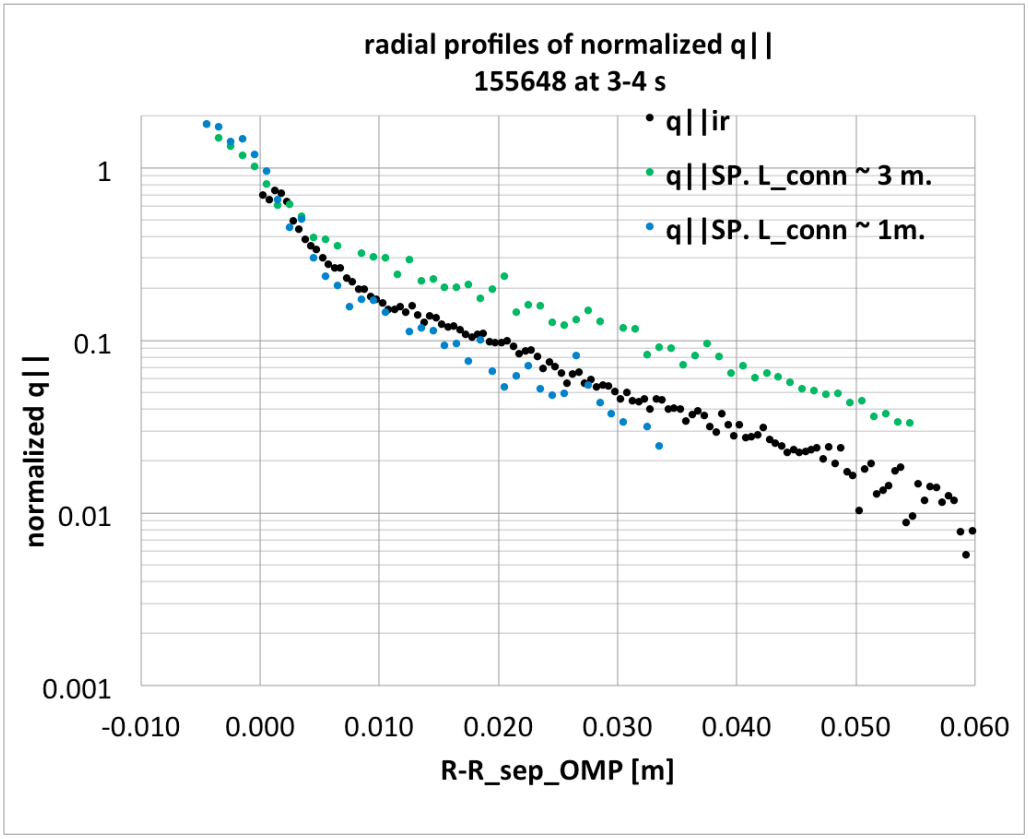


Fig 9-22: A comparison between the $q_{||}$ profiles for 155648 normalized to 1 at the LCFS. The normalized IR camera data are shown in black. The normalized Swing-Probe data are shown in green for the lower swing and blue for the upper swing, which is closer to the target. The two diagnostics show similar short features.

As the IR data can only be from outside the LCFS, this agreement confirms that the probe measurements of the short heat flux feature are not the result of positional errors.

9.5 Results from Other Diagnostics

The data from the core Low Field Side Thomson Scattering (LFS-TS), Divertor Thomson Scattering (DTS) and MidPlane Reciprocating Probe (MP RCP) have more noise than the Swing-Probe data, so no clear conclusions can be drawn from them. We expect little variation with poloidal location, so we generally expect all four diagnostics to agree. We present the data for comparison here for the shots with Swing-Probe data. Fig 9-23 through Fig 9-25 show the first three shots with I_p = 1.20 MA. Fig 9-26 to Fig 9-28 show the comparisons for the latter three shots with I_p = 0.61 MA.

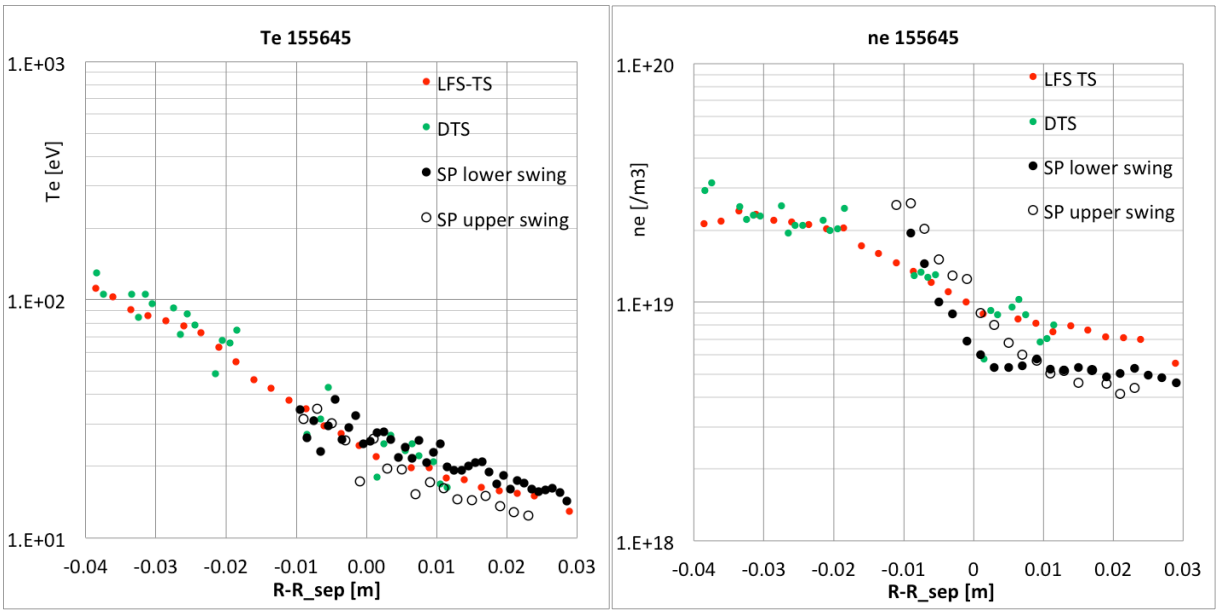


Fig 9-23: The electron temperature and density data from the Core Thomson, Divertor Thomson, and Swing-Probe for shot 155645 at ~2000 ms (elongated shape). We note the excellent agreement in T_e and that the density has good agreement in magnitude but not necessarily in the profile shape.

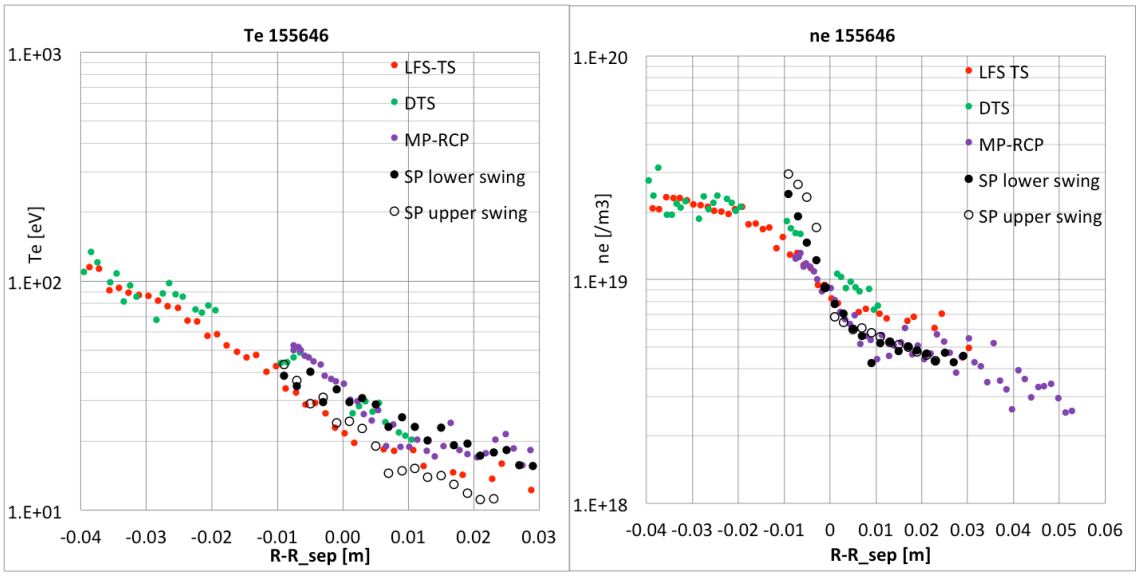


Fig 9-24: The electron temperature and density data from the Core Thomson, Divertor Thomson, Midplane Reciprocating Probe, and Swing-Probe for shot 155646 at ~2000 ms (elongated shape). We see good agreement among all of the diagnostics and do see signs of a short feature in the MP-RCP and perhaps in the LFS TS. However, without the SP data, the data from the other diagnostics would have too much uncertainty to confirm the existence of the short exponential component.

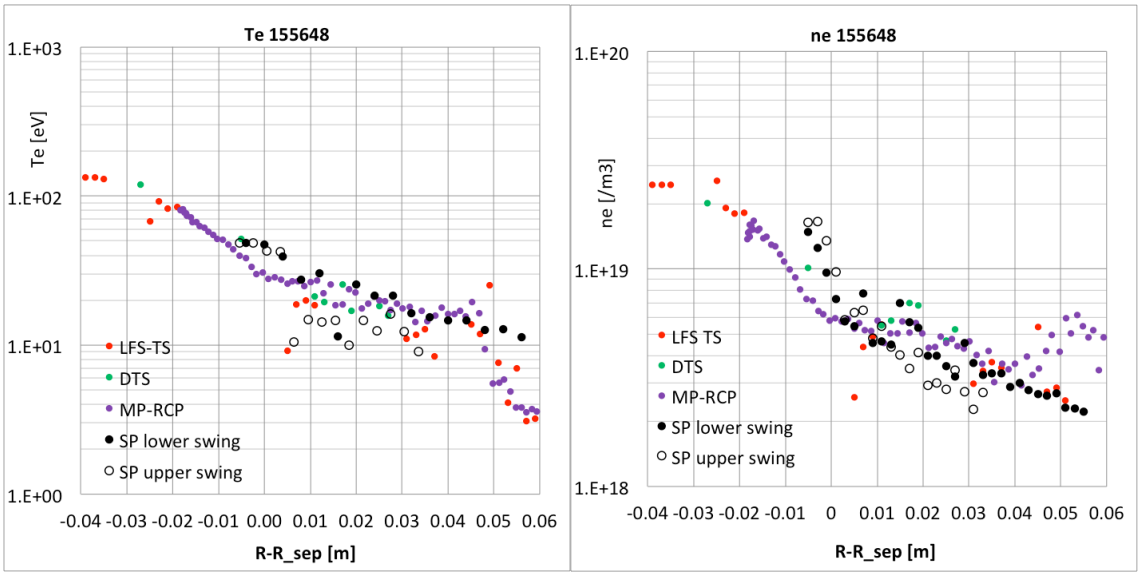


Fig 9-25: The electron temperature and density data from the Core Thomson, Divertor Thomson, Midplane Reciprocating Probe, and Swing-Probe for shot 155648 at ~3500 ms (round shape). The TS data is sparse for this shot. The MP-RCP data has an inflection point, but it is well inside the LCFS

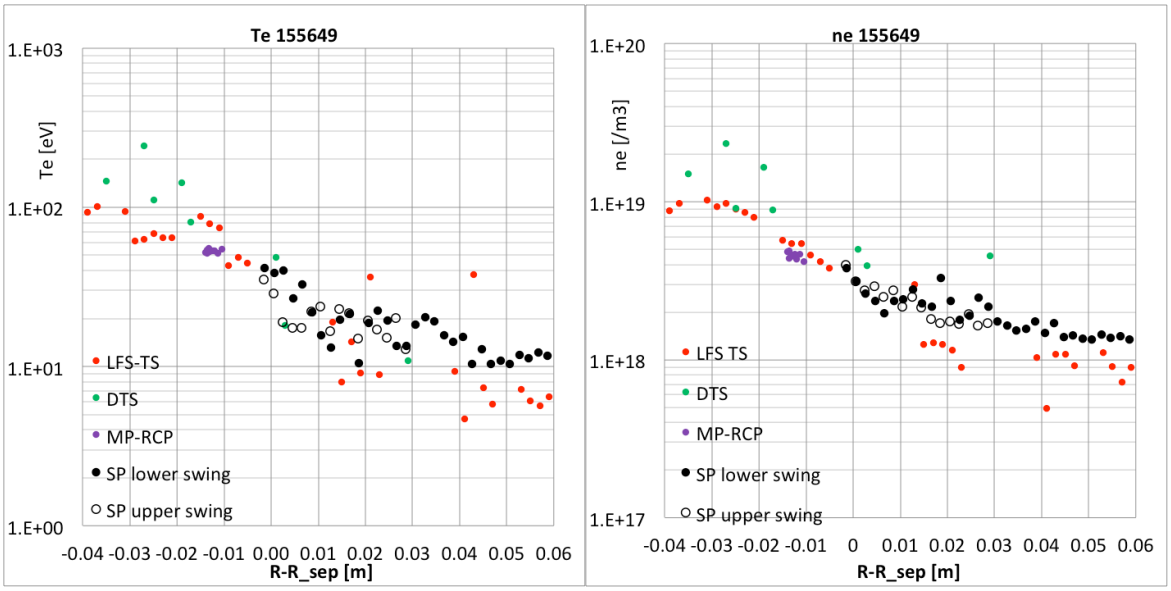


Fig 9-26: The electron temperature and density data from the Core Thomson, Divertor Thomson, Midplane Reciprocating Probe, and Swing-Probe for shot 155649 at ~3500 ms (round shape). The data from the TS and MP-RCP are sparse but in excellent agreement with the SP data.

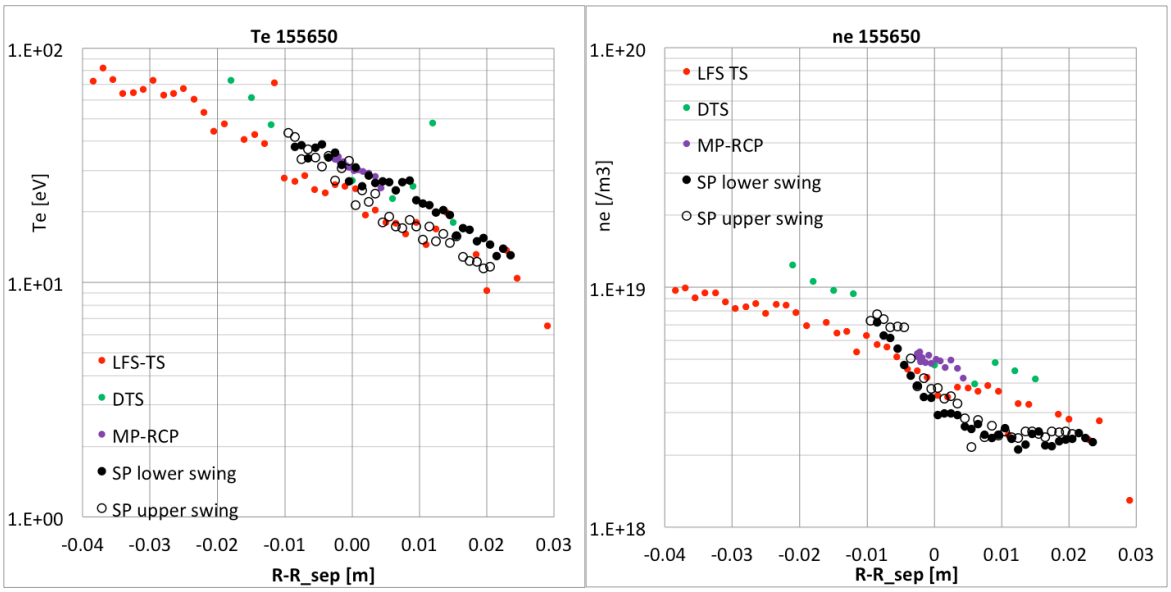


Fig 9-27: The electron temperature and density data from the Core Thomson, Divertor Thomson, Midplane Reciprocating Probe, and Swing-Probe for shot 155650 at ~2000 ms (elongated shape). Only the Swing-Probe shows a short feature.

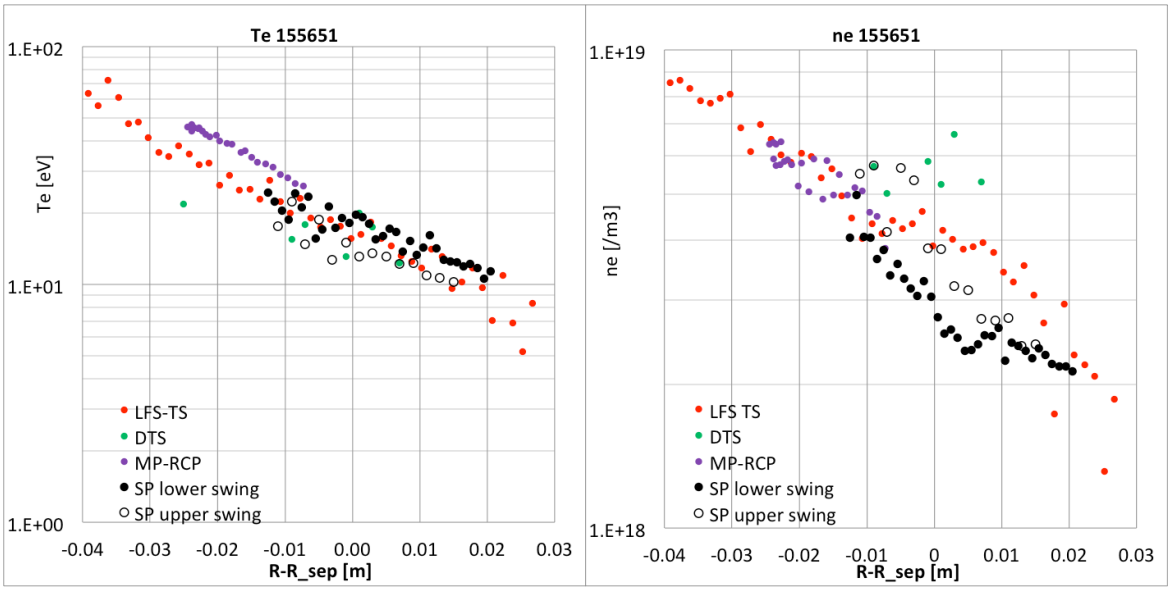


Fig 9-28: The electron temperature and density data from the Core Thomson, Divertor Thomson, Midplane Reciprocating Probe, and Swing-Probe for shot 155651 at ~2000 ms (elongated shape). There is insufficient data in the far-SOL to infer the shape of the parallel heat flux profile for this shot.

The core Thomson Scattering, Divertor Thomson Scattering, and Midplane Probe provide results that are in agreement with the Swing-Probe to within the scatter and systematic uncertainties. While a short component is observed some of the time, it is not seen consistently enough to form any solid conclusions. As in measurements made in other machines, it is possible to observe the short feature on the inboard side, but it is rarely apparent on the outboard side, either because outboard side fluctuations mask the short feature, or because it is only present on the inboard side.

9.6 Core Thomson Scattering Results

In two examples, the core Thomson Scattering data are dense enough to merit a closer look. We take 1mm bin averages of the T_e and n_e data where they are sufficiently dense. See example in Fig 9-29.

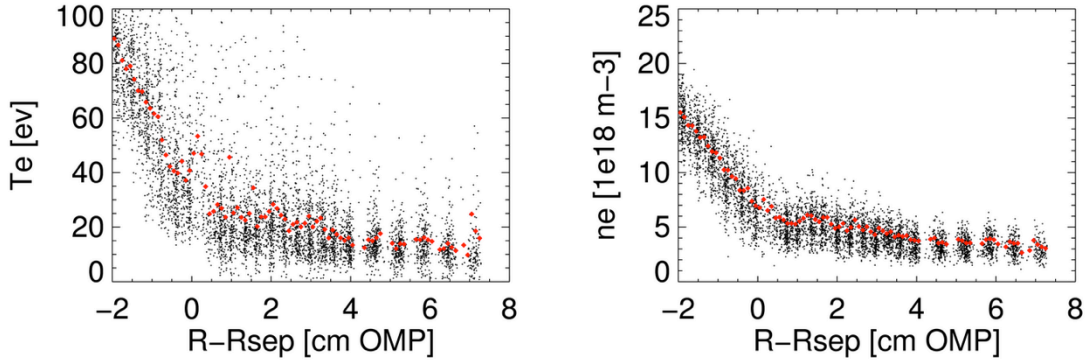


Fig 9-29: Core Thomson Scattering T_e and n_e data 1 mm bin averages for 155645 from 2800-4100 ms

Here again, we arbitrarily assume the sheath heat transfer coefficient γ = 7 to calculate $q_{\parallel}$.

$$q_{\parallel} = \gamma e T_e n_{se} C_s = \gamma e T_e \frac{n_e}{2} C_s = 5.484e-15 T_e^{3/2} n_e \qquad \textbf{eq. 9-5}$$

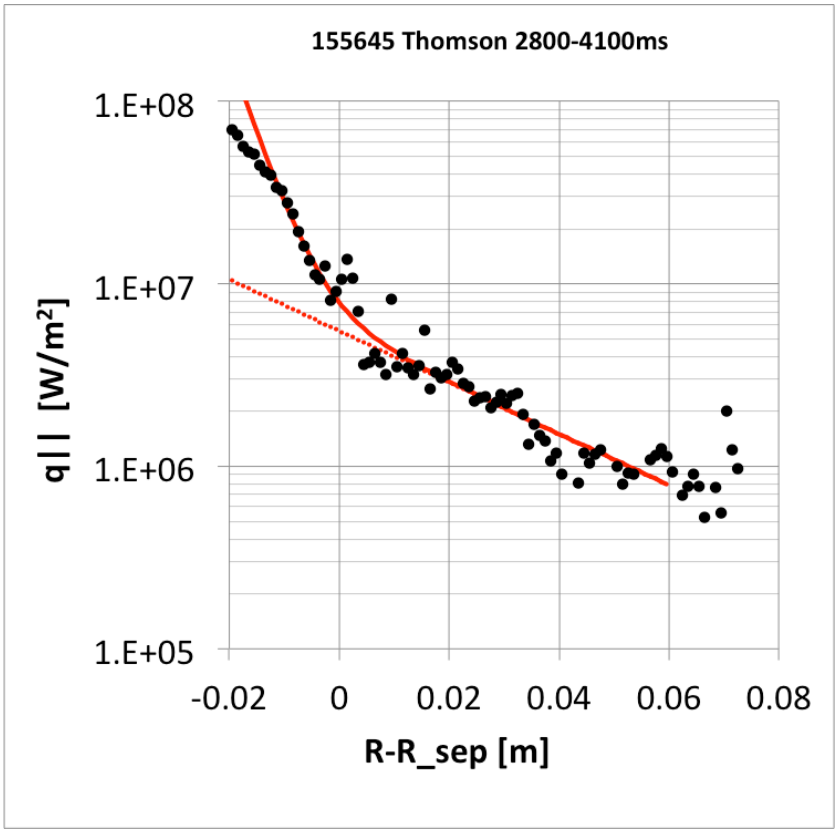


Fig 9-30: The 1 mm binned $q_{||}$ values for 155643 from 2800-4100 ms.

$q_{||0}^{long}$ = 5.52 MW/m², λ_{long} = 30 mm

$q_{||0}^{short}$ = 2.46 MW/m², λ_{short} = 4.7 mm

r_q = 0.44

r_{double}=1.38

In this case (Fig 9-30), we see a significant short feature in the SOL, However, the unfortunate location of the gap in the data (See Fig 9-29 between 0.0 cm < R-R_{sep} < 0.5 cm) decreases our confidence in being able to reliably identify this feature.

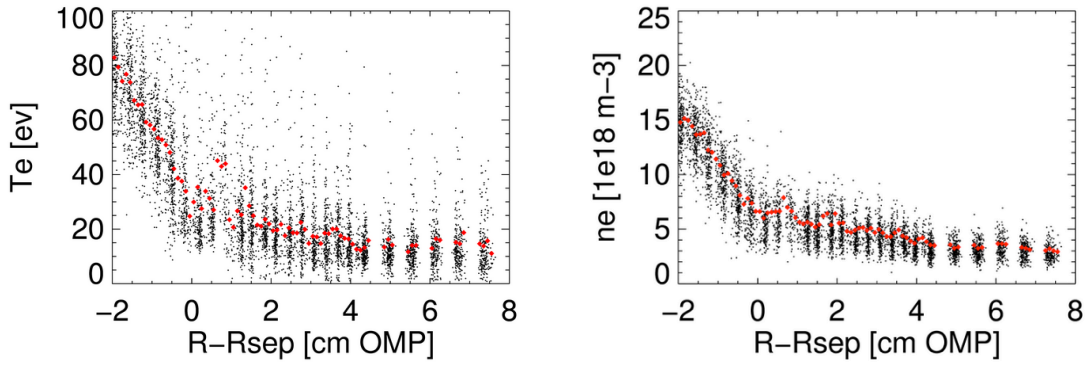


Fig 9-31: T_e and n_e bin averages for 155646 from 2800-4100 ms. There is a hot channel at R-Rmaj = 0.75 cm which we ignore for the $q_{||}$ calculation.

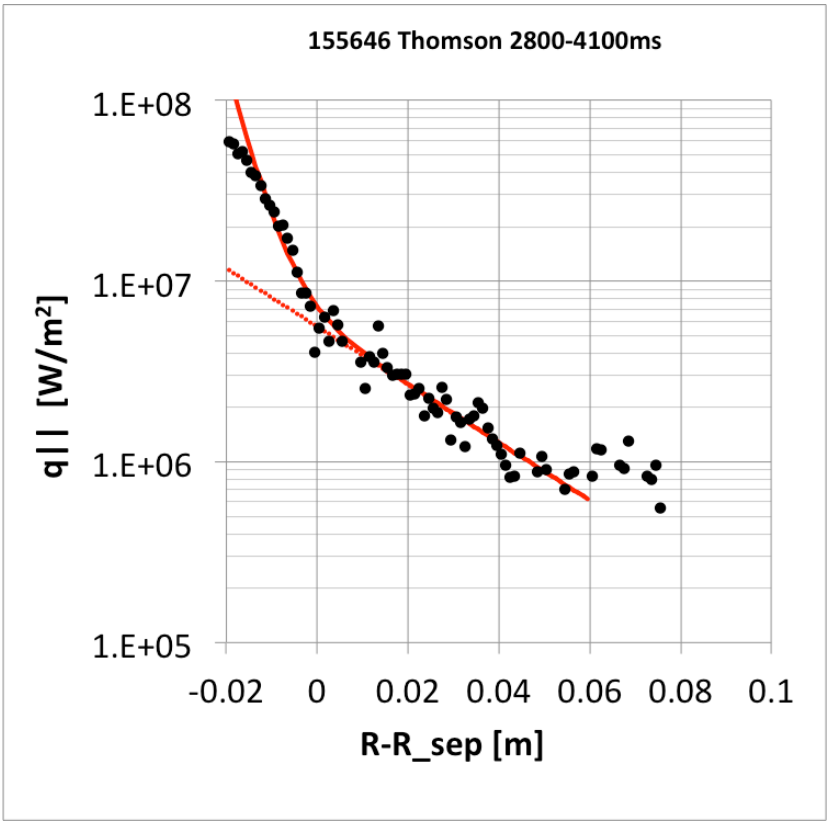


Fig 9-32: The 1mm binned $q_{||}$ values for 155643 from 2800-4100ms.

$q_{||0}^{long}$ = 5.60MW/m², λ_{long} = 27mm

$q_{||0}^{short}$ = 1.69MW/m², λ_{short} = 4.5mm

r_q = 0.30

r_{double}=1.56

For shot 155646 at 4500 ms, it is not clear if there is a short feature within the SOL; See Fig 9-31 and Fig 9-32. The double-exponential fit does show the presence of a significant short feature, but the scatter in the data is sufficient to obscure whether the deviation from the long exponential occurs in the SOL or if it only occurs in the confined plasma.

While the scatter in the TS data is large, it is unique in that it takes data for the entire duration of the shot and for every shot. This adds a level of consistency that we do not observe with the other diagnostics. We can therefore combine the TS data for four similar sets of conditions. See Fig 9-33.

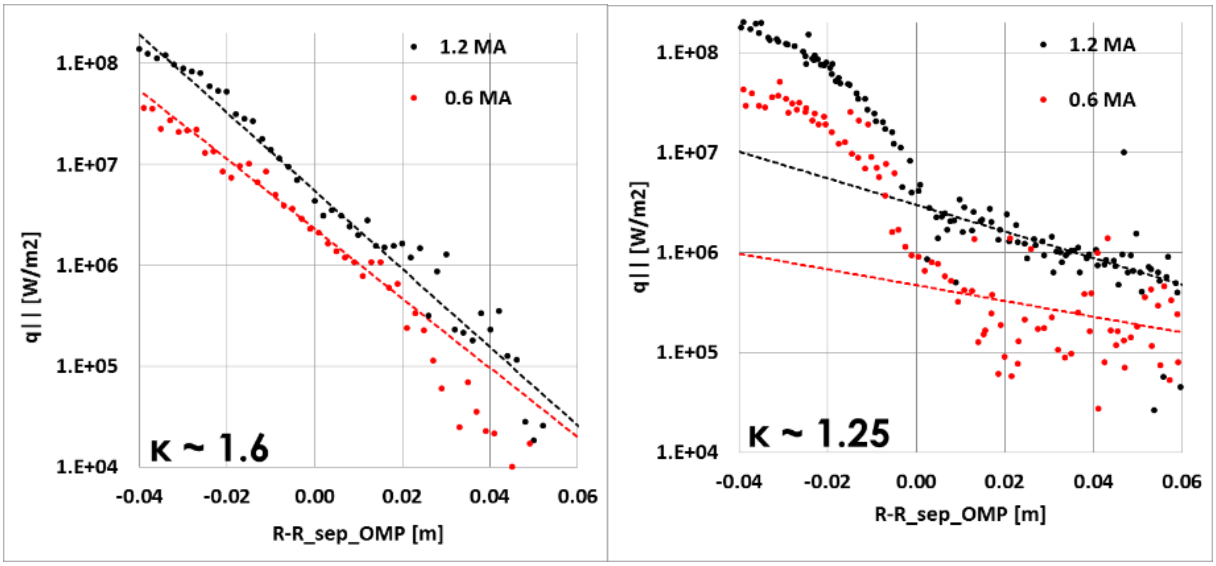


Fig 9-33: The Core Thomson scattering $q_{\parallel}$ data combined over multiple shots (4 shots for the 1.2 MA data, 3 shots for the 0.6 MA data) and bin averaged. For κ ~ 1.6 (left) the dashed lines are the best-fit single exponential to the data from -2 cm to 2 cm. For κ ~1.25 (right) the dashed lines are the best-fit single exponential to the data from 0 mm to 6 cm.

For κ ~ 1.6, the $q_{\parallel}$ data are clearly described by a single exponential across the SOL and much of the confined plasma. This is despite the fact that the Swing-Probe consistently observes a short component for all measured cases with I_p ~ 1.2 MA and κ ~ 1.6. For I_p ~ 1.2 MA and κ ~ 1.25 there does not appear to be a short feature and the $q_{\parallel}$ data

increases rapidly just inside the LCFS. The scatter in the $I_p \sim 0.6$ MA and $\kappa \sim 1.25$ data makes it unclear if a short feature is present.

In future IWL experiments in DIII-D, attempts should be made to move the IWL plasma along the centerpost slowly enough to provide overlapping data collection from multiple TS channels across the LCFS and SOL regions.

9.7 Conclusion for ITER

The Swing-probe measures $q_{\parallel}$ profiles that are consistent with earlier measurements made for the multi-machine scaling using the outboard-side reciprocating probe. Good agreement between the core Thomson, divertor Thomson, the Midplane Reciprocating Probe and the lower Swing-Probe gives us confidence in the quality of the data. While it is difficult to detect within the scatter of the Swing-Probe measurement, we observe that most of the $q_{\parallel}$ profile show an enhanced heat flux feature just outside the LCFS with r_q on the order of unity. These measurements are consistent with the DIII-D IR camera, and we therefore conclude that the enhanced heat flux features observed in JET, COMPASS, and DIII-D IR camera data are likely real and are probably not an artefact of the camera systems.

Since the 2013 DIII-D inner wall limited experiment, more results from TCV [58] and Alcator C-Mod [59] show the existence of the enhanced heat flux component. Like the dataset presented here, the short heat flux component was observable using IR cameras, and centerpost reciprocating probes (C-Mod), but was generally less apparent on the outboard side. While more work is needed to quantify the scaling and magnitude of the short heat flux component, the datasets from 5 machines suggest that it would be beneficial to modify the ITER inner wall limiter design to account for this feature, and a decision was made to do so in March 2014.

9.8 Discussion and Hypotheses

It is now clear that a region of enhanced parallel heat flux exists just outside the LCFS compared to the expected broad, single-exponential shape of the heat flux across the SOL. This feature appears to be well characterized by a double exponential with two decay lengths. It is equally valid to say that there appears to be two distinct regions with different cross-field decay lengths for $q_{\parallel}$. One region encompasses much of the confined plasma and extends past the LCFS by ~2-5mm. The other spans the remainder of the SOL. While uncertainties in the location of the LCFS can vary, we believe it to be on the order of 2mm for this experiment. The agreement between the IR camera data and the Swing-Probe data tends to confirm that the LCFS is defined accurately since the $q_{||IR}$ profiles are not directly sensitive to uncertainties in the location of the LCFS from EFIT.

We have observed that the short feature can be observed for IWL plasmas using IR cameras on the centerpost and inboard side probes. This feature is sometimes, though rarely observed on the outboard side probes and Thomson Scattering, though the reasons for this are not clear. The short feature is not observed for plasmas limited on the outboard limiter [54].

It is generally thought that λ_{SOL} is set by plasma turbulence AKA anomalous transport, so it can be hypothesized that the λ_{long} heat-flux feature, being the dominant feature, is due to turbulence driven cross-field transport, and that the short feature is caused by a different mechanism.

The poloidal ion gyro radius $r_{D^+}^{pol}$ is defined as [41]:

$$r_{D^+}^{pol} \approx \frac{m_D v_T}{eB_p} \qquad \textbf{eq. 9-6}$$

where m_D is the deuteron mass, v_T is the thermal velocity of the ion, and B_p is the magnitude of the poloidal magnetic field. v_T is calculated from the ion temperature T_i

which is not measured in our experiment. Typically, T_i at the LCFS is 1-6 times higher than T_e [60], so we will assume a typical ratio $T_i = 3T_e$. Fig 9-34 shows a comparison between the fitted values for λ_{short} and the poloidal gyro radius calculated at the OMP using $T_i = 3T_e$ measured by the Swing-Probe at the LCFS. The scatter in the data is too wide to confirm a correlation, but we do find that the λ_{short} is on the scale length of $r_{D^+}^{pol}$.

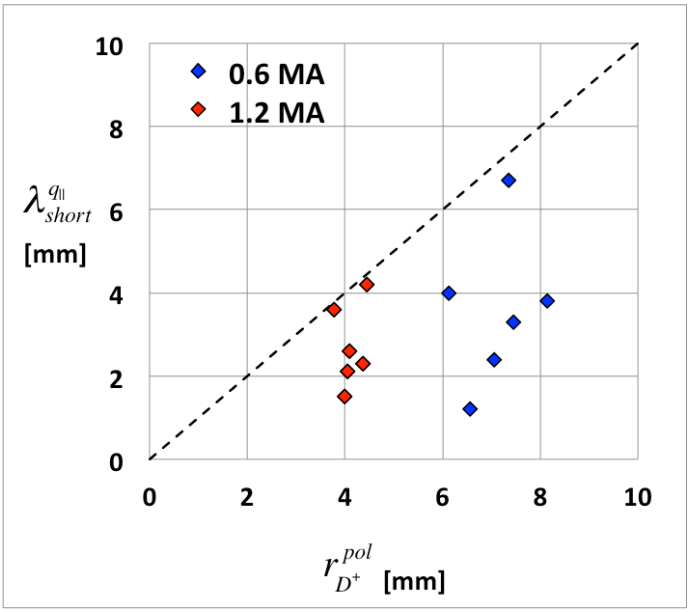


Fig 9-34: A comparison of the λ_{short} values from Swing-Probe $q_{||}$ profiles and the $r_{D^+}^{pol}$ values at the OMP calculated assuming Ti=3Te.

Due to the similar feature size, the tempting hypothesis is that the short feature is caused by neoclassical banana orbit transport, which should be active in the confined plasma and extending out one banana orbit radius beyond the LCFS [61].

$$\lambda_{sol} = r_{D^+}^{banana} = (\varepsilon)^{1/2} r_{D^+}^{pol} \qquad \textbf{eq. 9-7}$$

where the inverse aspect ratio $\epsilon = a/R_0$. Outside of this region, only anomalous transport remains, explaining the smaller cross-field gradient in the far-SOL. That is to say, this theory would describe the $q_{||}$ profiles as a two part piecewise function and that the boundary between the parts would be one banana radius outside the LCFS.

However, for neoclassical banana transport, the banana orbit frequency must be higher than the effective collision frequency for detrapping from a banana orbit, $\nu_{bi} > \nu_{eff}$. This can be true for H-mode plasmas with very low collisionality, but for the conditions of the current experiment ν_{bi} is several kHz, while ν_{eff} is several MHz [41,62], so neoclassical banana transport should not occur.

Another theory [63], appropriate for more collisional conditions, predicts that the SOL width is set by the $\nabla\vec{B}$ drift, the $curv\vec{B}$ drift, and the particle residence time in the SOL, giving

$$\lambda_{sol} = (2\varepsilon) r_{D^+}^{pol} \qquad \textbf{eq. 9-8}$$

[63] predicts the width of the whole SOL in H-mode, not the width of one component of the SOL in L-mode. Furthermore, [63] attempts to calculate the temperature at the LCFS whereas we must estimate Ti from measurements.

In addition to the these two models, arguments have been made for decades that the SOL width can be expected to be about one ion poloidal gyro-radius due to various neoclassical or drift effects [64–67]. This seems to be consistent with our λ_{short} measurements of a few mm, albeit with the uncertainty involved in having no direct measurement of T_i, and the uncertainties in our fits for λ_{short}. We hypothesize that the enhanced heat flux feature is caused by one of these effects, though we are not able to distinguish among them since they all predict essentially the same width, i.e. ~ $r_{D^+}^{pol}$.

9.9 Further Work

In order to further characterize the enhanced heat flux feature, a more comprehensive dataset is required. The DIII-D IWL experiment should be repeated with a number of improvements. Larger gaps must be maintained between the LCFS and secondary limiters in order to more accurately characterize λ_{long} and $q_{\|0}^{long}$. Gaps should be 4cm

minimum, though 6cm is ideal to provide enough data to clearly differentiate the long and short components of the double exponential. More attention needs to be paid to the rounded plasma shape $\kappa \cong 1.25$, as it appears to have the stronger enhanced heat flux features. Better IR camera data needs to be taken by considering the alignment of the centerpost tiles before the experiment. Plasma position should be managed so that the contact point is on part of the centerpost where the tile alignments are known to produce clear results. The plasma position should also be managed to provide dense swing-probe data near the LCFS. i.e. the conditions for shot 155648 @3500 ms should be replicated. Finally, the core and divertor Thomson can produce very detailed profiles if the plasma is slowly shifted vertically to provide overlapping coverage of the viewchords.

Chapter 10 Conclusions and Contributions

10.1 Conclusions

The Swing-Probes have proven themselves to be robust and reliable tools, and they are expected to continue to provide valuable data from the inboard side SOL of DIII-D plasmas in future. These probes represent a major technical achievement and are unique to DIII-D. They are the only complex moving parts in the DIII-D vessel, and are the first system to use grade A boron nitride, gall-resistant vacuum gears, zirconia ball bearings, vacuum lubricant, and Nextel insulation over copper wires.

The Swing-Probe systems were very conservatively designed, and are carefully integrated with the DIII-D control system to minimize the risk to probe and machine alike. Redundant electrical isolation and thick carbon armour give the probe an excellent chance of surviving the worst DIII-D disruptions. An open-loop control system accurately controls the probe and its position as a function of time is accurately calculated from the back-EMF of the actuator coils.

The data taken from the inboard SOL can have very low levels of turbulence and some of the I-V characteristics show a surprisingly good match to the classical Langmuir I-V function. It was therefore possible to quantitatively describe the sheath expansion process for these electrodes. A novel I-V fitter was developed that accounts for sheath expansion in a theoretically consistent way for protruding electrodes. Not only does this fitter accurately calculate the magnitude of sheath expansion effect, it improves the reliability of data analysis even in noisy conditions and significantly decreases the amount of intervention necessary for Langmuir probe analysis. This fitter can be expected to similarly improve the reliability of data analysis on other probe systems.

While clean theoretical I-V data gives good confidence in the accuracy of Langmuir probe data, further confidence is gained by validating the Swing-Probe plasma

parameters against those from existing diagnostics. The Swing-Probe data matches those from the fixed Langmuir probes, the core Thomson scattering and the divertor Thomson scattering in conditions where poloidal variations of the plasma properties are expected to be small.

Swing-Probe data were taken over a wide range of plasma conditions, from simple-as-possible ohmic L-mode, to conditions with high-power ELMs, to high-powered H-mode conditions. The harshest conditions include a 10MW shot with a line average density of $7e^{19}$ m^{-3}. The lower Swing-Probe reached the LCFS with a parallel heat flux around 60 MW/m^2. These measurements are providing a useful insight into the critically under-diagnosed inboard SOL of DIII-D plasmas. In L-mode conditions, we find that the electron pressure measured by the lower Swing-Probe at the separatrix is higher than that measured at the separatrix by the core Thomson. This elevated downstream electron pressure could be caused by ExB circulation in the divertor, but further study is needed.

In an inner wall limited plasma experiment in DIII-D requested by ITER, we have made the first probe measurements of a narrow heat flux feature. We see good agreement in T_e and n_e data among the lower Swing-Probe, infrared measurements of the centerpost, the core Thomson, divertor Thomson and the Midplane Reciprocating Probe, but the short heat flux feature is only regularly and clearly observed in the Swing-Probe data. This could be due to the low level of scatter in the Swing-Probe data, or because the short heat flux feature is weaker/absent on the outboard side SOL. The Swing-Probe data, in combination with measurements of the short heat flux feature made in other machines, lead us to conclude that the narrow heat flux feature just outside the LCFS is very likely real and that it poses a significant risk to ITER centerpost plasma facing modules. Based on these data, and similar data from other tokamaks, the ITER design team decided in April 2014 to redesign these plasma facing modules to accommodate for the short heat flux feature.

10.2 Original Contributions

This thesis contains a number of significant contributions to the field of fusion research related to physics, analysis methodology and engineering. These include:

- The first Langmuir I-V fitter to account for sheath-expansion for protruding electrodes in a theoretically consistent way. This analysis method can be applied to any protruding electrodes to improve the reliability of Langmuir probe data analysis.
- The first active measurements of the inboard-side scrape-off layer of DIII-D plasmas. These measurements are providing insight about the DIII-D SOL and have been used to benchmark edge codes. They are particularly useful in conditions where the inboard target is detached and where high neutral recycling perturbs measurements made by the inboard fixed Langmuir probes.
- The first active measurements of the narrow heat flux feature in inner wall limited plasmas. These measurements were central to the ITER design team's decision to reshape the plasma facing modules being built for the ITER centerpost.
- The development, design, and deployment of a novel probe system in DIII-D. The Swing-Probes provide a significant enhancement to the diagnostic capability of DIII-D. These probe assemblies are an impressive feat of engineering as they are:
    - The first in-vessel probes that swing vertically through the edge plasma to make measurements across varying poloidal locations.
    - The first precision moving parts in DIII-D to be actuated internally.
    - The first use of magnetic coil actuators/encoders in DIII-D.
    - The first use of zirconia ball bearings, grade A boron nitride, vacuum lubricants and gall resistant gears inside of the DIII-D vessel.

As the Swing-Probes were designed with the help of many collaborators, it's important to specify what my main contributions were.

- Overall concept and design
- Determination of the safe operating parameters, probe arming/firing interlock requirements, and plasma dwell time. Disaster scenarios were carefully considered, and high priority was placed on the smooth operation of the DIII-D program.
- Determination of the necessary levels of electrical isolation between the probe and vessel ground as well as between the pyrolytic graphite washers and the probe arm.
- Probe head and electrode design (with help from many)
- Risk assessment and development of the in-vacuum moving parts, including the vacuum lubrication certification, testing of the gearboxes, and the zirconia ball bearings.
- Position determination software using coil back-EMF.
- The physical model, including the eddy current drag, and the programming of the actuator coil control waveforms.
- All in-vessel maintenance, upgrades and repairs.

References

[1] M. Greenwald, Verification and validation for magnetic fusion, Phys. Plasmas. 17 (2010) 058101.

[2] N. Smick, B. LaBombard, C.S.S. Pitcher, Plasma profiles and flows in the high-field side scrape-off layer in Alcator C-Mod, J. Nucl. Mater. 337-339 (2005) 281–285.

[3] N. Smick, B. LaBombard, Wall scanning probe for high-field side plasma measurements on Alcator C-Mod, Rev. Sci. Instrum. 80 (2009) 23502.

[4] J. Luxon, A design retrospective of the DIII-D tokamak, Nucl. Fusion. 614 (2002).

[5] C.S. Pitcher, H. Bosch, A. Carlson, C. Dora, A. Field, A. Hermann, et al., First results with the in-vessel probe on ASDEX-UPGRADE, Proc. 20th EPS Conf. Control. Fusion Plasma Physics, Lisboa. 17C (1993) p. 291.

[6] C. Pitcher, H. Bosch, K. Büchl, A. Field, The effect of density on divertor conditions in ASDEX-Upgrade, J. Nucl. Mater. 222 (1995) 213–217.

[7] A. Schmid, A. Herrmann, V. Rohde, M. Maraschek, H.W.M.A.U. Team, H.W. Müller, Magnetically driven filament probe, Rev. Sci. Instrum. 78 (2007) 53502.

[8] J.P. Gunn, J.Y. Pascal, A magnetically driven reciprocating probe for tokamak scrape-off layer, Rev. Sci. Instrum. 82 (2011) 123505.

[9] K. Holtrop, P. Taylor, P. Anderson, W. Cary, M. Dunlap, R. Callis, et al., Vacuum Requirements for the DIII-D Vessel and Associated Diagnostics (DIII-D Internal Report), 2012.

[10] J.G. Watkins, X-point Probe Disruption Memo, 1994.

[11] T. Hender, J. Wesley, J. Bialek, A. Bondeson, A. Boozer, R. Buttery, et al., Progress in the ITER Physics Basis Chapter 3: MHD stability, operational limits and disruptions, Nucl. Fusion. 47 (2007) S128–S202.

[12] J.G. Watkins, J. Salmonson, D.N.N. Hill, R. Moyer, R. Doerner, R. Lehmer, et al., A fast scanning probe for DIII-D, Rev. Sci. Instrum. 63 (1992) 4728–4730.

[13] J.G. Watkins, J. Hunter, B. Tafoya, M. Ulrickson, R.D. Watson, R.A. Moyer, et al., Fast reciprocating Langmuir probe for the DIII-D divertor, Rev. Sci. Instrum. 68 (1997) 373–376.

[14] J.A. Boedo, N. Crocker, L. Chousal, R. Hernandez, J. Chalfant, H. Kugel, et al., Fast scanning probe for the NSTX spherical tokamak, Rev. Sci. Instrum. 80 (2009) 123506.

[15] Graphite Machining Inc. Brochure, (2013).

[16] H.D. Young Sears, Francis Weston,,, University physics., Addison-Wesley Pub. Co., Reading, Mass., 1992.

[17] Combat Solid Boron Nitride Technical Overview Datasheet, 2003.

[18] E. David R Lide, CRC Handbook of Chemistry and Physics, Internet Version 2007, (87th Edition), Taylor Fr. Boca Rat. FL. (2007) 0.

[19] D.H. Buckley, Friction, Wear, And Lubrication in Vacuum, NASA TN. E-5628 (1970).

[20] G.W. Brown, J.A. Koski, R.D. Watson, Engineering design of the ALT-II pump limiter for TEXTOR, Fusion Technol. 8 (1985) 1729–1734.

[21] G.W. Brown, C.D.D. Croessmann, J.G. Watkins, J.C. Banks, Environmental Effects on Friction and Wear of the ALT-II Limiter Blade Positioning Mechanisms, (1988).

[22] G.W. Brown, J.G. Watkins, A remotely controlled positioning system for the ALT-II limiter blades, Fusion Eng. Des. 13 (1990) 299–305.

[23] J.G. Watkins, Private Communication, (2011).

[24] W.A. Brainard, The Thermal Stability and Friction of the Disulfides, Diselenides, and Ditellurides of Molybdenum and Tungsten in Vacuum (10^-9 to 10^-6 TORR), (1969).

[25] A.R. Lansdown, Molybdenum disulphide lubrication, Elsevier, New York, 1999.

[26] D.G. Flom, A.J. Halter, Lubricants for the Space Environment., in: AIChE Mater. Conf. Symp. Mater. Re-Entry Spacecr. Syst. - Spacecr. Mater. Part 2, Philadelphia, 1968.

[27] O. Sauter, S.Y. Medvedev, Tokamak coordinate conventions:, Comput. Phys. Commun. 184 (2013) 293–302.

[28] D. Halliday, R. Resnick, J. Walker, Fundamentals of physics extended, John Wiley & Sons, 2010.

[29] C.K. Tsui, Swing-Probe Operator's Manual (DIII-D Internal Report), 2014.

[30] C.S. Pitcher, B. LaBombard, R. Danforth, W. Pina, M. Silveira, B. Parkin, Divertor bypass in the Alcator C-Mod Tokamak, Rev. Sci. Instrum. 72 (2001) 103.

[31] C. Markwardt, Non-linear least squares fitting in IDL with MPFIT, arXiv Prepr. arXiv0902.2850. (2009).

[32] J. Moré, The Levenberg-Marquardt algorithm: implementation and theory, Numer. Anal. (1978).

[33] J.A. Boedo, D.L. Rudakov, J.G. Watkins, X-Point Probe Tips History, 2011.

[34] J.A. Boedo, D.L. Rudakov, J.G. Watkins, Midplane Probe Tips History, 2011.

[35] S.A. Self, Exact Solution of the Collisionless Plasma-Sheath Equation, Phys. Fluids. 6 (1963) 1762.

[36] R. Dejarnac, J.P. Gunn, J. Stöckel, J. Adámek, J. Brotánková, C. Ioniþã, Study of ion sheath expansion and anisotropy of the electron parallel energy distribution in the CASTOR tokamak, Plasma Phys. Control. Fusion. 49 (2007) 1791–1808.

[37] U. Wolters, T. Daube, K.-U. Riemann, K. Wiesemann, An experimental study of ion currents to a flush mounted Langmuir probe in an ECR discharge, Plasma Phys. Control. Fusion. 41 (1999) 721–732.

[38] M. Weinlich, a. Carlson, Flush mounted Langmuir probes in an oblique magnetic field, Phys. Plasmas. 4 (1997) 2151.

[39] J. Gunn, C. Boucher, Flush-mounted Probes in the TdeV Tokamak: Theory and Experiment, Contrib. Plasma Phys. 36 (1996) 45–52.

[40] P.C. Stangeby, The Plasma Boundary of Magnetic Fusion Devices, Taylor & Francis Group, 2000.

[41] T.J. Dolan, Fusion Research: Principles, Experiments and Technology, Pergamon Pr, 1981.

[42] I.H. Hutchinson, Principles of Plasma Diagnostics, Cambridge University Press, 2005.

[43] G.D. Porter, J. Moller, M. Brown, C. Lasnier, Characterization of the separatrix plasma parameters in DIII-D, Phys. Plasmas. 5 (1998) 1410.

[44] G.D. Porter, Private Communication: UEDGE 152835_1800 Internal Memo, (2014).

[45] J.D. Elder, P.C. Stangeby, C.K. Tsui, J.G. Watkins, A.W. Leonard, B.D. Bray, et al., OEDGE Modeling of DIII-D Density Scan Discharges Leading to Detachment, APS DPP Poster. (2013) 1–17.

[46] P.C. Stangeby, J.M. Canik, D. Elder, C.J. Lasnier, A.W. Leonard, D. Eldond, et al., Identifying the location of the OMP separatrix in DIII-D using power accounting, Nucl. Fusion. (n.d.).

[47] A.W. Leonard, Private Communication, (2014).

[48] J.A. Boedo, M.J. Schaffer, R. Maingi, C.J. Lasnier, Electric field-induced plasma convection in tokamak divertors, Phys. Plasmas. 7 (2000) 1075.

[49] M.J. Schaffer, B.D. Bray, J. a. Boedo, T.N. Carlstrom, R.J. Colchin, C.-L. Hsieh, et al., E×B circulation at the tokamak divertor X point, Phys. Plasmas. 8 (2001) 2118.

[50] S. Potzel, M. Wischmeier, M. Bernert, R. Dux, H.W. Müller, a. Scarabosio, A new experimental classification of divertor detachment in ASDEX Upgrade, Nucl. Fusion. 54 (2014) 013001.

[51] S. Potzel, M. Wischmeier, R. Dux, H.W. Muller, F. Reimold, A. Scarabosio, et al., Formation of the high density front in the inner far SOL at ASDEX Upgrade and JET, Plasma Surf. Interact. Conf. Present. (2014).

[52] R. a. Pitts, S. Carpentier, F. Escourbiac, Physics basis and design of the ITER plasma-facing components, J. Nucl. Mater. 415 (2011) S957–S964.

[53] T. Casper, Y. Gribov, a. Kavin, V. Lukash, R. Khayrutdinov, H. Fujieda, et al., Development of the ITER baseline inductive scenario, Nucl. Fusion. 54 (2014) 013005.

[54] G. Arnoux, T. Farley, C. Silva, S. Devaux, M. Firdaouss, D. Frigione, et al., Scrape-off layer properties of ITER-like limiter start-up plasmas in JET, Nucl. Fusion. 53 (2013) 073016.

[55] J. Horacek, What did we learn out from 300 COMPASS ITER-like HFS limiter experiments with IRcam and probes?, COMPASS IRCAM Meeting Presentation. (2013).

[56] P. Vondracek, J. Horacek, R. Panek, Recessed double roof limiter experiment, Compass Present. (2014).

[57] D.L. Rudakov, J. a. Boedo, R. a. Pitts, G.L. Jackson, C.J. Lasnier, a. W. Leonard, et al., SOL width in limited versus diverted discharges in DIII-D, J. Nucl. Mater. 415 (2011) S387–S390.

[58] F. Nespoli, G. Canal, B. Labit, I. Furno, Heat load profiles in limited L-mode plasmas, Ecole Polytechnique Federal De Lausanne Presentation. (2014) 1–23.

[59] B. LaBombard, J.L. Terry, C. Theiler, S. Wolfe, Initial measurements of SOL heat flux profiles in inner-wall limited plasmas (in support of ITER urgent need), C-Mod Sci. Meet. Present. (2014).

[60] M. Kočan, J.P. Gunn, S. Carpentier-Chouchana, a. Herrmann, a. Kirk, M. Komm, et al., Measurements of ion energies in the tokamak plasma boundary, J. Nucl. Mater. 415 (2011) S1133–S1138.

[61] F. Hinton, M. Chu, Neoclassical ion transport through the separatrix in divertor tokamaks, Nucl. Fusion. 345 (1985).

[62] J. Wesson, Tokamaks (Oxford Engineering Science Series), Oxford University Press, USA, 1997.

[63] R.J. Goldston, Heuristic drift-based model of the power scrape-off width in low-gas-puff H-mode tokamaks, Nucl. Fusion. 52 (2012) 013009.

[64] V. Petrov, Characteristics of steady-state plasma flow in the tokamak limiter scrape-off layer, Nucl. Fusion. 259 (1984).

[65] A. Nedospasov, V. Petrov, G. Fidel'man, Plasma convection in the poloidal limiter shadow of a tokamak, Nucl. Fusion. 21 (1985).

[66] A.V. Chankin, V.A. Vershkov, Non-ambipolarity of perpendicular plasma transport and asymmetry of particle flow onto the tokamak rail limiter, J. Nucl. Mater. 162-164 (1989) 208–213.

[67] M. Kovanen, R. Reichle, Modelling of the observed particle and heat fluxes in the X-point region in JET, Nucl. Fusion. 863 (1992).